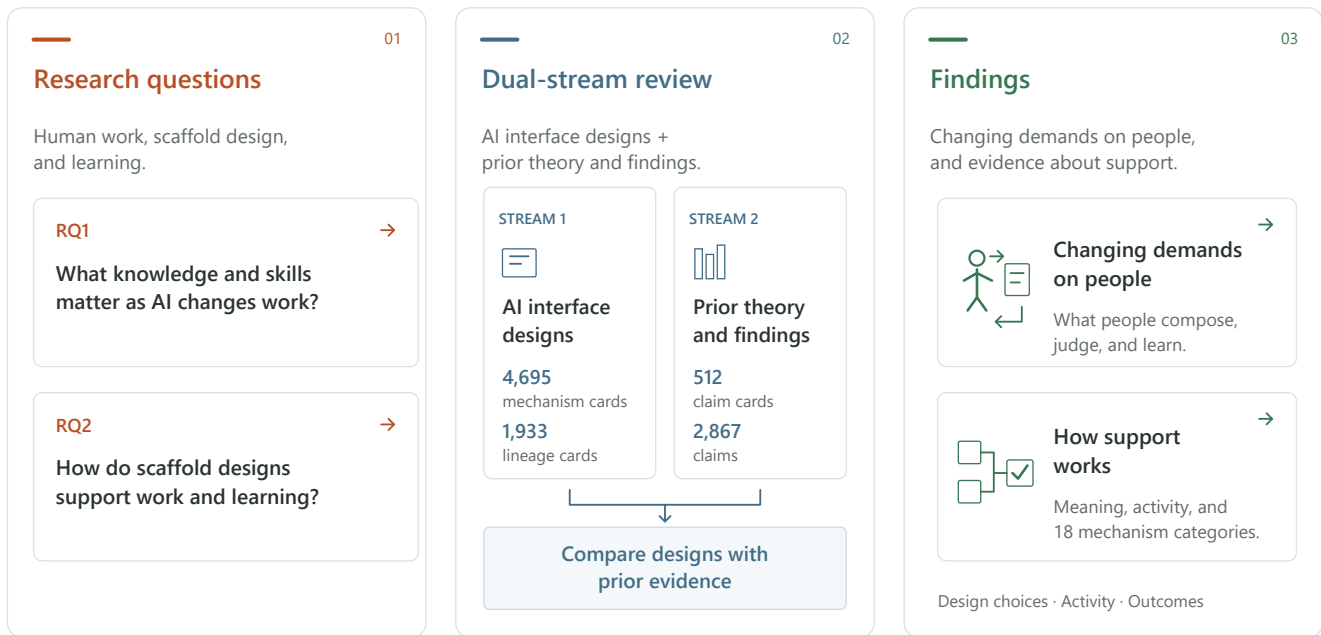


# Beyond Getting Work Done: A Landscape Review of Semantic Scaffolding for Learning with AI

Sirui Tao  
University of California San Diego  
La Jolla, California, USA

Steven P. Dow  
University of California San Diego  
La Jolla, California, USA



**Figure 1: The review at a glance. Two research questions guide a dual-stream review of AI interface designs and prior theory and findings. The findings examine how work changes and how support is designed and evaluated. Card counts cover the selected AI papers in Stream 1 and retained sources in Stream 2.**

## ABSTRACT

As AI changes what people can delegate, it also changes what they have opportunities to understand, practice, and learn. We examine how interfaces can support development within useful work through *semantic scaffolding*: designed structures that make task meaning available for human reasoning and action. Our dual-stream landscape review connects 4,638 AI-involved papers from 40 venues (2019–2026) with 512 sources on prior theory and empirical findings. GenAI papers more often delegate generation while supporting people in creating, editing, and repairing artifacts. The examples reveal demands on interpreting material, judging its grounds, and making reasoning usable by others. Eighteen mechanism categories map how represented meaning connects to human activity. Their comparisons show why supplying material can create practice as well as remove it, and why what a representation makes inspectable matters to the judgment performed. Evaluations also separate better work from learning: CodeVoyager improved immediate closed-book comprehension without a completion advantage, while required reflection reduced completed practice from 3.4 to 2.0 problems within a fixed time. We argue for choosing capabilities with people,

designing support that evolves with their work and development, and connecting evidence about process, artifacts, and learning. The review provides concrete design alternatives and a research agenda for learning within everyday AI tools.

## CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in interaction design**; *Graphical user interfaces*; *Collaborative interaction*; *Natural language interfaces*.

## KEYWORDS

Semantic scaffolding, learning, human–AI interaction, external cognition, landscape review, interface mechanisms

## 1 INTRODUCTION

AI tools help people write, program, analyze information, and make decisions. They can supply material people work with: alternative passages, intermediate plans, explanations, and visual representations [63, 72, 154]. Interfaces let people inspect, steer, and revise this material as a task develops. A writer can work from generated

alternatives; a programmer can inspect a proposed solution; a team can reason through a shared model. What people need to know changes with what these tools make possible. This raises a design question beyond completing the current task: what should people have opportunities to understand, practice, and learn?

We take the position that people should be able to develop through their work even as tools become more capable. The capability worth developing is a moving target: producing an artifact, judging alternatives, and directing delegated work can matter differently to different people. Helpful output alone gives little guidance about which of these to practice. The assistance dilemma captures part of this tension: support can enable progress while taking over an activity the learner needs to practice [73]. Learning depends on how people engage with the material, not simply on how much interaction an interface requires [28]. A critical review of HCI research also finds that claims about critical thinking often outrun clear definitions and assessments [62]. Our review finds a related gap in evidence over time: among 396 AI-involved sources in the selected outcome appraisal, only 31 assessed capability in a later session. We therefore need to connect the work an interface makes possible with opportunities to develop and evidence of what carries into later work.

*Semantic scaffolding* is our account of how designed structures make task meaning available for people to reason and act with. A concept map relates ideas; a comparison makes differences available for judgment; an editable plan connects actions to a goal. Their learning potential lies in what people do through them: explaining a relation, choosing a criterion, or revising a plan in response to evidence. This gives support a dual role: helping someone make progress now and creating opportunities to develop understanding, skill, and judgment through that work. Scaffolding theory explains responsive support that fades as responsibility passes to the learner [141, 151]; cognitive-artifact research explains representations that remain useful as people become more skilled [107, 115]. Generative and malleable interfaces make it possible to create, reshape, and preserve these structures during use [23, 67, 100]. The design challenge is to make support fit both the decision at hand and the person's development: some assistance can fade, while useful structures persist or evolve. Section 3 develops this account.

We investigate this challenge through an LLM-assisted, dual-stream landscape review. Language models processed the papers into structured evidence cards with supporting passages and page references. Stream 1 screened 35,480 abstract-bearing records from 40 venues spanning 2019–2026, selecting 7,372 for full-text processing. The current card corpus contains 4,638 AI-involved papers. Inclusion requires AI involvement, an implemented or studied structure with task meaning, and a person who inspects or acts through it. Stream 2 starts from 795 candidate works, of which 543 have full-text claim cards, and contributes 512 retained sources on prior theory and empirical findings, represented by 2,867 claims. We connect the designs to human activity, their theoretical rationale, and evidence about performance and learning. We ask:

- **RQ1.** What knowledge and skills matter as AI changes work?
- **RQ2.** How do scaffold designs support work and learning?

We approach RQ1 through the workflows and cognitive work that the reviewed designs support. GenAI papers more often delegate generation to the tool while also supporting people in creating, editing, and repairing artifacts. The examples reveal demands on knowledge: interpreting generated material, judging its grounds, and making one's reasoning usable by another person. These suggest learning priorities to consider alongside people's purposes. For RQ2, 18 mechanism categories distinguish how interfaces make task meaning available and how people work with it. The comparisons show why an account of learning needs both: CodeVoyager improved immediate comprehension without a detected advantage in assisted completion, while required reflection used practice time without establishing a transfer advantage [7, 70]. These findings motivate support that creates opportunities for meaningful practice and changes as people develop, while retaining resources useful to their work.

This paper contributes:

- a landscape of changing human work with AI, connecting quantitative technology comparisons to the knowledge and judgment that particular designs call for;
- an illustrated account of semantic-scaffolding mechanisms that connects represented task meaning, human activity, and concrete examples;
- an evidence-grounded position on learning within everyday tools, linking design choices and prior theory to process, artifact, and learning outcomes and to questions for future systems and evaluations.

## 2 BACKGROUND

Research on AI assistance explains ways to distribute work between people and tools. Learning research asks how people develop through that activity, and interface research offers ways to change its representations. Bringing these accounts together helps us ask which opportunities for development belong within tools for everyday work.

### 2.1 Assisting Work and Supporting Learning

Distributed cognition follows work across people, tools, and representations [52]. External resources can become part of how people remember and reason [32, 61]. Offloading also changes what people retain: expected access to information shifts memory toward where to find it [134], while automation can leave operators with the hardest cases and little practice [8]. These accounts motivate examining the activity left to the person alongside the output of the person-tool system.

AI interaction research makes that division of work a design choice. Mixed-initiative interfaces combine automation and direct manipulation [56]; interactive machine learning centers the user-system loop [3]. Explainable-AI frameworks separate understanding, trust, debugging, and decision support [105], while teaming and learning-support research adds capability and transfer [22, 49]. Generative systems expose prompts, plans, alternatives, and evidence for people to inspect and steer [24, 72, 137, 153]. When systems also supply criteria and explanations, people may need to judge the guidance as well as the answer.

Learning research explains why reducing effort is an incomplete design goal. The assistance dilemma asks when giving or withholding help enables learning [73]; ICAP distinguishes the engagement an activity invites [28]. Guidance useful to novices can become redundant with expertise [65], and difficulty during practice can support later retention [15]. People also trade accuracy against effort [114]. Together, these accounts direct attention to which work assistance removes, which work remains worth doing, and how that balance changes as a person develops.

## 2.2 Interfaces People Can Inspect and Reshape

Recent design spaces make representational choices explicit. Min et al. separate an abstraction's schema from its representation and examine granularity, presentation, and transformation [101]. Kim et al. organize how generative interfaces elicit preferences along six axes, including when a request appears and which part of an artifact it concerns [68]. These accounts show how interface structure can help people express and work with task knowledge.

Interfaces can adapt controls to a person's abilities [45, 149]. Generative systems extend what can be constructed during use [23]: Google's Generative UI work creates interactive pages and simulations in response to a prompt [81, 82]. Malleable overview-detail interfaces let people change the content, composition, and layout of views [100]; Meridian provides a specification and tools for building those transformations [102]. Maru preserves a user's information-architecture rules across generation [67]. Generating, reshaping, and preserving support offer different ways an interface might respond to changing work and developing expertise.

## 2.3 What Prior Reviews Establish and Leave Open

Scaffolding syntheses connect contingent support and learner activity to learning [11, 141]. Self-explanation research identifies practice exposure and redundant prompting as consequential conditions [122]. Krakauer distinguishes artifacts whose advantage remains only during use from those that leave capability behind [74]. Inie et al.'s critical scoping review examines how HCI research about AI defines and assesses critical thinking [62]. This work supplies both substantive learning principles and reasons to examine whether a study's measure tests its proposed benefit.

HCI landscapes offer complementary ways to compare designs. Creativity support reviews relate tools to practice and evaluation [42], organize traces of creative activity [51], and examine generative tools in artists' work [72]. Design spaces organize recurring choices and their trade-offs [94]; their visual form shapes what designers explore [14, 79]. Human-AI guidelines already address intelligibility and correction [4]. Citeology, Royal Constellations, and HCI Research Trails make relationships among cited works navigable [19, 98, 104].

These bodies of work supply different parts of the design problem. Learning theory explains why an activity might develop a capability; interface research shows ways to make that activity possible; evaluations establish what happened in a particular setting. Our review connects these parts across domains. An explanation, an editable plan, and a shared view can all support reasoning, but leave different work to the person. The next section develops semantic

scaffolding as an account of that relationship and its potential for learning.

## 3 WHAT IS SEMANTIC SCAFFOLDING?

A tool changes both what a person can accomplish and what they have occasion to learn. These changes need not move together. A plan can help someone finish an unfamiliar task; constructing or revising that plan can help them understand how its steps fit together. Which support is useful depends on the work, what the person already understands, and what they want to become able to do. We develop semantic scaffolding from this relationship, asking what earlier accounts explain, what generative AI changes, and what should persist as the work itself changes.

### 3.1 What Is Old: Tools Change the Work of Thinking

Cognitive artifacts reorganize a task by storing, displaying, or transforming information [107]. A checklist turns remembering into preparing, consulting, and interpreting a list. A calculator performs arithmetic while the person formulates a problem and judges its result. Such tools change the knowledge needed to work well with them. External-representation research explains why even the same information supports different operations when arranged differently [126, 160]. People can also change their surroundings to aid reasoning and coordinate several representations as they work [2, 71]. The design question is therefore which thinking a structure performs and which thinking it makes possible for the person.

That relationship develops through use. Wendy Mackay's account of *co-adaptation* describes people repurposing email filters from sorting messages before reading to filing them afterwards, and developers responding to that practice [93]. People learn ways of working with a tool and change its role in their work. A scaffold designed around an initial task or an assumed novice can consequently outlast the need that motivated it, acquire a new purpose, or need to be changed.

McLuhan's hot and cool media distinguish the detail a medium supplies and the participation it asks of an audience [99]. Our proposed hot/cold information lens extends this idea to *supplied inference*: how much interpretation a representation provides and what its user works out. Observations, a structured comparison, and a recommendation may concern the same problem while doing different parts of the reasoning for the reader. This extension helps articulate a design choice left open by visual form alone. Supplying an inference can free attention for another judgment; withholding it can create an occasion to reason. Its learning value depends on which activity the person undertakes.

### 3.2 What Is New: The Support Can Be Generated and Reworked

Generative AI can supply both material for a task and structures for working with it. AI Chains exposes intermediate steps that people can inspect and change [154]; Luminate supplies dimensions for comparing ideas; VIVID generates rubrics and expected answers alongside candidate dialogues [30, 137]. Design rationale has long connected questions, options, and criteria [94]. Generating these together places a consequential choice in ordinary use: a system can

help formulate the problem and its standards as well as propose an answer. A person may need to judge the plan or criterion through which the answer was produced, then decide what to preserve or change.

The representation can also change during the work. Generative interfaces create interactive material; malleable interfaces let people reshape views; and persistent information architectures retain organizing rules across generation [23, 67, 100, 102]. Together these possibilities widen the choices for support: what it supplies, who shapes it, and when it changes. Fitting a new task and responding to a person's development become related design problems. As a person learns to plan, for example, a tool could shift from supplying steps to helping them compare and revise their own plans, while continuing to organize a difficult project. Changing the interface makes such designs possible; responding well requires understanding the person's goals, knowledge, and experience.

### 3.3 What Persists: People Need Opportunities to Develop

The purpose of scaffolding is to make meaningful participation possible. Wood, Bruner, and Ross describe a tutor taking over parts of a task beyond a learner's current capacity so that the learner can undertake the rest [151]. Developmental scaffolding responds to the learner, reduces assistance as capability grows, and transfers responsibility [141]. Reiser distinguishes structuring a task from making its important problems available for reasoning [120]; Quintana et al. develop these concerns into supports for inquiry [119]. These accounts make the activity worth undertaking the starting point for design. An easier task and a productive opportunity to learn can require different kinds of help.

Reflection, comparison, and analogy offer enduring ways to engage with that activity. Schön's reflective practice connects action to reconsidering a situation in light of its response [128]. Structural comparison can make relevant differences noticeable [48]; analogy brings relationships from a familiar situation to a new one [47]; variation theory asks which distinctions a learner needs to discern [96]. These approaches inform both narrowing and opening possibilities: a contrast can help someone discriminate between alternatives, while an analogy can suggest a different way to frame the problem. A tool can supply examples, organize the comparison, or prompt a prediction while leaving the target reasoning available to the person.

What deserves that practice changes with people's purposes and their tools. A writer may want to develop prose, editorial judgment, or both. Choosing one need not make the other obsolete. The persistent commitment is to people having opportunities to develop capabilities they value and can use later. Those capabilities may include working more effectively with continuing resources. Pea distinguishes developmental scaffolding that fades from continuing distributed intelligence [115]; useful support can remain distributed across people and tools [10]. We therefore need to specify what a person is becoming able to do, and with which resources, rather than making the removal of every tool the endpoint of learning.

### 3.4 Our Position: Design the Relationship between Meaning and Activity

We use *semantic scaffolding* for support that makes task meaning available through a designed structure a person can reason or act with. The meaning may be a relationship between concepts, a criterion for a choice, an intermediate step toward a goal, or evidence for a claim. The scaffold is the relationship between that structure and the person's activity. A concept map, for example, gives form to relations; exploring, explaining, or revising those relations are different uses of it. Graphologue lets people explore concepts and their connections [63]. Zong et al. use the term more specifically for explanations of domain groupings to blind and low-vision readers [165]. Our account connects these forms through the task meaning they provide and the activity they support.

This focus complements affordances and signifiers, which concern available actions and how people discover them [108]. HaLL-Mark's highlight both signals authorship and connects a passage to its production history [55]. In Codestrates v2, a tree both organized code and governed execution order, a consequence a participant discovered during use [18]. Learning to operate a control and understanding the task relationship it exposes are connected but distinguishable achievements.

For learning, the design problem is how to use this relationship to help someone develop through the work. A person might follow a supplied plan, explain why a step is needed, or construct a plan with feedback. The representation, the activity, and the opportunity to learn differ even when the final task is the same. The two roles of support—accomplishing work and developing capability—can coexist in one tool and change over time. Prompts may fade, a useful representation may remain, and the work a person chooses to undertake may grow more demanding. The broad corpus criteria in Section 4.1.3 let us examine those possibilities without assuming that every included design has already demonstrated learning. This preserves the developmental question raised by scaffolding research [115, 118, 131].

Our position is that interfaces should make room for this development within useful work. To inform their design, we connect the meaning a structure makes available, the activity documented in its use, and the consequence assessed. The review asks what knowledge and skills that work calls for, how mechanisms support it, and which proposed learning opportunities the evaluations support, challenge, or leave untested.

## 4 METHODS

We reviewed two bodies of work (Figure 2). Stream 1 is a systematic corpus of AI-involved systems and studies from fixed venues and years; it supplies the designs we map. Stream 2 collects theories and empirical studies of how support works, including studies without AI; we interpret the designs against it. The two are counted separately. Scoping-review guidance shaped selection, extraction, and reporting [6, 80, 139], and earlier HCI landscape reviews shaped the venue frame and synthesis [42, 51].

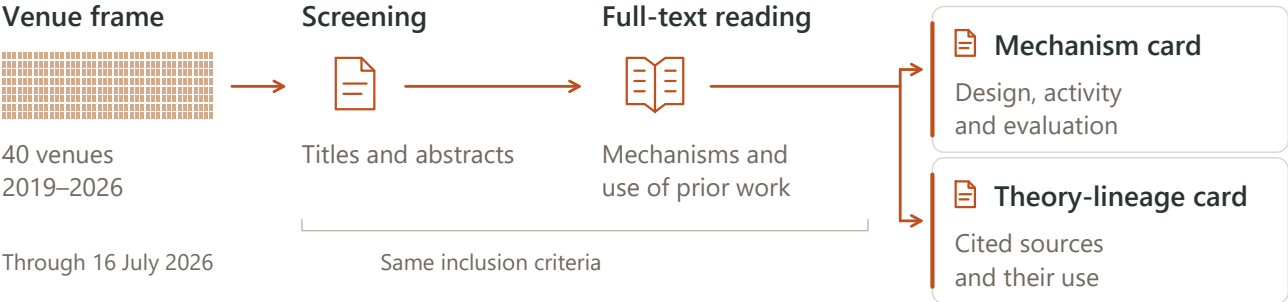
### 4.1 Literature Review

**4.1.1 Venues.** Stream 1 covers 40 venues in HCI, design and creativity, visualization, graphics, learning and education, software

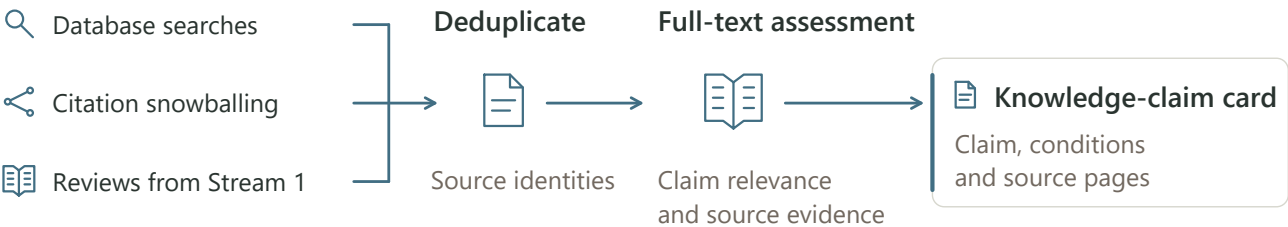


# Literature selection and evidence extraction

## Stream 1 · AI-involved designs



## Stream 2 · Prior theory and findings



**Figure 2: Literature selection and evidence extraction.** Stream 1 runs from a venue frame through screening and full-text reading to mechanism and theory-lineage cards. Stream 2 combines three discovery routes, removes duplicates, and records each source’s claims on a knowledge-claim card.

engineering, and AI and language, published from 2019 to 2026 (Figure 3). Papers are dated by their version of record; the selection figure, the design space, and the trend figures cover papers published by 16 July 2026. Ineligible publication types were excluded by rule, and citation counts played no part.

**4.1.2 Screening.** Two language models separately screened the 35,480 records with an abstract; the 2,498 without one were not screened. Their verdicts, a review of their disagreements, and (for the first reading list) PDF availability determined the 7,372 records selected for full-text reading. One language model processed each full text and recorded whether it met the criteria, with supporting passages and page references.

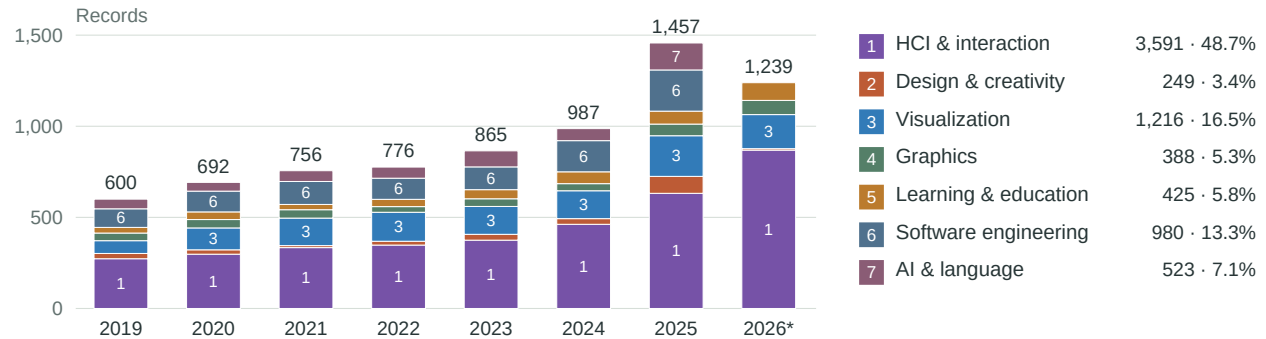
The AI requirement was added partway through full-text reading, so three separate model passes rechecked the 1,154 earlier cards that left the question open. A paper left the corpus only when all three found no AI, with low uncertainty and a quoted passage (492 cards). The corpus holds 4,638 papers, and 1,076 of them form a learning subset: their support has a documented learning purpose, or the study examines its learning consequences. Another 475 papers without AI form a labeled comparison set outside the systematic corpus counts; the text marks any it uses as an example.

**4.1.3 Inclusion Criteria.** Stream 1 includes a paper when three conditions hold, at screening and at full text (Figure 4): AI is part of the system or the study; the paper implements or studies a designed structure with task meaning; and a person inspects or acts through that structure. AI is part of a paper when an AI system is built, used, evaluated, or simulated (including Wizard-of-Oz systems presented as AI), or when AI is what the study designs, explains, or labels. Re-wording a chatbot’s replies (hedges, tone, length) is not a designed structure. Workshop templates, cards, storyboards, and probes used to elicit views count only when the paper builds and evaluates them as its contribution. Reviews, surveys, and meta-analyses go to Stream 2. A design can qualify without a user study; its benefits then count as intended uses, not measured effects. Learning purpose, measured outcomes, adaptation, fading, and the transfer of responsibility are recorded but do not decide inclusion.

**4.1.4 Theories and Empirical Studies.** Stream 2 gathers theories, design principles, and empirical findings about how support works. Its sources came from database searches, backward and forward citation snowballing [150], and reviews found in Stream 1, and every source received the same full-text assessment of relevance and support. Of 795 candidate works, language models processed 543

## Selected records by year and venue group

7,372 records selected for full-text reading; venue groups describe disciplines, not application domains.



### Each venue: selected records and share of the 7,372 selections

|         |               |                  |            |                  |            |          |            |
|---------|---------------|------------------|------------|------------------|------------|----------|------------|
| 1 CHI   | 1,548 · 21.0% | 1 ECSCW          | 10 · 0.1%  | 2 Design Studies | 13 · 0.2%  | 5 AIED   | 58 · 0.8%  |
| 1 UIST  | 306 · 4.2%    | 1 GROUP          | 6 · 0.1%   | 2 IDETC          | 151 · 2.0% | 5 IJAIED | 37 · 0.5%  |
| 1 IUI   | 359 · 4.9%    | 1 ASSETS         | 84 · 1.1%  | 3 VIS            | 394 · 5.3% | 5 EDM    | 103 · 1.4% |
| 1 DIS   | 232 · 3.1%    | 1 HRI            | 187 · 2.5% | 3 EuroVis        | 111 · 1.5% | 6 CHASE  | 55 · 0.7%  |
| 1 CSCW  | 258 · 3.5%    | 1 IVA            | 82 · 1.1%  | 3 TVCG           | 635 · 8.6% | 6 ICSE   | 321 · 4.4% |
| 1 C&C   | 67 · 0.9%     | 1 CHIIR          | 56 · 0.8%  | 3 VL/HCC         | 76 · 1.0%  | 6 FSE    | 238 · 3.2% |
| 1 TOCHI | 45 · 0.6%     | 1 HCOMP          | 36 · 0.5%  | 4 SIGGRAPH       | 295 · 4.0% | 6 ASE    | 366 · 5.0% |
| 1 TiiS  | 102 · 1.4%    | 1 IJHCS          | 127 · 1.7% | 4 SIGGRAPH Asia  | 93 · 1.3%  | 7 ACL    | 191 · 2.6% |
| 1 EICS  | 33 · 0.4%     | 2 ICC            | 53 · 0.7%  | 5 L@S            | 41 · 0.6%  | 7 EMNLP  | 255 · 3.5% |
| 1 HAI   | 53 · 0.7%     | 2 Design Science | 32 · 0.4%  | 5 LAK            | 186 · 2.5% | 7 RecSys | 77 · 1.0%  |

\*2026 is partial: publication cutoff 16 July. Zero selections can reflect unavailable proceedings.

**Figure 3: Records selected for full-text reading, by publication year and venue group. The lower panel gives each venue's count and share of the 7,372 selections. The 2026 bar covers papers published by 16 July.**

full texts into claim cards; the model-assisted assessment excluded 31 for recorded reasons, such as falling outside the review questions or appearing in a questionable venue. The analysis uses the remaining 512 sources and their 2,867 claims. Books, foundational theory, null findings, and studies without AI are eligible.

## 4.2 Analysis

RQ1 asks what knowledge and skills matter as AI changes work. We approach this through the activities interfaces support, the operations assigned to tools and people, and their variation by technology, year, and context. Quantitative comparisons locate changes in the literature; examples explain the knowledge and judgment involved in performing the supported work. Together they identify candidate learning priorities to consider in relation to people's goals. This is an interpretation of documented designs and use, not a measurement of society-wide skill value. RQ2 asks how scaffold designs support work and learning. We connect represented task meaning to the activity a mechanism makes possible, then examine its theoretical rationale and the process, artifact, and learning outcomes assessed by studies.

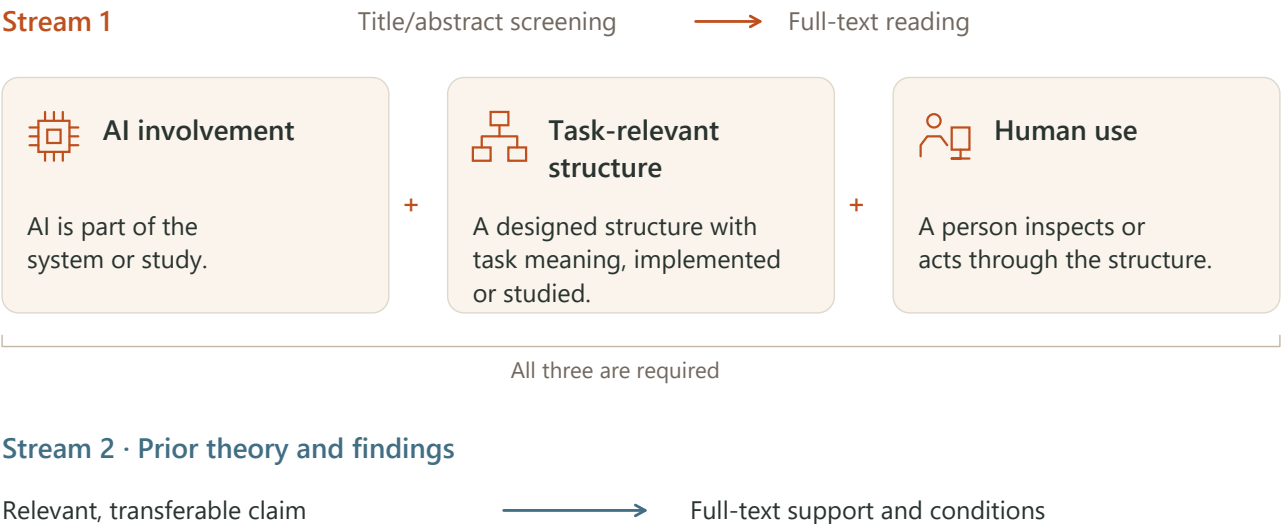
**4.2.1 What We Recorded.** A language model processed each full text into a structured evidence card, quoting the passages and pages

behind its entries.<sup>1</sup> Scientific-document benchmarks and expert-assisted literature mining offer precedents for such passage-linked records [142, 147]. The Stream 1 unit is a *mechanism*, an operation involving a designed structure with task meaning: AI Chains' workflow overview and its step editor are two mechanisms [154]. For each mechanism, the card records what the interface represents, how and when it presents that meaning, and what people can do. It also records how the work changed, what the tool took over and what the person kept, and how the design was evaluated: comparison, measures, duration, participants, and results. A second card records how the paper uses each cited source: to frame a problem, explain a mechanism, motivate the design, inform a measure, critique an approach, or in passing. The Stream 2 unit is a *knowledge claim*, a finding or theoretical argument about how support works, for whom, and when; each card holds one to eight claims with their evidence, conditions, and limits. Missing information is recorded as not reported or unclear, and author claims, measured results, and reviewer inferences stay apart (fields in Appendix A).

**4.2.2 Building the Design Space.** Closed fields take predefined values, such as whether learning was measured and for how long; open fields describe mechanisms and human work in each paper's terms. We merged the 142,877 open-field observations into 1,290

<sup>1</sup>Evidence cards: mechanism, theory-lineage, and knowledge-claim cards.

# Inclusion criteria



**Figure 4: Inclusion criteria.** Stream 1 requires all three conditions at screening and full text. Stream 2 assesses whether a claim is relevant and supported by its full text. Learning purpose and measured outcomes are recorded during extraction.

codes and grouped these into 223 categories and 92 families across 18 fields (Appendix A.3). Each grouping links to its observations, so the grouping and the assignments can be checked separately [58]. The mechanism field’s hierarchy is the design space: 5 families and 18 categories (Section 6; codebook in Appendix B). For the four-function view (Figure 12), we assigned each of the 110 mechanism codes to the functions it serves; exploring alternatives serves both representing and evaluating.

We built the vocabulary from the cards completed by 28 August 2026 and coded later cards with it. The design-space counts describe the 2,847 papers of that first set still in the corpus (2,891 for the closed fields); the annual and technology figures draw on the 4,133 papers with usable coding (Appendix A.6). A paper counts once per category, however many mechanisms it has. We classified each paper’s AI as generative or traditional AI/ML by the technology it used, not its year, and adjusted technology comparisons for year and venue, since venue mix alone can create differences (Appendix D).

**4.2.3 Comparing Designs and Studies.** We used the codes to find cases and the source passages to compare them. For each question, we compared the task, the representation supplied, the actions available, the reported use, and the assessed consequence, recording contrary and unclear cases beside supporting ones. We chose examples that fit a mechanism, varying domain and year; citation counts mattered only after fit. We retrieved prior findings and counterexamples through the seven fields coded on Stream 2 claims, such as the form of support and a claim’s conditions. The lineage cards trace how a cited theory or finding entered a design or a measure (Section 6.5); reference lists and citation counts come from Semantic Scholar (Appendix A.9.2).

In the selected lineage comparisons, we examined the connection between the source’s claim, the implemented feature, the activity attributed to the person, and the assessment. This distinguishes a design rationale from evidence that the proposed activity occurred or helped. Unavailable lineage evidence remains unknown, rather than evidence that a design has no theoretical grounding.

For study results, each contrast keeps its population, comparator, and outcome, timing, and the assistance available during assessment, and positive, mixed, null, harmful, and unmeasured results stay distinct. A whole-system result is credited to the whole system, and a learning claim counts as tested only when a study measured learning or transfer. We sorted outcomes into process, artifact, and learning (PAL), each assessed by the study or reported by participants (Section 6.6). We did not pool effects across studies (Section 9).

**4.2.4 Checking the Coding.** Scripts checked the cards’ required fields and matched their quotations against the source text, and quotations the text layer could not match were checked against page images; counts and plots are computed from the saved data. To test the open-code grouping, a separate model pass coded 1,571 extracted observations in six fields with the existing codes hidden. Agreement was low (adjusted Rand index .079 to .445), and the blind pass grouped more coarsely in five of the six fields (Appendix A.4). The mechanism vocabulary gained 12 codes across the 1,754 papers read after it was built (Appendix E).

## 4.3 AI Use Disclosure

We used language models to screen records, process full texts into evidence cards, code observations from those cards, group codes,

develop and critique the cross-case comparisons, write analysis code, and draft and revise text, tables, and figures. Claude Opus 5 and GPT-5.6 Sol screened titles and abstracts. GPT-5.6 Sol, Claude Opus 5, and Claude Opus 5.5 processed full texts into the Stream 1 cards; Claude Opus 5 and GPT-5.6 Sol wrote the Stream 2 cards; Claude Opus 5.5 ran the three-pass AI check. Claude Opus 5 merged labels into codes, and model-assisted review, including Claude Fable 5.1, grouped the codes into categories and families. GPT-6 Sol linked card evidence for the cross-case comparisons, and GPT-6 Astra developed and critiqued them. GPT-6 Astra, working through Codex, drafted the merged manuscript, and Claude Opus 5.5 revised it. Prompts and run records are kept. The authors set the research questions, scope, criteria, and card formats, directed the synthesis, and are responsible for its conclusions; people have not yet checked the coding or the cross-case interpretations (Section 9).

## 5 FINDINGS: KNOWLEDGE AND SKILLS IN CHANGING WORK

Our first question asks what knowledge and skills matter as AI changes work. The reviewed workflows suggest three connected priorities: composing with generated material, judging a result and the grounds offered for it, and interpreting work shared with others. Quantitative comparisons locate these activities in the literature; concrete designs show what people need to understand and decide within them. These are candidate learning priorities, whose value depends on people's purposes and future work. The comparisons that follow explain their relevance; Section 6 examines designs that support them and the capabilities studies actually assess.

### 5.1 Composing with generated material

Research on cognitive artifacts distinguishes what a person and tool achieve together from the task left to the person [107]. Creativity-support research likewise examines how tools support activities such as exploring possibilities and developing ideas [42, 72]. The AI designs in our review make this distinction concrete. CorpusStudio retrieves examples and patterns from prior writing for authors to interpret before composing their prose [33]. ABScribe generates alternative passages that authors keep, edit, and compare [121]. The material supplied changes the writer's starting point and the choices available next. Writing knowledge remains relevant to both designs: authors need it to judge which examples or passages fit their purpose. The opportunity to learn shifts with the work they do with that material.

**GenAI moves generation to the tool.** In 2,119 papers from 2019–2025 with coded delegated work, the tool generates the artifact in 19.3% of GenAI and 1.9% of traditional AI/ML papers (+16.6 points after adjusting for year and venue; 95% CI 12.7 to 20.5), and it more often generates responses and revisions (+9.4). Analysis moves the other way: the tool less often calculates and models (−11.5), evaluates and classifies (−9.7), or compares and finds patterns (−6.7) (Appendix D.2).

The human activity codes show a corresponding emphasis on working with artifacts (Figure 5). Among 2,426 technology-classified papers with activity coding, creating, editing, or repairing artifacts appears in 359 of 1,213 GenAI papers (29.6%) and 183 of 1,213 traditional AI/ML papers (15.1%). Specifying or refining

system behavior appears in 37.6% and 30.0%, respectively; comparing alternatives or testing scenarios appears in 11.6% and 20.0%. These pooled shares describe the activities supported by designs across 2019–2026; the delegated-work comparison above separately adjusts for year and venue in 2019–2025. Across these separately coded subsets, GenAI papers more often describe tool generation and human creation, editing, or repair. The group-level patterns motivate examining how designs connect those operations within a task.

The technology contrast concerns supported operations. The separate domain analysis finds that none of the 16 trends meets the corrected significance threshold after accounting for venue between 2019 and 2025 (Table 12). The quantitative pool contains 4,133 AI-involved papers; each comparison uses those with the required codes. Technology classification follows the system used, including earlier generators and generative hybrids. GenAI accounts for 9.2% of classified papers in 2019 and 80.7% in 2025 (Figure 56). Appendix D gives the annual counts and domain distributions; 2026 covers publication through 16 July.

Figure 6 organizes the activities people can carry out with these designs. Activities can recur or combine within a task: an author may inspect examples, specify an intended change, edit a passage, and then reconsider the examples. The map describes kinds of activity; coverage of several kinds alone does not establish support for a complete workflow.

### 5.2 Judgment needs a basis for checking

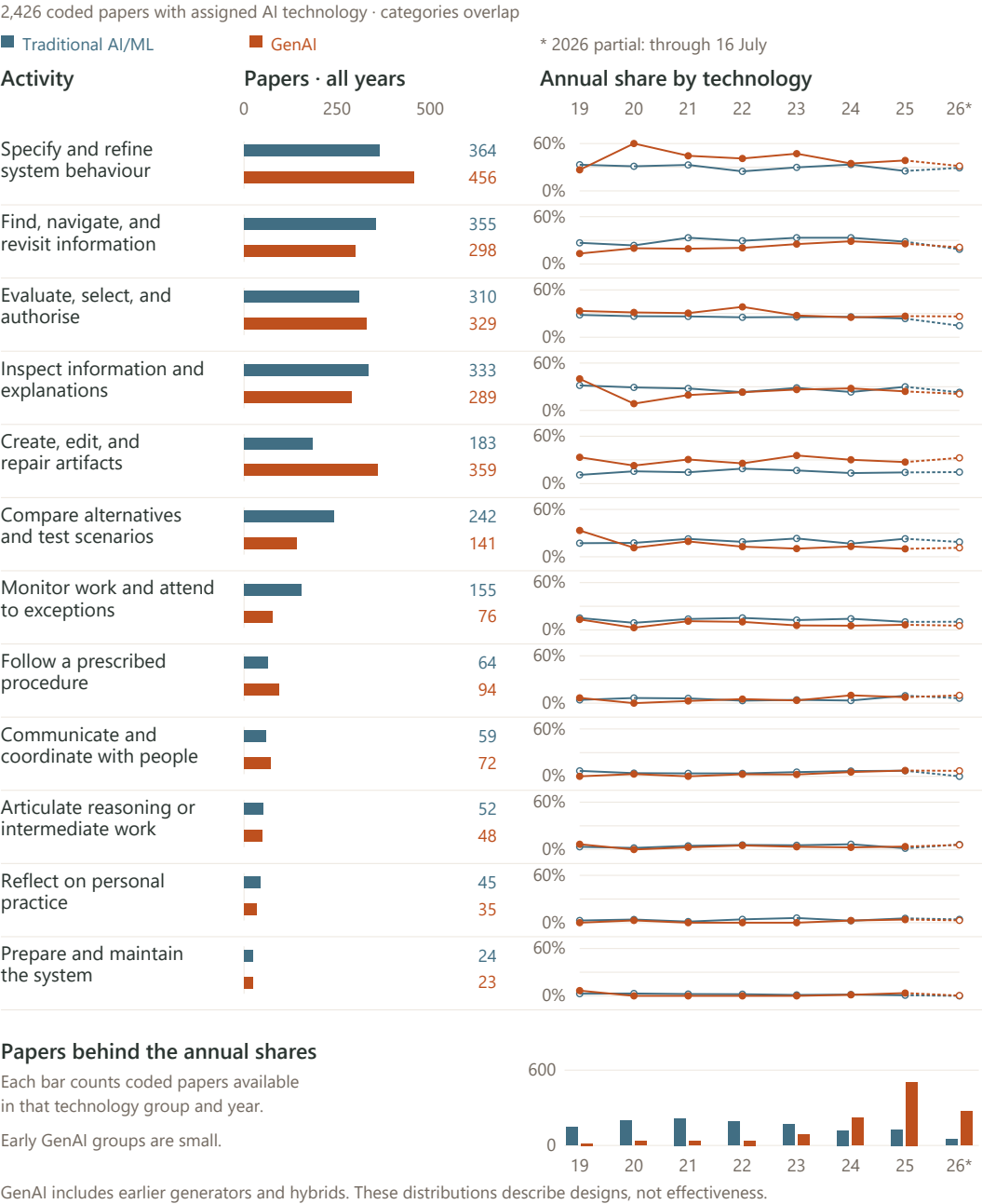
Generating material leaves consequential work to the person who will use it. The retained-role codes locate that work: deciding or authorizing appears in 740 of 3,570 papers (20.7%), specifying goals and inputs in 441 (12.4%), and checking or verifying in 309 (8.7%) (Figure 7). The tool and person panels have separate coding denominators and overlapping roles; they describe operations, not amounts of mental effort or improvements in capability.

Figure 8 shows how these operations fit together. A tool can compose material while a person judges its fit, revises it, or checks its evidence. The design question is which information makes that judgment possible.

Different references support different checks. Leap exposes values produced when generated code runs [41]; CorpusStudio offers precedents from a writing corpus [33]; ONYX shows how it interpreted an action the user taught it [124]; and Rehearsal simulates how a conversation partner might respond [129]. Execution, precedent, interpretation, and simulation answer different questions. A simulation can support rehearsal without predicting what a particular person will say; a corpus can supply examples without establishing that an idea is new.

Making a reference available is only part of the work. ONYX's full interface improved script accuracy, yet users missed or wrongly dismissed 44.8% of its follow-up questions [124]. Leap's live values did not significantly change validation success compared with a condition in which people could still run the code themselves [41]. Together these cases point to a concrete design problem: help people recognize when a check matters, interpret its information, and act on what it reveals. They make source evaluation, domain knowledge, and an understanding of the system's interpretation

### Workflow activities: volume and annual share



**Figure 5: Workflow activities by AI technology.** Bars show counts across all years; annual lines show shares within each technology group. Denominators count papers with usable activity codes and a known technology. Activities overlap, and 2026 covers papers published by 16 July. Shares are not adjusted for year or venue.

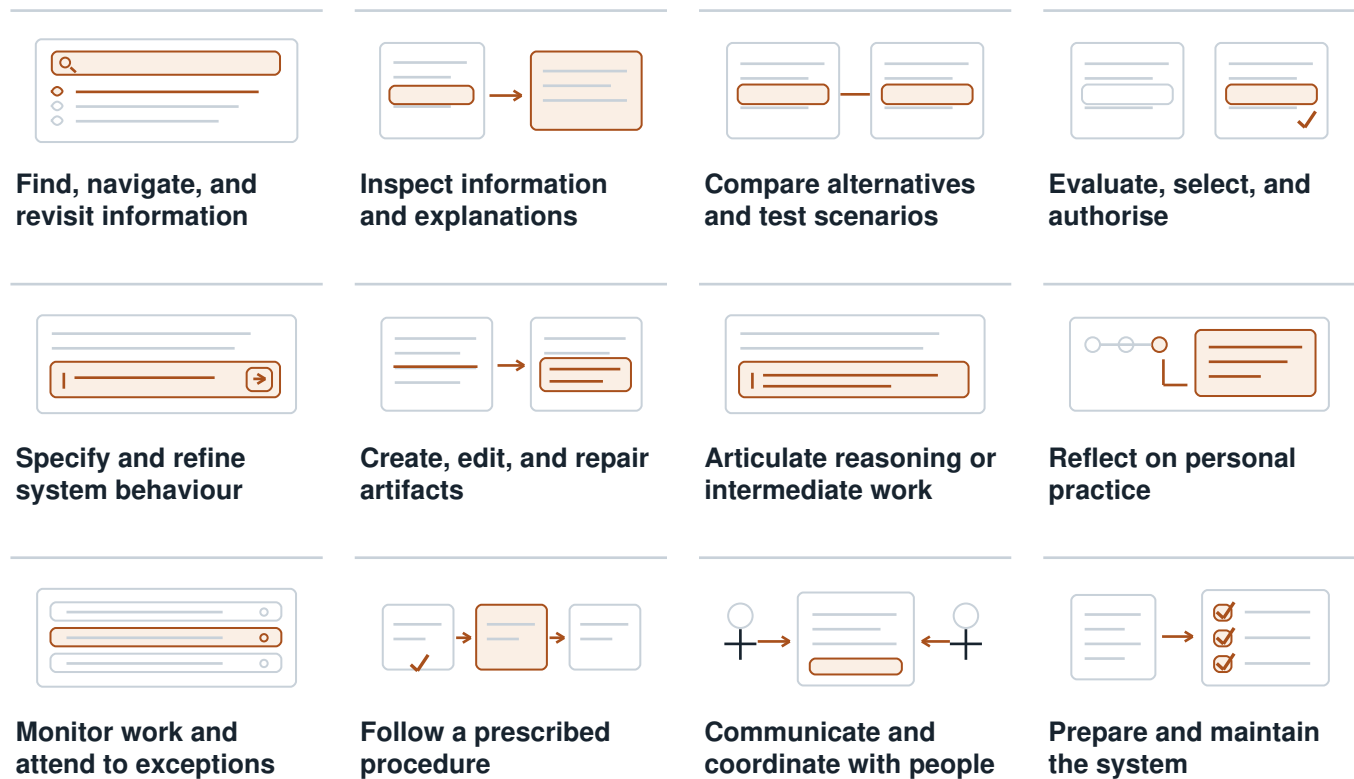
relevant to judgment. A verification step is useful only if the person can tell what it establishes about the result.

Generation can also preserve the activity someone needs to practice. ExPeerience uses AI to create problem contexts, tables,

and explanations while leaving construction of the SQL solution to the learner [163]. The placement of assistance matters: generating the exercise and generating its solution leave different opportunities

## What does the person do?

Twelve activities that interfaces can support



**Figure 6: What does the person do? Twelve activity groups organize the workflow codes. Each schematic illustrates an action through an interface: finding, inspecting, comparing, judging, directing, creating, articulating, reflecting, monitoring, following a procedure, coordinating, or preparing a system. These activities can occur together; the grid does not imply a sequence. Allocation, supplied support, and measured outcomes are coded separately.**

to think and act. Section 6 examines the mechanisms that make these choices possible and the evidence for their consequences.

### 5.3 Who can see the work changes who can help

Distributed-cognition and scaffolding accounts follow support across people and representations [52, 115]. The reviewed designs show why this matters: the person using an interface, the person whose activity it represents, and the person expected to benefit can be different. A useful handoff carries information that someone else can interpret and respond to.

In entrepreneurship coaching, novices describe a project and select risks; a mentor receives selected and omitted risks with source traces before a meeting [59]. HTCT turns tutor discourse into analytics that a coach selects and discusses with the tutor [17]. One carries the novice's priorities into a conversation; the other makes teaching activity available for feedback. The studies follow different parts of the response: the entrepreneurship study observed real

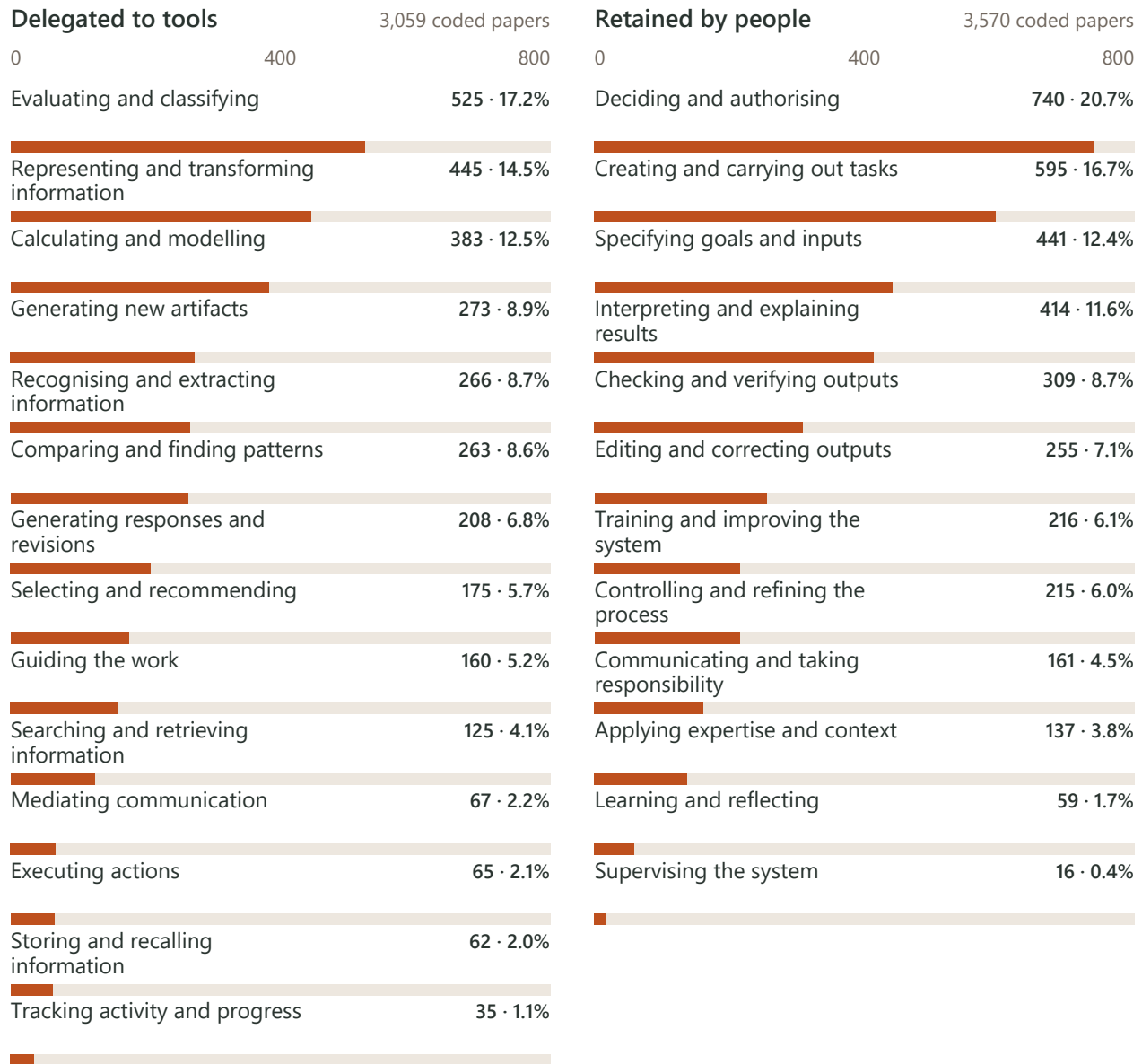
coaching meetings, whereas HTCT followed tutors' subsequent practice over seven weeks.

Software feedback can also make a learner's difficulty less visible to another person. In WriteAid (Stream 2), some recurring questions became less visible to the teacher when pupils sought software feedback, while the teacher gained time to circulate and new peer conversations occurred [106]. Assistance changed both access to information and the opportunities to respond. This suggests examining who no longer sees a question as well as who can now answer it.

The recipient's response distinguishes a proposed connection from an observed change in practice. eaSEL can share a child's reflective artifact with a parent and prompt a conversation, but its evaluation did not observe the whole parent-child exchange [130]. M-Powering Teachers, in contrast, provided mentors with feedback on their own talk; randomized access increased their uptake of students' contributions and reduced their share of talk

## Work delegated to tools and retained by people

Bar length: number of papers · percentages use each field's coded papers



Counts use all papers coded for each field. Allocation conditions and unspecified work are listed with the data.

**Figure 7: Delegated and retained work, ranked by prevalence. Bar lengths encode paper counts on a common scale; labels also give percentages within each panel's coding denominator. A paper can record several roles. Each panel uses papers with usable codes for that role; technology classification is not required.**

[34]. The latter establishes a change in teaching practice, while student learning remains a separate question.

The target of that response also matters. VizProg users diagnosed learners' code and selected feedback in a classroom replay, but the

feedback was not delivered to students [159]. QLens instead lets designers inspect prospective feedback; a recommended path derived from successful traces was judged unsuitable as direct instruction

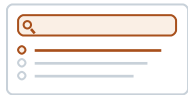


## How is the work divided?

The tool supplies an operation; the person still has work to do.

### What the tool does

#### Perception, search and memory



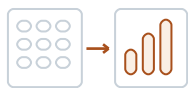
- Recognising and extracting information
- Searching and retrieving information
- Storing and recalling information
- Tracking activity and progress

#### Representation and translation



- Representing and transforming information
- Comparing and finding patterns

#### Computation and judgement



- Calculating and modelling
- Evaluating and classifying

#### Content generation



- Generating new artifacts
- Generating responses and revisions

#### Decision, guidance and action



- Guiding the work
- Executing actions
- Selecting and recommending
- Mediating communication

### What the person retains

#### Judgement and decision



- Interpreting and explaining results
- Deciding and authorising
- Specifying goals and inputs
- Applying expertise and context

#### Doing and revising the work



- Creating and carrying out tasks
- Editing and correcting outputs
- Controlling and refining the process
- Training and improving the system

#### Checking the output



- Checking and verifying outputs

#### Responsibility, social and self-directed work



- Communicating and taking responsibility
- Learning and reflecting
- Supervising the system

**Figure 8: How is the work divided? Five families describe operations supplied by tools; four describe work retained by people. The two columns are independent accounts of a design, not paired transfers. Labels distinguish fourteen tool operations and twelve human activities; learning and reflection are separate from supervising a system. Failed, withheld, absent, and unspecified delegation are recorded separately. Browse examples of tool and human work.**

[155]. Recognizing a difficulty, improving one's teaching, and deciding what will help someone else are different skills. The cases

connect each to a specific representation and response. For shared



support, the important sequence is what travels, who receives it, and what they do next.

Across these workflows, generation changes the material people start from, checking requires knowledge of both a result and its grounds, and handoffs require interpreting another person's situation. These are concrete targets for development: composing with alternatives, judging a criterion, and responding to someone else's reasoning. Their value as learning targets depends on the person's goals and future work. They can accompany continued execution skills rather than replace them. The next section asks how interface mechanisms make this work possible and what evaluations establish about performance and learning.

## 6 FINDINGS: SUPPORTING WORK AND LEARNING

Semantic scaffolding supports work by making task meaning available for people to inspect, judge, and change. The reviewed designs expose two choices relevant to learning: how a representation frames the task, and which operations it supplies or leaves for the person. We first map these design choices, then examine how people judge generated material, revise connected work, and learn through its use. Theory-lineage comparisons explain the rationales behind selected designs; evaluations establish their particular benefits and costs.

### 6.1 Representations make work available for inspection and action

A mechanism connects a representation of task meaning to a contribution the person can make. A goal expressed as an authored rule or an adjustable threshold makes different actions possible: stating an intention, changing a limit, or comparing its consequences. We therefore ask what task meaning is represented, who constructs it, and what the representation invites or enables a person to do. People, AI, or both can supply the structure. Figure 9 presents the 18 categories through these relationships, and Figure 10 compares their prevalence across technologies at the level of the 5 coded families; a paper can use several. Table 17 counts the papers in each category and how many of them concern or measure learning, and the category maps in Appendix C.1 lead to particular mechanisms and their source papers.

Prior design-space reviews similarly make interface choices visible: how abstract ideas take concrete form, or how an interface elicits a person's preferences [68, 101]. Scaffolding research adds the distinction between structuring a task and making important work available for the learner to confront [120]. The relationships below connect these design choices to the work people do and the outcomes studies assess.

Alternatives, relationships, and histories give people material to compare. Graphologue turns text into entities and links [63]; Crystalline gathers options, criteria, and evidence from the pages a developer reads into a comparison table whose cells lead back to their sources [85]. Sources, rationale, and state make different grounds inspectable. HaLLMark traces copied LLM text to its prompts but relies on writers to mark the passages they consider AI-influenced [55]; automatic provenance captures direct reuse more readily than inspiration.

Goals and editable material let people state intentions and change the work: Plume can lock text while it regenerates the explanation around it [84]. Recommendations, questions, and sequences instead give people a proposed action or a route through the task. OATutor selects practice and reveals hints [113], while WithYou shortens the pauses it inserts into speech as a learner's shadowing improves [161]; revealing more help and reducing help are different interventions.

A shared representation makes contributions and interpretations available to other people. Wikibench supports discussion that revises label definitions and inclusion rules [76]. The concerns combine: a shared comparison can help people understand alternatives, act on them, and discuss their reasons.

To use the taxonomy, start from a decision the person must make. Find the mechanism that supplies information or an action for it, then ask what response it requires and which outcome would show success. A provenance link, for example, gives a basis for checking a claim, but the person still judges whether the source supports the wording. The link shows traceability; accurate checking, or learning to judge sources, needs its own observation. Consider the same collection of passages: supplied labels give the person categories to interpret; asking the person to categorize passages elicits their distinctions; editable labels let them revise the scheme. Choosing among these designs begins with which operation matters to the person's work and development.

### 6.2 Generated candidates still need criteria and grounds for judgment

More opportunities to choose generated material make the basis of a choice consequential. The quantitative patterns locate this design problem; the cases show how an interface's alternatives and criteria shape what a person can examine and improve.

**Technology and annual patterns. GenAI systems more often expose generated candidates.** GenAI papers more often describe user control and expression (+13.8 points after adjusting for year and venue; 95% CI 8.9 to 18.7; Appendix D.2). In the 2,532 papers with facet codes, two facets in the control-and-expression family show a reliable gap, and both put generated candidates before the person: GenAI systems more often generate alternatives to explore (+4.8; 95% CI 3.0 to 6.6) and complete or continue the person's own content (+3.0; 1.4 to 4.6; Figure 9). The separate retained-role analysis shows a positive difference for steering the loop (+3.4; 1.2 to 5.6) and a negative difference for improving the system (-5.9; -7.7 to -4.0; Figure 8). Other controls have different estimates: editing machine output (+1.8; -1.4 to 5.1), direct manipulation of a semantic object (+0.9), and specifying criteria and rubrics (-2.8; -4.9 to -0.8). Explicit criteria and rubrics are coded in 4.4% of GenAI papers. The generated alternatives alone do not establish how people judge them. Section 5.2 and Figure 6.2 compare designs that supply them.

**GenAI systems rank items less often.** Organization and comparison is less common in GenAI papers (Figure 10). Its largest facet gap is ranked or prioritized surfacing, which orders items so the important ones come first (3.2% of GenAI and 11.2% of traditional AI/ML papers; -6.5; 95% CI -8.3 to -4.7), and no other facet in the family differs by more than 2.9 points. Appendix D gives the annual family profiles (Figure 57).

## Representations shape participation

Read the task meaning and the person's contribution together.



Representations may be constructed by AI, people or both. Activities shown are design possibilities.

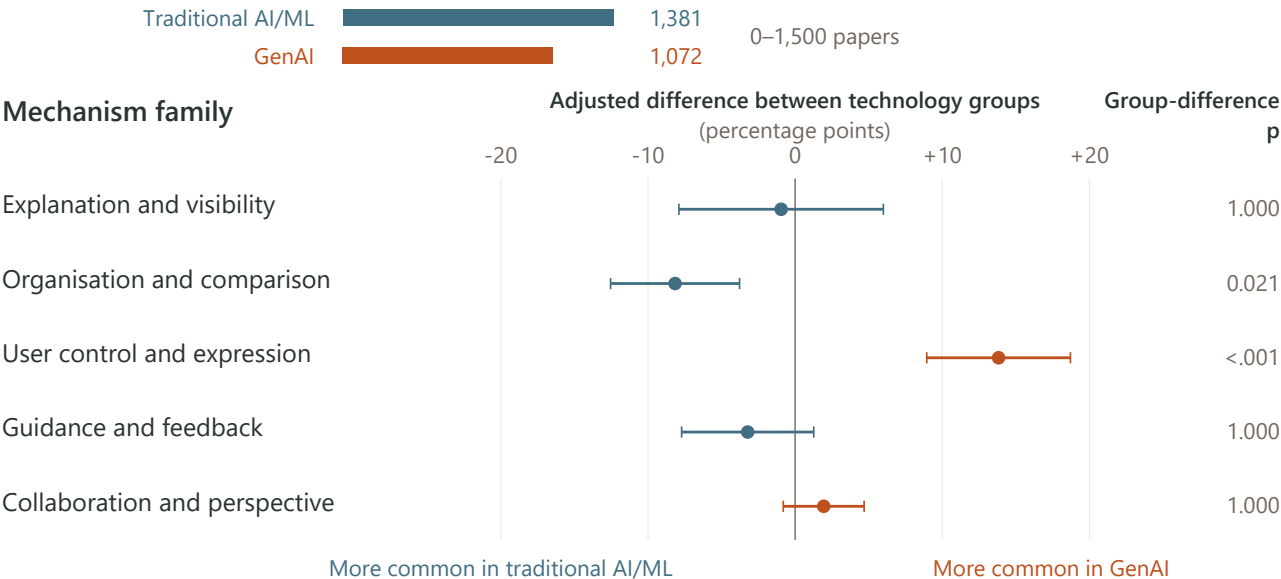
Marked categories combine operations with form/access, help timing or social conditions. These choices can vary across the other relations.

**Figure 9: Represented task meaning and possible human participation across eighteen mechanism categories.** Each card pairs a representation with the contribution it enables or invites; the central schematic illustrates the interface operation. People, AI or both may construct the representation. Marked categories expose form, timing and social conditions that can vary across relations. These are design possibilities, not observations of use or learning. Original category memberships and all 110 distinctions remain in Appendix C.1. Explore the design atlas; search all 110 distinctions.

# GenAI papers more often offer user control

GenAI versus traditional AI/ML, accounting for publication year and venue

2,453 papers with family coding and classified technology · 2019–2025



Lines: 95% confidence intervals · Group-difference p-values: Holm-adjusted

**Figure 10: Between-technology differences in mechanism families among 2,453 papers with usable family coding and classified technology, 2019–2025.** Dots show GenAI minus traditional AI/ML prevalence, adjusted for year and venue; lines show pointwise 95% confidence intervals. The *p*-values test these group differences, with Holm correction across 51 contrasts, rather than change over time. User control retains support in the 2023–2025 and partial-2026 sensitivity windows. Of 3,424 family-coded papers, 544 lack classified technology and 427 classified papers fall outside 2019–2025.

*Collaboration and perspective became more common over time.* Holding the technology mix fixed at the original analysis’s reference composition (43.75% GenAI), its average annual change was +1.13 percentage points (95% CI [0.53, 1.73], Holm *p* = 0.005). The 2023–2025 sensitivity also showed an increase (+2.84 points per year, CI [1.27, 4.41], Holm *p* = 0.007). Guidance and feedback increased in the primary window (+1.61 points per year, CI [0.72, 2.50], Holm *p* = 0.007), but its recent interval included zero ([−2.69, 6.54]). None of the five tests of change in the GenAI–traditional AI/ML difference survived Holm correction in either window. These temporal estimates (Figure 11) answer a different question from the overall technology-group differences in Figure 10; Appendix D.3 reports both temporal estimands for every family.

**People may need to judge both an alternative and its criterion.**

When a system generates the criteria as well as the alternatives, people must judge both. In Luminare, selecting an axis rearranges the responses, whereas adding a dimension revises the responses themselves [137], so a writer judges the dimensions as well as the ideas. VIVID generates expected answers and rubrics along with candidate dialogues, and instructors inspect and edit them

[30]. Design-rationale research already links questions, options, and criteria [94], and conventional tools sort alternatives into categories that designers choose. Generated criteria raise a narrower question: can people revise a criterion while holding the alternatives fixed, or does the revision also replace what they are judging?

Generated criteria can improve comparison. Selenite gives readers generated overviews of criteria and labels the relevant passages; in a twelve-person paired comparison with unaided reading, participants produced more valid criteria with the overview (12.93 versus 6.79), with higher precision and recall [86]. A criterion can also be easy to act on and still miss the quality it stands for. RECAST lets people revise text against a toxicity model; revisions lowered the model’s toxicity scores without reliably improving human ratings of toxicity [152]. When both the model’s response and human judgment matter, the evaluation has to measure both.

The supplied account can also direct attention away from what it omits. In a study with eighteen clinicians, automated suggestions shortened median label-finding time from six to three seconds. With prepopulated suggestions, clinicians annotated a smaller share of nontrivial spans the system omitted (medians of 58% versus 74% without suggestions) [83]. The aid helped one operation while

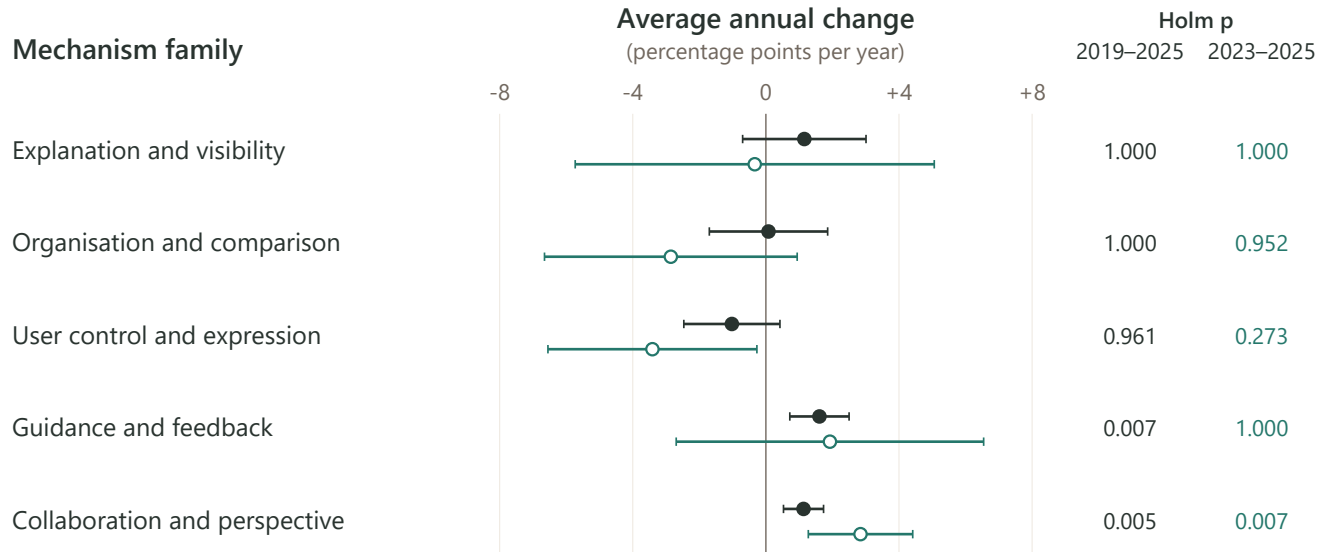
## Collaboration and perspective increase across years

Annual change in coded prevalence, with a fixed mix of AI technologies

Adjusted for venue · 39 venues · Partial 2026 excluded

● 2019–2025 · 2,453 papers

○ 2023–2025 · 1,391 papers



Lines: 95% confidence intervals · Holm correction across ten temporal tests per window

**Figure 11: Average annual change in coded mechanism-family prevalence: 2019–2025 (2,453 papers) and the 2023–2025 sensitivity (1,391).** Dots show percentage points per year; lines show pointwise 95% venue-clustered confidence intervals. The technology mix is held fixed at 43.75% GenAI in both windows. Positive values indicate increasing prevalence. These are temporal slopes, not the overall technology-group differences in Figure 10. Appendix D.3 gives the methods, changing-gap estimates and all ten tests used for Holm correction within each window.

changing attention to material outside its suggestions. Checking the supplied items and searching for missing ones are different contributions to the task.

A label can also leave the question itself ambiguous. In a prior sensemaking study (Stream 2), an unexplained “trust” lens led some researchers to consider a person’s trustworthiness and another to consider that person’s trust in sensor data [37]. The same displayed criterion can therefore invite different judgments; understanding its use requires following what people take it to mean. Across these examples, making a task inspectable also presents an account of what matters in it. The design choice is which criteria, relationships, and omissions the person can question; the learning target is the particular judgment needed to do so.

### 6.3 Iteration requires revising decisions and their consequences

Judgment continues as work changes. A revised criterion can call for new alternatives, and a corrected assumption can alter later results. The cases show why support for iteration depends on the links between operations as well as the individual controls.

Mechanisms can also be grouped by the activity they support: *specify intent*, *represent possibilities*, *evaluate alternatives*, and *act and revise* (Figure 12). Appendix C.1 maps all 110 facet codes to these activities; a code can serve several. Annotations, histories, shared views, and handoffs connect the activities across time and people. The functions describe dependencies, not a fixed order: comparing needs alternatives and criteria, and a comparison may prompt a change to either.

**Different mechanisms can serve the same function.** Intent elicitation, constraints, and shared criteria frame a task. Structured alternatives and source links make possibilities inspectable; comparisons, explanations, and reflection support judgment; selection, editing, and changed constraints enable action. One mechanism can serve two functions: a comparison exposes options and supports choice, and a constraint can frame a task or redirect it. Guidance can enter at any point, and histories keep context across transitions.

Luminate’s dimensions organize alternatives for exploration and evaluation, with return paths to earlier prompt-specific spaces [137]. AI Chains instead feeds one step’s output into the next step’s input, so people can inspect, edit, and re-run intermediate steps [154]. The

order in which a system computes need not dictate the order in which people inspect or repair its work.

**An ordered workflow can still require return paths.** ColabCoder opens shared comparison after independent coding, then supports discussion, decisions, and grouping while letting people restore their original codes [46]. Crystalline instead builds a route back into the material: a reminder marks related evidence on pages the developer has visited but not yet attended to, and three of twelve participants reported using it to guide their research [85]. ColabCoder sets an order for the work; Crystalline returns people to an earlier step when their comparison may have a gap. Transitions therefore deserve the same design care as individual mechanisms: keep the criterion that shaped an alternative, the evidence behind a draft, and a route back to an earlier decision, and test whether people understand a transition, change it, and recover its context.

Scholastic shows why the effect of a control also needs to be understandable. Researchers' codes constrain keyword clusters, while linked document maps, readers, and code views support their interpretation. In a formative study with two researchers, the participant who saw a model update expected documents with the same code to cluster together. They instead remained spread across clusters: the constraint bound keywords, not document membership [54]. The mismatch concerns how the control's effect was understood: it constrained keywords while the researcher expected it to reorganize documents. Exposing a model update alone does not make that relationship clear.

#### **What a revision preserves.**

An editable step does not guarantee that people can correct it and keep the later work they still want. Across the cases, the design question is what a human action authorizes the system to reuse or treat as accepted. Table 1 distinguishes saving an old version, preserving valid edits during regeneration, and confirming a result.

#### **Keeping an old version does not carry its edits forward.**

Kazemitabaar et al.'s data-analysis systems saved earlier versions in branches, but regeneration ignored later manual edits [66]. Participants described avoiding revisions and expecting to redo or reread work; one wanted invalidated parts highlighted instead of regenerated, and another valued the interconnected updates. Recovering an old version differs from preserving the still-valid parts of that work in a new one.

LineageD's pilot showed the cost of predicting a whole lineage at once: correcting one pairing invalidated the rest of the path. The redesign predicts one level at a time, but moving to the next level implicitly confirms pairings the person has not individually confirmed [53]. Plume keeps locked or manually edited text and uses it as context for later generation [84], so a lock preserves wording without limiting its influence on other text or vouching for its correctness. Relating a local step to the whole process is hard in itself: in AI Chains, four of twenty participants asked how a local change would affect the final result [154].

**Interfaces promise to keep different things.** Automatic propagation can itself help. CoPrompt participants valued automatic coordination, and eight used history and difference views to resolve conflicts or restore versions [40]. Pail's code differences, summaries, and design panel supported inspection, but participants reported difficulty following cascades across them, less so when they paused to read the changes [157]. Selecting part of an artifact does not by

itself fix what stays unchanged: FusAI leaves unselected image regions untouched, whereas Textoshop can change text beyond the selected passage [97, 116]. Blace keeps earlier states for rollback, and DesignManager records the process behind an asset [88, 156]; both support recovery and inspection of past work, while consistency with a revised goal still requires checking the downstream artifacts. A design should name the property it protects and the change that could violate it, and an evaluation should test whether people anticipate that change, detect a violation, and recover the work they want to keep.

**Edits can change the specification, the model, or the output.** A correction can change what the person asked for, how the system understood it, or what it produced. FusAI lets designers compose and revise a visual specification [116], and Falx synthesizes candidate charts from examples, which participants described checking against input values they knew [144]. CoAutoML lets novices inspect and edit the task, inputs, target, and evaluation assumptions it proposes before they confirm a machine-learning task [136]. Plume's locks retain part of what the system produced [84]. Outside generative chat interfaces, corrections can also update the model: SmartEye uses a corrected crop as a positive training example for later recommendations [92], and DeepSI fine-tunes BERT from spatial feedback and recomputes its document map, though its authors note limited visibility into the model's internal state [13]. The design question is what a correction changes and which later operation uses it, not whether every control comes before generation.

## **6.4 What AI supplies changes what people practice**

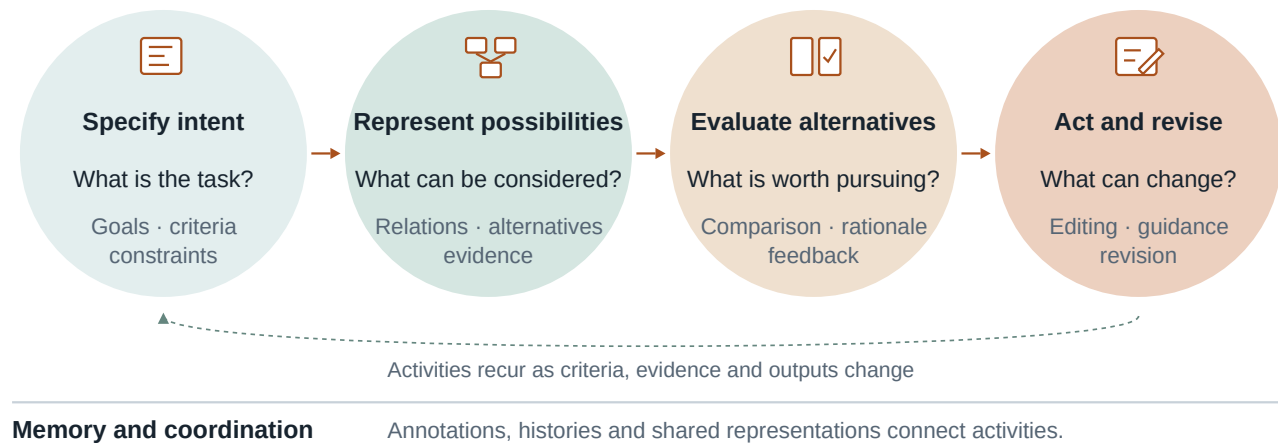
Support can supply a representation, ask people to construct one, or use it as the basis for another activity. The reviewed comparisons show why the operation matters: receiving material can enable useful practice, while an extra requirement to reason or reflect can change both practice and its cost. Which arrangement supports learning depends on the particular capability being developed.

Learners need not build the material they learn with. VocabEncounter supplies phrases in context and reports better vocabulary two days later than its word-only condition [5]. Rehearsal supplies conversational alternatives, practice, and feedback; its participants used more cooperative strategies in a later unaided conversation than participants given the same static materials without interactive practice [129]. In a study of AI assistance for incidental learning, explanations without a recommended answer improved immediate concept learning over minimal feedback [44]. Supplied material also leaves room for different uses: CoNotate participants reported both running its suggested queries and borrowing their vocabulary to write new ones [111]. The design question is which operation the supplied material helps people perform, and whether the study tests that ability.

Explanation content, required reflection, dialogue, and optional recommendations change different parts of practice. Table 2 links four AI-involved comparisons to their assessed outcomes.

The practice cost of reflection does not explain every null learning result. In Xiaomai's advanced-mathematics study, requiring error reflection produced neither a detected learning advantage nor

## From task intent to action



**Figure 12: From task intent to action.** Four overlapping functions offer an activity-oriented view of the mechanism taxonomy. Memory and coordination connect them across time and people. Arrows show possible dependencies and return paths, not measured transition frequencies. The accompanying source examples illustrate these relationships [46, 85, 137, 154].

**Table 1: Revision and downstream effects. What three interfaces preserve, update, or treat as confirmed.**

| Design choice                    | Comparison and finding                                                                                                           | Conditions and contrary evidence                                                                                             | Design implication                                                                           |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| What survives regeneration?      | In task decomposition, editing an assumption, plan or code component can regenerate later work while omitting manual edits [66]. | The earlier version remains recoverable in a branch; preservation within the revised result is a different property.         | Make the fate of earlier edits inspectable when an upstream decision changes.                |
| What does locking text preserve? | Plume retains locked or manually edited dashboard text while using it to inform further generation [84].                         | An unchanged passage can still influence other text; locking does not isolate its downstream effects.                        | Distinguish preserving wording from restricting its influence on new content.                |
| What counts as confirmation?     | LineageD permits corrections before predicting the next level; advancing implicitly confirms remaining pairings [53].            | Explicit and implicit confirmations share the green marking, so the display does not distinguish how a pairing was accepted. | Preserve whether an item was checked explicitly when that distinction matters to later work. |

**Table 2: Selected AI-involved comparisons of explanation and practice designs.**

| Comparison                                         | Assessed result                                                                                           | Conditions                                                                                 |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Predicted-misconception explanations [21]          | Better immediate exercise-prescription decisions after AI removal than unilateral explanations or no AI   | Advantage over random foils unresolved; no delayed test                                    |
| Required reflection after AI feedback [7]          | 2.0 versus 3.4 problems completed in eight minutes; transfer and code-evaluation advantages unestablished | Fixed time budget; fewer attempts do not establish learning harm                           |
| QuizBot versus flashcards [123]                    | Higher recognition (87.2% versus 65.8%), with an advantage at two months                                  | Equal repetitions; 46 versus 14.3 minutes of practice; dialogue bundled with other support |
| Adaptive versus generic video recommendations [77] | Small algebra-ability advantage during normal schooling                                                   | Low optional uptake; policy effect; closure-period advantage unresolved                    |

a detected difference in attempts [39]. Together with the fixed-time comparison above, this makes the content and uptake of reflection worth examining alongside the practice it displaces.

Explanation design also concerns sequence. Incremental XAI supplies factors progressively during model-prediction practice,

then tests prediction on unseen cases without the explanation display [16]. This separates the presentation of an explanation from what a person can do once it is removed.

Prior learning research separates creating an explanation from using it. In a randomized study of 120 learners, everyone wrote

subgoal labels; those who then practiced with their own labels scored 21.6 of 25 on an immediate test, against 17.1 with expert labels and 18.8 with none. Instructional materials were removed for the test, while the programming environment remained available [95]. This comparison sharpens ExPeerience’s question (Section 5.2): which part of the task should the person learn to perform? Learning research has long stressed what support asks learners to do [28, 120]; the interface choice is what to generate and what to leave for the learner.

CoWriting Kazakh tests a related division of work: children either try writing a word in the new Latin script before seeing the robot’s correct spelling, or write it in Cyrillic and let the robot convert it. Both conditions show the robot’s model spelling. Across 67 children, letter knowledge increased during the single session, but the study found no overall advantage for either conversion condition [125]. The design question is which operation the learner needs to practice and which feedback makes that practice useful.

NoteCopilot makes the amount of supplied structure a concrete comparison. In its thirty-person, within-person study, intermediate assistance yielded higher immediate understanding scores than either minimal or automated assistance when learners first answered without their notes [27]. The packages changed both the supplied material and the work of organizing it. This supports examining the division of work, without identifying engagement as the cause or establishing a delayed benefit.

**Writing feedback can describe a revision or judge its quality.** ArgRewrite classifies the purposes of a writer’s earlier revisions without judging their quality, and asks writers to agree or disagree with those labels before and after revising again [1]. ArgueTutor pairs claim and premise labels with recommendations based on their balance [143], and LegalWriter adds advice on argument components while leaving the semantic fit between claim and conclusion to the writer [148]; both improved the quality of text written with them over their comparison interfaces. A description of a revision and advice about quality make different assertions, and one interface can offer both. A learning claim needs a test of the judgment or revision skill the feedback is meant to build.

## 6.5 Prior knowledge connects design choices to testable accounts

Prior knowledge can motivate a design, specify an activity, or inform an assessment. The same principle can therefore support different interface choices, and a concrete implementation can still leave its proposed learning benefit open. The lineage comparisons trace these connections: which prior claim informs the feature, what the person does with it, and which comparison tests the resulting account.

The source-use map (Figure 14) links 1,697 matched lineage cards to 43,733 source–paper pairs in the current analysis. A source recorded in two papers contributes two pairs. The records distinguish concrete uses in a design or measure, conceptual influences, and citation-only mentions. These roles describe how prior work enters a paper; they are not a ranking of theoretical quality. Papers without matched lineage evidence remain outside this comparison.

**A rationale can leave the design choice open.** DataDive draws on statistical thinking and critical reading to motivate contextualizing numerical claims. Its three design goals come from a seven-person formative study; the candidate contexts and ranking criteria are author-specified choices [69]. This establishes why context matters while leaving which generated questions help readers reason as an empirical design question. Its 21-person comparison found no detected advantage in data or data-interpretation recall over ordinary search; readers recalled more facts from outside the text in the baseline condition. The rationale, the interaction, and the assessed consequence support different parts of the account.

**Making correspondence inspectable.** Kori turns the concern about split attention into a visible link between prose and chart marks [78]. The system proposes links, and authors preview, accept, discard, or draw them by hand; once a link is accepted, hovering over its phrase highlights the chart marks it refers to (Figure 13). The author checks each correspondence, and the reader uses it to interpret the data.

**Operationalizing a theory is not testing learning.** QuizBot cites ICAP [28] as its learning rationale, whereas the Khanmigo methodology turns ICAP into an LLM-judged indicator of cognitive engagement [123, 140]. The studies assess different capabilities: QuizBot tests recognition two months later, while Khanmigo tracks correctness on the next attempted item. A direct test in diagnostic reasoning makes the distinction sharper: problematizing elicited more constructive responses than structuring, yet no condition advantage on diagnostic strategies was detected in subsequent cases after the pedagogical module was removed [50]. Both groups received active scaffolds, and the tests occurred in the same session. A theory can identify a promising activity; observing more of it still leaves its contribution to capability to be established.

**Prior knowledge can shape the evaluation as well as the design.** Sibyl adopts Doshi-Velez and Kim’s application-grounded evaluation [35, 166], testing explanations with screeners in child-welfare work. Screeners often rated case-specific explanations helpful and found other forms less useful during those cases, and their feedback exposed problems of terminology and presentation.

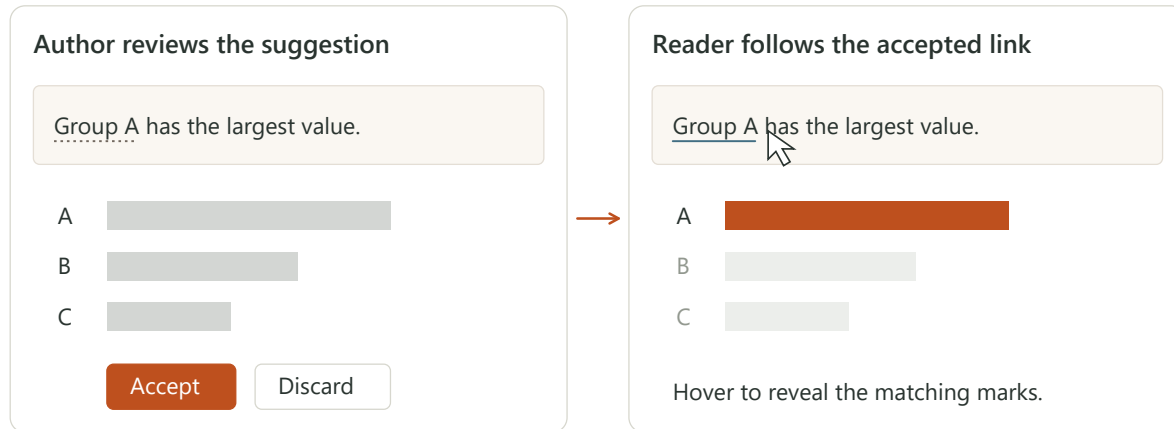
**A theory’s evidence travels with its comparison.** Codigen cites cognitive mapping to motivate active learning and supplies generated programming material on an editable whiteboard [38]. The cited medical study compares maps with notes organized around the same five problem-solving elements, with the assigned support available during the diagnostic case [146]. It does not compare self-authored with generated material. Codigen’s novice pre/post quiz gains likewise do not isolate the whiteboard’s contribution. The connection suggests a design, while testing its learning rationale would require comparing how learners construct and use the material.

The comparison can change when a whole workflow is implemented. Dow et al. compared making three prototypes before critique with receiving critique after each one, holding prototype count, feedback, and time equal [36]. VisiBlends cites parallel prototyping within a rationale for process scaffolding, then supports generation, evaluation, and iteration of visual blends [29]. Its thirteen-person crossover study found more successful blends with the tool, without establishing an advantage in the next no-tool block. This tests a bundled production workflow; it does not isolate the earlier



## Kori makes text–chart links inspectable

AI proposes a connection. The author decides whether to keep it.



The author checks the correspondence; the reader uses it to navigate the data.

**Figure 13: From an AI suggestion to an interactive reference in Kori. The author reviews a proposed text–chart connection. After acceptance, hovering over the linked phrase highlights the matching marks by fading the others [78]. Schematic redrawn from the paper’s interface description, with illustrative data.**

study’s critique order. The source suggests a design choice whose contribution still needs a comparison within the new workflow.

These cases locate different gaps in the connection from theory to evidence. Kori makes a design response to split attention inspectable; the ICAP comparison separates a theory-motivated design, an engagement indicator, and evidence of later capability. Codigen and VisiBlends carry a prior idea into a different activity or a larger package of support. These connections make the next empirical question concrete: which design choice enables the intended activity, and whether that activity develops the intended capability.

### 6.6 Studies establish different kinds of success

The reviewed studies report three kinds of outcomes: process, or what people actually did; artifacts, or the work produced with support; and learning, or what a person could do on a subsequent attempt. Required reflection left fewer attempts within a fixed practice interval, a process finding rather than proof of learning harm [7]. BELVEDERE’s map and prose results below concern artifacts; the capability comparisons that follow test specific activities under specified assistance. Each result retains its own comparison and timing. Rating a finished artifact later still measures that artifact, not what its author has learned.

**Product and capability measures answer different questions.** In BELVEDERE, the maps showed more explicit links between evidence and hypotheses, but no difference in final prose conclusions was detected between conditions, and students could still consult their maps while writing [138]. Both outcomes describe

products made with support; neither tests what students can do without the maps.

**A benefit depends on the comparison and the situation.** Cognitive forcing asks people to engage more deliberately with AI advice. Among 199 participants, fully correct decisions averaged 35% with simple explanations and 33% with forcing, a nonsignificant difference; on items with incorrect AI advice, forcing raised correctness from 3% to 9%, while the no-AI condition reached 18% [20]. Adaptive help raises the same issue: predicting whether a person will answer correctly differs from showing that advice based on that prediction helps them [90, 135]. Appendix A.7 gives the learning and duration profiles of the reviewed studies.

Assistance also defines the capability being tested: independence from a helper can coexist with continued use of a device. We return to that design choice in Section 7.2.

**Performance and subsequent capability.** An interface can improve one assessed outcome while leaving another unresolved. Table 3 distinguishes comprehension, revision quality, enacted behavior, and application of supported skills.

These comparisons apply Norman’s distinction between performance with an artifact and the capability attributed to the person [107]. DBox’s decomposition and feedback also improved performance on an immediate similar problem solved without assistance, although effort rose [91].

**Assessment conditions specify the capability demonstrated.** Among the 396 AI-involved sources in the outcome appraisal, 223 reported a separate capability assessment, 31 assessed capability in a later session, 71 used new material or a

**Table 3: Assessed outcomes in four AI-involved comparisons.**

| Comparison                                   | Assessed result                                                                                                         | Conditions                                                                    |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| CodeVoyager versus same-model chat [70]      | Higher immediate closed-book comprehension (8.06 versus 5.44/20); no detected assisted-completion or accuracy advantage | 16 developers; interface package; no delayed test                             |
| FeedbackWriter versus usual TA feedback [87] | Better revisions ( $d = 0.5$ ); knowledge-test advantage unestablished                                                  | Randomized 354-student trial; optional posttest; test-time assistance unclear |
| Rehearsal added to static training [129]     | Better immediate conflict-response behavior; recall and recognition advantages unestablished                            | Behavior and recall assess different capabilities                             |
| CodeTailor versus generated solutions [57]   | Better immediate unsupported application of supported elements; total posttest advantage unestablished                  | Selected 14 of 18 participants; tested elements depend on practice difficulty |

changed task, and 67 reduced or removed a named training aid. These overlapping features describe the selected appraisal set (Appendix A.8).

An assessment’s task, timing, and available support each change what it shows: an immediate new problem can test application without testing retention, and a later test can still allow help. WriteAid tested pupils’ writing two to three weeks later without AI or teacher support, but on the same topic and without an untreated comparison or a pretreatment skill measure [106].

**The promised use determines the informative test.** Immediate performance, later retention, and application to new material need not move together. Two classroom experiments with structured LLM reflection found no significant advantage on exams two weeks later [75]. Preference is not capability either: across three AI Margin Notes studies of reader interfaces, note selection, and human and AI note-taking, comprehension two hours after reading did not differ significantly, even where preferences or task demands did [64].

An informative evaluation therefore starts from the design’s intended use. An explanation meant to prepare a new decision needs a decision task; feedback meant to change the next attempt needs that attempt observed; a practice partner meant to build interpersonal skill needs evidence of enacted behavior, while recall answers a different question. Effectiveness reviews already show that scaffolding can improve cognitive outcomes [11]; the comparisons here show which alternative and which outcome a designer can use as evidence.

## 7 DISCUSSION

People should be able to use increasingly capable tools while continuing to develop understanding, skill, and judgment. What they need to learn will change with their work, goals, and available resources. Our review makes this design problem concrete: interfaces supply different material, leave different activities to people, and create different occasions for practice. We argue that semantic scaffolding should make those occasions meaningful, adapt them as people and tasks change, and connect claims about learning to the activity and evidence that support them.

### 7.1 Make developing capability a design objective

The workflow findings show why deciding what to automate also requires deciding what people should have opportunities to learn.

CorpusStudio gives writers examples to interpret before composing; ABScribe gives them passages to compare and revise [33, 121]. Both support writing, but they place different demands on the writer. Norman’s distinction between system accomplishment and the person’s activity helps explain this change [107]. As the starting material changes, the person may need to develop a different combination of composition, interpretation, and editorial judgment. Which combination matters depends on the work they want to do. A learning objective should therefore be chosen with the person, alongside the task the tool helps them complete.

Generation can create practice in that capability. ExPeerience supplies problem contexts, tables, and explanations while leaving SQL construction to the learner [163]. Rehearsal supplies a counterpart and feedback through which a person practices responding to conflict. Compared with participants given the same static training materials, Rehearsal participants used more cooperative strategies in an immediate subsequent unaided conversation [129]. These designs use generated material to make an activity available: constructing a solution or trying a response and considering its consequences. The location of assistance matters because generating a problem, a response, or feedback gives the person different work to do next.

Meaningful practice needs both a worthwhile operation and support for performing it. Reiser distinguishes organizing a task from drawing attention to problems that require the learner to reason [120]. A worked example can make an unfamiliar problem approachable; a comparison can expose a distinction; feedback can help someone reconsider a decision. The design task is to connect these resources to the understanding or skill being developed. Simply leaving more work to the person supplies no such connection. In CoWriting Kazakh, children either tried writing in the new script or let the robot convert an entry in the familiar script; both conditions showed the robot’s model spelling. Among 67 children, the conditions produced no detected overall difference in letter learning during the single session [125]. The target operation, its difficulty, and the feedback available together define the learning opportunity.

This gives the mechanism taxonomy a purpose beyond choosing an interface feature. Start with an activity through which the person could develop: explaining a relationship, comparing possible solutions, diagnosing a problem, or revising a judgment. Then choose the representations and actions that make that activity possible within the work. A generated explanation may support reasoning

about an unfamiliar case; editable alternatives may support exploring the consequences of a choice. Semantic scaffolding connects those forms of support to a deliberate opportunity for the person to think and act.

## 7.2 Adapt scaffolds to changing work and developing people

An opportunity that is useful now may need to change as a person develops. Expertise-reversal research shows why guidance useful to a novice can become redundant as knowledge grows [65]. Work also changes within and across tasks: a new goal, audience, or source can make an earlier decision worth reconsidering. Scaffolds therefore need to respond to two forms of change. They should help people address the difficulty in the current task and support the capability they are developing across attempts. Successful completion alone gives an incomplete basis for deciding which change is needed.

Continuing access to a resource can be part of that development. Pea distinguishes developmental scaffolding with fading from enduring distributed intelligence [115]. The distinction directs attention to what responsibility changes and what resources remain useful. In vocational prompting, completing a task with a device counted differently from asking an immediate helper for assistance [25]. The person could develop a specified form of independence while retaining the device. A learning trajectory might similarly move from interpreting worked examples toward constructing and explaining a solution, with references available throughout. When the aim is to transfer that responsibility, its corresponding assistance can diminish while other resources remain.

The revision findings show what continuity requires from a representation. Plume can preserve selected wording during regeneration, while the reviewed data-analysis systems can retain a recoverable earlier version yet omit its manual edits from regenerated work [66, 84]. These differences matter when people return to a decision: they need to recover their contribution, understand what changed, and decide which parts still fit. Malleable interfaces and persistent organizing rules offer ways to reshape a working representation [67, 100, 102]. Maru's persistence can also carry a rule from an earlier subtask into a setting where it no longer fits [67]. Support for development should let people revisit their reasoning, preserve useful work, and revise or retire an organizing rule when its purpose changes.

Adaptation also involves the people who help someone learn. The workflow findings trace support through novice-mentor and tutor-coach exchanges [17, 59]. A representation can make a difficulty available to someone able to respond. It needs to carry enough context for that person to understand the learner's priorities. Designing its evolution therefore includes deciding what travels between people and what each can question. Together, these cases motivate support that preserves enough context for people to reconsider earlier work while making room for greater capability, changed goals, and new contributions from others.

## 7.3 Connect learning claims to activity and evidence

The proposed design commitment makes three consequences relevant: what people do with support, the work they produce, and the

capability they develop. The Findings show why these need distinct evidence. Among sixteen developers, CodeVoyager improved immediate closed-book comprehension relative to same-model chat, with no detected assisted-completion or accuracy advantage. FeedbackWriter's randomized 354-student comparison with usual TA feedback improved revisions; its optional knowledge test left a learning advantage unestablished, with test-time assistance unclear [70, 87]. An evaluation can use these different outcomes to understand what a design changes and whether it serves the intended learning objective.

Theory can explain the connection to look for. Kori translates a concern about split attention into inspectable correspondences between text and chart marks [78]. That connection identifies a feature to examine and an interpretive activity it is intended to support. The diagnostic-reasoning comparison follows the account further: problematizing elicited more constructive responses than structuring, while the two active-scaffold conditions showed no detected difference in diagnostic strategies on subsequent same-session cases after the pedagogical module was removed [50]. The activity observation and capability assessment locate different parts of the explanation. The theory lineage is useful when it makes these relations explicit: which construct guides the design, how the interface engages it, and what evidence would test the proposed consequence.

Process, artifact, and learning measures can then be chosen to examine that account. Process observations show how people use a representation, including the practice and checking it elicits. Artifact assessments show whether the work meets its purpose. Capability assessments examine the intended activity on another attempt, under specified conditions of task, timing, and available help. Participant accounts describe how people interpreted the task, what they valued, and what they found difficult. Combining them can explain why support was taken up, ignored, or used in an unexpected way.

We propose two further measurement choices: whether evidence is collected during or after the task, and whether it comes from observation or external assessment, or from participants' reports. Table 4 presents these choices as a guide for planning evaluations.

Collection time and the subject of a measure answer different questions. A retrospective report of effort describes an earlier activity; a later quality rating still concerns the artifact. A learning claim needs an assessment of the intended capability, with the comparison and remaining resources specified. External ratings depend on scoring criteria, and a numerical self-rating remains a participant report. For each measure, the construct, instrument, rater, and assessment conditions explain what evidence it contributes. The proposed matrix organizes these choices without adding new classifications of the reviewed studies.

This combined account helps designers judge a learning opportunity in relation to useful work. Extra explanation or practice takes time; its value depends on what the person learns and what the task requires. An evaluation that follows activity, assesses its products, and examines subsequent capability can inform whether to retain a scaffold, change its form, or offer it at a different moment. It also gives future studies a specific explanation to extend or challenge.

**Table 4: Proposed evaluation guide by collection time and evidence source.**

| Evidence source                 | During the task                          | After the task                                                      |
|---------------------------------|------------------------------------------|---------------------------------------------------------------------|
| Observed or externally assessed | Actions, revisions, errors, task time    | Artifact quality ratings; scored understanding, retention, transfer |
| Reported by participants        | In-the-moment effort, confidence, agency | Retrospective effort, usefulness, perceived learning, self-efficacy |

## 8 RESEARCH OPPORTUNITIES

The review opens a research agenda around creating meaningful learning opportunities as work changes. Four questions concern the designs that make an activity possible, explanations of how that activity supports learning, the formation of judgment across people, and the place of learning in everyday tools. Together they connect interface development with an account of how people develop through use.

### 8.1 Design representations around meaningful learning activities

The design space offers ingredients for supporting comparison, explanation, diagnosis, construction, and revision. Research can investigate how to combine them around a capability a person wants to develop. For example, two worked solutions can make a relationship available for comparison; a partial representation can invite someone to supply the missing relation; a history can let them revisit why they changed their interpretation. Each arrangement connects task meaning to a different operation. The opportunity is to develop a repertoire of such connections, including the feedback and prior knowledge that make the operation productive.

The existing comparisons expose design choices within that repertoire. In the subgoal-label experiment, all learners created labels; those who then practiced with their own labels did better on an immediate test than those practicing with expert labels or no labels. Instructional materials were removed for the test, while the programming environment remained available [95]. The comparison concerns what people use after construction as well as what they construct. Likewise, a ready-made criterion can reveal a useful distinction, as in Selenite [86], while revising a criterion asks the person to decide what matters. The assistance dilemma makes the broader question explicit: which support enables progress, and which operation is worth leaving for the person to practice [73, 120]?

Research through design can explore these choices with people working on problems they care about. Prototypes could let a person move between an example, their own construction, feedback, and a revised attempt while retaining the relationships among them. Work with users can identify where a representation opens a useful question and where it makes that question harder to understand. Focused comparisons can then examine a particular choice, such as supplying an explanation before or after a first attempt. The contribution would be reusable designs with an account of the activity they support and the situations in which that activity is worth practicing.

### 8.2 Develop explanations of how scaffolds support learning

The theory lineage creates an opportunity to connect a construct, its implementation, and the evidence used to judge it. A principle can inspire several representations, and similar representations can engage different activities. Codigen’s cognitive-mapping rationale illustrates the work needed between these steps. Its cited medical study compared maps with notes organized around the same five problem-solving elements, with the assigned support available during assessment [146]. Codigen instead lets learners arrange and edit generated programming material; its novice pre/post quiz gains came from an activity without an unexposed control [38]. A useful next account would explain which part of constructing, organizing, or using a representation supports the capability in each setting, and identify a comparison that tests that explanation.

Practice conditions belong in that account. QuizBot compared equal numbers of repetitions while allowing substantially different practice time. Required reflection after AI feedback instead consumed part of a fixed eight-minute interval, leaving fewer completed problems [7, 123]. In Xiaomai, reflection produced neither a detected learning advantage nor a detected difference in attempts [39]. Together these comparisons motivate examining what people do during the added activity, how they use its feedback, and what other practice it changes. The explanation must distinguish these possibilities to inform another design.

Such research can combine close observation with focused interventions and comparisons across tasks. Observation can reveal how a representation is interpreted; an intervention can test an activity the explanation identifies; later assessments can examine the intended capability under the resources available in future use. Equal-time and equal-repetition comparisons address different practical questions where the activity changes exposure. For optional support, the offer and its uptake also need separate accounts [77]. The proposed evaluation guide in Table 4 helps select complementary evidence. Across a program of studies, the aim is an explanation that accounts for both productive use and failure, and helps anticipate when a scaffold will need a different form.

### 8.3 Study how people form and share judgments in use

Semantic scaffolds also offer a way to study how interpretations develop through collective work. A shared label or comparison brings people’s reasoning into contact, while leaving room for different meanings. The sensemaking study’s “trust” lens referred to different relationships for different participants [37]; CollabCoder brings individual codes into shared discussion [46]. Distributed cognition

provides a basis for following that reasoning across people and representations [52]. The research question is how an interpretation becomes useful, contested, or revised as people try to produce work together.

Handoffs make this process visible over time. Entrepreneurship coaching carries novices' selected and omitted risks into a mentor meeting; HTCT makes tutor discourse available to a coach who selects and discusses it [17, 59]. Studies can follow what a recipient infers, what the person represented corrects, and how their exchange changes the next attempt. WriteAid adds a different relationship: software feedback made some recurring questions less visible to the teacher while creating time to circulate and occasions for peer conversation [106]. A new route for assistance can therefore reorganize the opportunities for people to understand and respond to one another.

Studying these changes calls for observing ordinary work, including deadlines, disagreements, and the demands of producing an artifact. Traces can locate a revision or an abandoned alternative; observation and participants' accounts can explain its meaning. Follow whose interpretation enters the shared representation, who can challenge it, and what people carry into a later decision. Learning claims can then be examined for the people concerned, alongside changes in the group's practice and produced work. This would explain how scaffolds participate in the development of judgment within a social setting, including the conditions that make useful differences available for discussion.

## 8.4 Create learning opportunities within everyday tools

A tool for writing, programming, or analysis could help people develop capabilities through the work they already want to do. A consequential decision creates a reason to examine an explanation, compare alternatives, or try a revision. The incidental-learning study gives one bounded example: explanations without a recommended answer improved immediate concept learning relative to minimal feedback [44]. The broader opportunity is to make such moments available across ordinary use, where people also face immediate goals, limited time, and changing reasons to learn.

This requires designing a continuing relationship between the tool and the person's learning goals. A user might want to recognize misleading comparisons, diagnose a class of programming errors, or make stronger written arguments. A tool could help them notice relevant moments, retain earlier attempts and explanations, and return to an unresolved question when time permits. Malleable interfaces and persistent information architectures offer ways to change the invitation while keeping a useful working representation [67, 102]. People need to be able to choose, defer, and revise their goals as well as the support. Declining a practice opportunity during an urgent task is information about that situation; the tool would need further evidence to infer mastery.

Longitudinal design work can investigate how these opportunities accumulate and how support should evolve. Co-design can establish which capabilities people value and what resources their work requires. Field studies can then follow the opportunities encountered and taken up, interruptions, changes in work quality, and direct assessments of the intended capability on new cases.

Assessments should retain resources that normally belong to the task and identify the particular training aid being varied. Following people as their goals, tasks, and expertise change would reveal when a scaffold helps them develop, when it needs revision, and which resources remain part of capable practice. That knowledge could make learning a sustained possibility within useful tools.

## 9 LIMITATIONS

*Coverage.* We searched 40 venues over 8 years, which favors published, English-language research [6, 80]. Technology comparisons leave out 645 papers whose AI technology could not be determined, and each analysis uses the papers coded for its fields (Table 9). The theories and empirical studies we compare against are a selection, not a census of every relevant tradition.

*Reading and coding.* Language models read each paper against a fixed codebook, one model per paper, and grouped the coded descriptions into categories. People have not yet checked the coding or the cross-case interpretations. Models can miss evidence shown only in figures and can share biases, and other analysts could draw the boundaries between closely related mechanisms differently.

*Effects.* Studies differ in tasks, participants, comparators, outcomes, and duration, and many compare whole systems. Most coded changes in how people work rest on the authors' own attribution rather than a measured contrast (75.8% of coded process shifts). The GenAI and traditional AI/ML comparisons are associations among published designs, not effects of the technology. The design space cannot rank mechanisms by effectiveness or estimate an overall learning benefit.

## 10 CONCLUSION

As AI changes work, people need opportunities to develop the understanding, skills, and judgment that let them pursue what matters to them. This review connects the structures AI interfaces supply with the activities people carry out and the consequences studies assess. The design space makes concrete choices available for creating meaningful practice within useful work: what to provide, what to invite people to do, and how to support the next attempt. Developing these opportunities requires attention to both the person's changing capability and the demands of the task. The research agenda is to understand how semantic scaffolds can evolve with that relationship, helping people learn through everyday tools and with one another.

## REFERENCES

- [1] Tazin Afrin, Omid Kashefi, Christopher Olshefski, Diane Litman, Rebecca Hwa, and Amanda Godley. 2021. Effective Interfaces for Student-Driven Revision Sessions for Argumentative Writing. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. ACM, New York, NY, USA, 1–13. doi:10.1145/3411764.3445683
- [2] Shaaron Ainsworth. 2006. DeFT: A conceptual framework for considering learning with multiple representations. *Learning and Instruction* 16, 3 (June 2006), 183–198. doi:10.1016/j.learninstruc.2006.03.001
- [3] Saleema Amershi, Maya Cakmak, William Bradley Knox, and Todd Kulesza. 2014. Power to the people: The role of humans in interactive machine learning. *AI magazine* 35, 4 (2014), 105–120.
- [4] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul Bennett, Kori Inkpen, Jaime Teevan, Ruth Kikin-Gil, and Eric Horvitz. 2019. Guidelines for Human-AI Interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in*

- Computing Systems*. Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3290605.3300233
- [5] Riku Arakawa, Hiromu Yakura, and Sosuke Kobayashi. 2022. VocabEncounter: NMT-powered Vocabulary Learning by Presenting Computer-Generated Usages of Foreign Words into Users' Daily Lives. In *CHI Conference on Human Factors in Computing Systems (CHI '22)*. ACM, New York, NY, USA, 1–21. doi:10.1145/3491102.3501839
  - [6] Hilary Arksey and Lisa O'Malley. 2005. Scoping studies: towards a methodological framework. *International Journal of Social Research Methodology* 8, 1 (2005), 19–32. doi:10.1080/1364557032000119616
  - [7] Michael W. Asher, Gillian Gold, and Paulo F. Carvalho. 2026. Benefit or Bottleneck? Assessing the Impact of Structured Reflection on Learning from AI-Driven Explanatory Feedback. In *Proceedings of the Thirteenth ACM Conference on Learning @ Scale*. ACM, New York, NY, USA, 1–11. doi:10.1145/3774398.3811598
  - [8] Lisanne Bainbridge. 1983. Ironies of automation. *Automatica* 19, 6 (1983), 775–779. doi:10.1016/0005-1098(83)90046-8
  - [9] Alex Barrett, Fengfeng Ke, Nuodi Zhang, Chih-Pu Dai, Saptarshi Bhowmik, and Xin Yuan. 2025. Pattern analysis of ambitious science talk between preservice teachers and AI-powered student agents. In *Proceedings of the 15th International Learning Analytics and Knowledge Conference (LAK 2025)*. ACM, New York, NY, USA, 761–770. doi:10.1145/3706468.3706570
  - [10] Brian R. Belland. 2011. Distributed Cognition as a Lens to Understand the Effects of Scaffolds: The Role of Transfer of Responsibility. *Educational Psychology Review* 23, 4 (Sept. 2011), 577–600. doi:10.1007/s10648-011-9176-5
  - [11] Brian R Belland, Andrew E Walker, Nam Ju Kim, and Mason Lefler. 2017. Synthesizing results from empirical research on computer-based scaffolding in STEM education: A meta-analysis. *Review of educational research* 87, 2 (2017), 309–344.
  - [12] Manon Berney, Abdessalam Ouazaki, Vladimir Macko, Bruno Kocher, and Adrian Holzer. 2024. Care-Based Eco-Feedback Augmented with Generative AI: Fostering Pro-Environmental Behavior through Emotional Attachment. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–15. doi:10.1145/3613904.3642296
  - [13] Yali Bian and Chris North. 2021. DeepSI: Interactive Deep Learning for Semantic Interaction. In *26th International Conference on Intelligent User Interfaces (IUI '21)*. ACM, New York, NY, USA, 197–207. doi:10.1145/3397481.3450670
  - [14] Michael Mose Biskjaer, Peter Dalsgaard, and Kim Halskov. 2014. A constraint-based understanding of design spaces. In *Proceedings of the 2014 conference on Designing interactive systems (DIS '14)*. ACM, New York, NY, USA, 453–462. doi:10.1145/2598510.2598533
  - [15] Robert A. Bjork. 1994. Memory and Metamemory Considerations in the Training of Human Beings. In *Metacognition: Knowing about Knowing*, Janet Metcalfe and Arthur P. Shimamura (Eds.). The MIT Press, Cambridge, MA, 185–206. doi:10.7551/mitpress/4561.003.0011
  - [16] Jessica Y Bo, Pan Hao, and Brian Y Lim. 2024. Incremental XAI: Memorable Understanding of AI with Incremental Explanations. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–17. doi:10.1145/3613904.3642689
  - [17] Brandon M. Booth, Jennifer Jacobs, Jeffrey B. Bush, Brent Milne, Tom Fischhaber, and Sidney K. DMello. 2024. Human-tutor Coaching Technology (HTCT): Automated Discourse Analytics in a Coached Tutoring Model. In *Proceedings of the 14th Learning Analytics and Knowledge Conference (LAK '24)*. ACM, New York, NY, USA, 725–735. doi:10.1145/3636555.3636937
  - [18] Marcel Borowski, Bjarke V. Fog, Carla F. Griggio, James R. Eagan, and Clemens N. Klokmoose. 2023. Between Principle and Pragmatism: Reflections on Prototyping Computational Media with Webstrates. *ACM Transactions on Computer-Human Interaction* 30, 4 (Aug. 2023), 1–53. doi:10.1145/3569895
  - [19] Nadieh Bremer and Shirley Wu. 2021. Royal Constellations. In *Data Sketches*. A K Peters/CRC Press, 116–131. doi:10.1201/9780429445019-11
  - [20] Zana Bućinca, Maja Barbara Malaya, and Krzysztof Z. Gajos. 2021. To Trust or to Think: Cognitive Forcing Functions Can Reduce Overreliance on AI in AI-assisted Decision-making. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW1 (April 2021), 1–21. doi:10.1145/3449287
  - [21] Zana Bućinca, Siddharth Swaroop, Amanda E. Paluch, Finale Doshi-Velez, and Krzysztof Z. Gajos. 2025. Contrastive Explanations That Anticipate Human Misconceptions Can Improve Human Decision-Making Skills. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–25. doi:10.1145/3706598.3713229
  - [22] Sabrina Caldwell, Penny Sweetser, Nicholas O'Donnell, Matthew J. Knight, Matthew Aitchison, Tom Gedeon, Daniel Johnson, Margot Brereton, Marcus Gallagher, and David Conroy. 2022. An Agile New Research Framework for Hybrid Human-AI Teaming: Trust, Transparency, and Transferability. *ACM Transactions on Interactive Intelligent Systems* 12, 3, Article 17 (jul 2022), 36 pages. doi:10.1145/3514257
  - [23] Yining Cao, Peiling Jiang, and Haijun Xia. 2025. Generative and Malleable User Interfaces with Generative and Evolving Task-Driven Data Model. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–20. doi:10.1145/3706598.3713285
  - [24] Joseph Chee Chang, Nathan Hahn, and Aniket Kittur. 2020. Mesh: Scaffolding Comparison Tables for Online Decision Making. In *Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology (UIST '20)*. ACM, New York, NY, USA, 391–405. doi:10.1145/3379337.3415865
  - [25] Yao-Jen Chang, Wan Chih Chang, and Tsen-Yung Wang. 2009. Context-aware prompting to transition autonomously through vocational tasks for individuals with cognitive impairments. In *Proceedings of the 11th international ACM SIGACCESS conference on Computers and accessibility (ASSETS09)*. ACM, New York, NY, USA, 19–26. doi:10.1145/1639642.1639648
  - [26] Meng Chen, Akhil Iyer, and Amy Pavel. 2025. Surfacing Variations to Calibrate Perceived Reliability of MLLM-generated Image Descriptions. In *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '25)*. ACM, New York, NY, USA, 1–17. doi:10.1145/3663547.3746393
  - [27] Xinyue Chen, Kunlin Ruan, Kexin Phyllis Ju, Nathan Yap, and Xu Wang. 2025. More AI Assistance Reduces Cognitive Engagement: Examining the AI Assistance Dilemma in AI-Supported Note-Taking. *Proceedings of the ACM on Human-Computer Interaction* 9, 7 (Oct. 2025), 1–29. doi:10.1145/3757632
  - [28] Michelene T. H. Chi and Ruth Wylie. 2014. The ICAP Framework: Linking Cognitive Engagement to Active Learning Outcomes. *Educational Psychologist* 49, 4 (2014), 219–243. doi:10.1080/00461520.2014.965823
  - [29] Lydia B. Chilton, Savvas Petridis, and Maneesh Agrawala. 2019. VisiBlends: A Flexible Workflow for Visual Blends. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*. ACM, New York, NY, USA, 1–14. doi:10.1145/3290605.3300402
  - [30] Seulgi Choi, Hyewon Lee, Yoonjoo Lee, and Juho Kim. 2024. VIVID: Human-AI Collaborative Authoring of Vicarious Dialogues from Lecture Videos. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–26. doi:10.1145/3613904.3642867
  - [31] Yoonseo Choi, Eun Jeong Kang, Seulgi Choi, Min Kyung Lee, and Juho Kim. 2025. Proxona: Supporting Creators' Sensemaking and Ideation with LLM-Powered Audience Personas. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–32. doi:10.1145/3706598.3714034
  - [32] Andy Clark and David Chalmers. 1998. The extended mind. *Analysis* 58, 1 (1998), 7–19. doi:10.1093/analysis/58.1.7
  - [33] Hai Dang, Chelse Swoopes, Daniel Buschek, and Elena L. Glassman. 2025. CorpusStudio: Surfacing Emergent Patterns In A Corpus Of Prior Work While Writing. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–19. doi:10.1145/3706598.3713974
  - [34] Dorottya Demszky and Jing Liu. 2023. M-Powering Teachers: Natural Language Processing Powered Feedback Improves 1:1 Instruction and Student Outcomes. In *Proceedings of the Tenth ACM Conference on Learning @ Scale (L@S '23)*. ACM, New York, NY, USA, 59–69. doi:10.1145/3573051.3593379
  - [35] Finale Doshi-Velez and Been Kim. 2017. Towards A Rigorous Science of Interpretable Machine Learning. doi:10.48550/ARXIV.1702.08608
  - [36] Steven P. Dow, Alana Glassco, Jonathan Kass, Melissa Schwarz, Daniel L. Schwartz, and Scott R. Klemmer. 2010. Parallel prototyping leads to better design results, more divergence, and increased self-efficacy. *ACM Transactions on Computer-Human Interaction* 17, 4 (Dec. 2010), 1–24. doi:10.1145/1879831.1879836
  - [37] Dimitra Dritsa and Steven Houben. 2024. How Design Researchers Make Sense of Data Visualizations in Data-Driven Design: An Uncertainty-Aware Sensemaking Model. *ACM Transactions on Computer-Human Interaction* 31, 6 (Dec. 2024), 1–53. doi:10.1145/3685268
  - [38] Mohamed Ez-Zaouia and Yazid Boumarafi. 2025. Codigen: Enhancing Personalized Programming Education with Visual Authoring and LLMs. In *2025 IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)*. IEEE, 272–285. doi:10.1109/vl-hcc65237.2025.00038
  - [39] Ying Fang, Bo He, Zhi Liu, Sannyuya Liu, Zhonghua Yan, and Jianwen Sun. 2024. Evaluating the Design Features of an Intelligent Tutoring System for Advanced Mathematics Learning. In *Artificial Intelligence in Education (Lecture Notes in Computer Science, Vol. 14829)*. Springer, Cham, 338–350. doi:10.1007/978-3-031-64302-6\_24
  - [40] Li Feng, Ryan Yen, Yuzhe You, Mingming Fan, Jian Zhao, and Zhicong Lu. 2024. CoPrompt: Supporting Prompt Sharing and Referring in Collaborative Natural Language Programming. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–21. doi:10.1145/3613904.3642212
  - [41] Kasra Ferdowsi, Ruanqianqian (Lisa) Huang, Michael B. James, Nadia Polikarpova, and Sorin Lerner. 2024. Validating AI-Generated Code with Live Programming. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–8. doi:10.1145/3613904.3642495
  - [42] Jonas Frich, Lindsay MacDonald Vermeulen, Christian Remy, Michael Mose Biskjaer, and Peter Dalsgaard. 2019. Mapping the Landscape of Creativity Support Tools in HCI. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*. ACM, New York, NY, USA, 1–18. doi:10.1145/3290605.3300619

- [43] Y. Fu, D. Bromley, and V. Setlur. 2025. Data Weaver : Authoring Data-Driven Narratives through the Integrated Composition of Visualization and Text. *Computer Graphics Forum* 44, 3 (May 2025). doi:10.1111/cgf.70098
- [44] Krzysztof Z. Gajos and Lena Mamykina. 2022. Do People Engage Cognitively with AI? Impact of AI Assistance on Incidental Learning. In *27th International Conference on Intelligent User Interfaces (IUI '22)*. ACM, New York, NY, USA, 794–806. doi:10.1145/3490099.3511138
- [45] Krzysztof Z. Gajos, Daniel S. Weld, and Jacob O. Wobbrock. 2010. Automatically generating personalized user interfaces with Supple. *Artificial Intelligence* 174, 12–13 (2010), 910–950. doi:10.1016/j.artint.2010.05.005
- [46] Jie Gao, Yuchen Guo, Gionnieve Lim, Tianqin Zhang, Zheng Zhang, Toby Jia-Jun Li, and Simon Tangi Perrault. 2024. CollabCoder: A Lower-barrier, Rigorous Workflow for Inductive Collaborative Qualitative Analysis with Large Language Models. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–29. doi:10.1145/3613904.3642002
- [47] Dedre Gentner. 1983. Structure-Mapping: A Theoretical Framework for Analogy\*. *Cognitive Science* 7, 2 (April 1983), 155–170. doi:10.1207/s15516709cog0702\_3
- [48] Dedre Gentner and Virginia Gunn. 2001. Structural alignment facilitates the noticing of differences. *Memory & Cognition* 29, 4 (June 2001), 565–577. doi:10.3758/bf03200458
- [49] Frederic Gmeiner, Jamie Lynn Conlin, Eric Handa Tang, Nikolas Martelaro, and Kenneth Holstein. 2024. An Evidence-based Workflow for Studying and Designing Learning Supports for Human-AI Co-creation. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–15. doi:10.1145/3613905.3650763
- [50] Fatma Betül Güres, Tanya Nazaretsky, Seyed Parsa Neshaei, and Tanja Käser. 2026. Structuring versus Problematising: How LLM-based Agents Scaffold Learning in Diagnostic Reasoning. In *Proceedings of the LAK26: 16th International Learning Analytics and Knowledge Conference (LAK 2026)*. ACM, New York, NY, USA, 514–525. doi:10.1145/3785022.3785105
- [51] Noor Hammad, David Chuan-En Lin, Amy Smith, Max Kreminski, Erik Harpstead, and Jessica Hammer. 2026. Tracing Creativity: A Design Space For Creative Activity Traces in HCI. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)*. ACM, New York, NY, USA, 1–24. doi:10.1145/3772318.3791263
- [52] James Hollan, Edwin Hutchins, and David Kirsh. 2000. Distributed cognition: toward a new foundation for human-computer interaction research. *ACM Transactions on Computer-Human Interaction (TOCHI)* 7, 2 (2000), 174–196.
- [53] Jiayi Hong, Alain Trubuil, and Tobias Isenberg. 2022. LineageD: An Interactive Visual System for Plant Cell Lineage Assignments based on Correctable Machine Learning. *Computer Graphics Forum* 41, 3 (June 2022), 195–207. doi:10.1111/cgf.14533
- [54] Matt-Heun Hong, Lauren A. Marsh, Jessica L. Feuston, Janet Ruppert, Jed R. Brubaker, and Danielle Albers Szafr. 2022. Scholastic: Graphical Human-AI Collaboration for Inductive and Interpretive Text Analysis. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology (UIST '22)*. ACM, New York, NY, USA, 1–12. doi:10.1145/3526113.3545681
- [55] Md Naimul Hoque, Tasfia Mashiat, Bhavya Ghai, Cecilia D. Shelton, Fanny Chevalier, Kari Kraus, and Niklas Elmqvist. 2024. The HaLLMark Effect: Supporting Provenance and Transparent Use of Large Language Models in Writing with Interactive Visualization. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–15. doi:10.1145/3613904.3641895
- [56] Eric Horvitz. 1999. Principles of mixed-initiative user interfaces. In *Proceedings of the SIGCHI conference on Human factors in computing systems the CHI is the limit - CHI '99 (CHI '99)*. ACM Press, New York, NY, USA, 159–166. doi:10.1145/302979.303030
- [57] Xinying Hou, Zihan Wu, Xu Wang, and Barbara J. Ericson. 2024. CodeTailor: LLM-Powered Personalized Parsons Puzzles for Engaging Support While Learning Programming. In *Proceedings of the Eleventh ACM Conference on Learning @ Scale (L@S '24)*. ACM, New York, NY, USA, 51–62. doi:10.1145/3657604.3662032
- [58] Chao-Chun Hsu, Erin Bransom, Jenna Sparks, Bailey Kuehl, Chenhao Tan, David Wadden, Lucy Wang, and Aakanksha Naik. 2024. CHIME: LLM-Assisted Hierarchical Organization of Scientific Studies for Literature Review Support. In *Findings of the Association for Computational Linguistics: ACL 2024*, Lun-Wei Ku, Andre Martins, and Vivek Srikumar (Eds.). Association for Computational Linguistics, Bangkok, Thailand, 118–132. doi:10.18653/v1/2024.findings-acl.8
- [59] Evey Jiaxin Huang, Matthew Easterday, and Elizabeth Gerber. 2025. AI That Helps Us Help Each Other: A Proactive System for Scaffolding Mentor-Novice Collaboration in Entrepreneurship Coaching. *Proceedings of the ACM on Human-Computer Interaction* 9, 7 (Oct. 2025), 1–30. doi:10.1145/3757549
- [60] Jeremy Zhengqi Huang, Caluã de Lacerda Patata, Liang-Yuan Wu, and Dhruv Jain. 2025. CapTune: Adapting Non-Speech Captions With Anchored Generative Models. In *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '25)*. ACM, New York, NY, USA, 1–18. doi:10.1145/3663547.3746346
- [61] Edwin Hutchins. 2005. Material anchors for conceptual blends. *Journal of pragmatics* 37, 10 (2005), 1555–1577.
- [62] Nanna Inie, Anne Marie Kristensen, Morten Misfeldt, and Peter Dalsgaard. 2026. A Critical Review of Critical Thinking in HCI Research About AI. *International Journal of Human-Computer Interaction* (Sept. 2026), 1–36. doi:10.1080/10447318.2026.2722522
- [63] Peiling Jiang, Jude Rayan, Steven P. Dow, and Haijun Xia. 2023. Graphologue: Exploring Large Language Model Responses with Interactive Diagrams. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23)*. ACM, New York, NY, USA, 1–20. doi:10.1145/3586183.3606737
- [64] Nikhita Joshi and Daniel Vogel. 2026. Designing and Evaluating AI Margin Notes in Document Reader Software. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)*. ACM, New York, NY, USA, 1–18. doi:10.1145/3772318.3790786
- [65] Slava Kalyuga, Paul Ayres, Paul Chandler, and John Sweller. 2003. The Expertise Reversal Effect. *Educational Psychologist* 38, 1 (2003), 23–31. doi:10.1207/S15326985EP3801\_4
- [66] Majeed Kazemitabaar, Jack Williams, Ian Drosos, Tovi Grossman, Austin Zachary Henley, Carina Negreanu, and Advait Sarkar. 2024. Improving Steering and Verification in AI-Assisted Data Analysis with Interactive Task Decomposition. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology (UIST '24)*. ACM, New York, NY, USA, 1–19. doi:10.1145/3654777.3676345
- [67] Eunhye Kim, DaEun Choi, Bryan Min, Hyunjung Yi, Yue Jiang, and Juho Kim. 2026. Maru: Information Architecture as a Shared Language for Generating Aligned and Persistent User Interfaces. arXiv:2608.25565v1. doi:10.48550/arXiv.2608.25565 Author manuscript for UIST '26.
- [68] Eunhye Kim, Bryan Min, Haijun Xia, and Juho Kim. 2026. Elicitive User Interfaces: Designing How Users Shape Generative Interfaces. doi:10.48550/arXiv.2609.23642 arXiv:2609.23642 Preprint, version 1, September 20, 2026.
- [69] Hyunwoo Kim, Khanh Duy Le, Gionnieve Lim, Dae Hyun Kim, Yoo Jin Hong, and Juho Kim. 2024. DataDive: Supporting Readers' Contextualization of Statistical Statements with Data Exploration. In *Proceedings of the 29th International Conference on Intelligent User Interfaces (IUI '24)*. ACM, New York, NY, USA, 623–639. doi:10.1145/3640543.3645155
- [70] Yeonjoon Kim, Kyochul Jang, Hyungwoo Song, Minjeong Shin, and Bongwon Suh. 2026. CodeVoyager: Integrating Interactive Visual Aids with LLMs for Code Comprehension. In *Proceedings of the 31st International Conference on Intelligent User Interfaces (IUI '26)*. ACM, New York, NY, USA, 1053–1084. doi:10.1145/3742413.3789057
- [71] David Kirsh and Paul Maglio. 1994. On Distinguishing Epistemic from Pragmatic Action. *Cognitive Science* 18, 4 (Oct. 1994), 513–549. doi:10.1207/s15516709cog1804\_1
- [72] Hyung-Kwon Ko, Gwanmo Park, Hyeon Jeon, Jaemin Jo, Juho Kim, and Jinwook Seo. 2023. Large-scale Text-to-Image Generation Models for Visual Artists' Creative Works. In *Proceedings of the 28th International Conference on Intelligent User Interfaces (IUI '23)*. ACM, New York, NY, USA, 919–933. doi:10.1145/3581641.3584078
- [73] Kenneth R. Koedinger and Vincent Aleven. 2007. Exploring the Assistance Dilemma in Experiments with Cognitive Tutors. *Educational Psychology Review* 19, 3 (2007), 239–264. doi:10.1007/s10648-007-9049-0
- [74] David C. Krakauer. 2016. Will A.I. Harm Us? Better to Ask How We'll Reckon With Our Hybrid Nature. *Nautilus*. <https://nautilus.us/will-ai-harm-us-better-to-ask-how-well-reckon-with-our-hybrid-nature-236098> Published September 5, 2016. Accessed September 26, 2026.
- [75] Harsh Kumar, Ruiwei Xiao, Benjamin Lawson, Ilya Musabirov, Jiakai Shi, Xinyuan Wang, Huayin Luo, Joseph Jay Williams, Anna N. Rafferty, John Stamper, and Michael Liut. 2024. Supporting Self-Reflection at Scale with Large Language Models: Insights from Randomized Field Experiments in Classrooms. In *Proceedings of the Eleventh ACM Conference on Learning @ Scale (L@S '24)*. ACM, New York, NY, USA, 86–97. doi:10.1145/3657604.3662042
- [76] Tzu-Sheng Kuo, Aaron Lee Halfaker, Zirui Cheng, Jiwoo Kim, Meng-Hsin Wu, Tongshuang Wu, Kenneth Holstein, and Haiyi Zhu. 2024. Wikibench: Community-Driven Data Curation for AI Evaluation on Wikipedia. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–24. doi:10.1145/3613904.3642278
- [77] Walter L. Leite, Samrat Roy, Nilanjana Chakraborty, George Michailidis, A. Corinne Huggins-Manley, Sidney D' Mello, Mohamad Kazem Shirani Faradonbeh, Emily Jensen, Huan Kuang, and Zeyuan Jing. 2022. A novel video recommendation system for algebra: An effectiveness evaluation study. In *LAK22: 12th International Learning Analytics and Knowledge Conference (LAK22)*. ACM, New York, NY, USA, 294–303. doi:10.1145/3506860.3506906
- [78] Shahid Latif, Zheng Zhou, Yoon Kim, Fabian Beck, and Nam Wook Kim. 2022. Kori: Interactive Synthesis of Text and Charts in Data Documents. *IEEE Transactions on Visualization and Computer Graphics* 28, 1 (Jan. 2022), 184–194. doi:10.1109/tvcg.2021.3114802



- [79] Bryan Lawson. 2006. *How Designers Think: The Design Process Demystified*. Routledge. [https://books.google.com/books/about/How\\_Designers\\_Think.html?id=IPvqZJNAdG8C](https://books.google.com/books/about/How_Designers_Think.html?id=IPvqZJNAdG8C)
- [80] Danielle Levac, Heather Colquhoun, and Kelly K. O'Brien. 2010. Scoping studies: advancing the methodology. *Implementation Science* 5, 1 (2010), 69. doi:10.1186/1748-5908-5-69
- [81] Yaniv Leviathan, Dani Valevski, Matan Kalman, Danny Lumen, Eyal Segalis, Eyal Molad, Shlomi Pasternak, Vishnu Natchu, Valerie Nygaard, Srinivasan Venkatachary, James Manyika, and Yossi Matias. 2026. Generative UI: LLMs are Effective UI Generators. arXiv:2604.09577. <https://arxiv.org/abs/2604.09577>
- [82] Yaniv Leviathan, Dani Valevski, Vishnu Natchu, and Yossi Matias. 2025. Generative UI: A rich, custom, visual interactive user experience for any prompt. Google Research. <https://research.google/blog/generative-ui-a-rich-custom-visual-interactive-user-experience-for-any-prompt/> Published November 18, 2025.
- [83] Ariel Levy, Monica Agrawal, Arvind Satyanarayan, and David Sontag. 2021. Assessing the Impact of Automated Suggestions on Decision Making: Domain Experts Mediate Model Errors but Take Less Initiative. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. ACM, New York, NY, USA, 1–13. doi:10.1145/3411764.3445522
- [84] Maxim Lisnic, Vidya Setlur, and Nicole Sultanum. 2025. Plume: Scaffolding Text Composition in Dashboards. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–18. doi:10.1145/3706598.3713580
- [85] Michael Xieyang Liu, Aniket Kittur, and Brad A. Myers. 2022. Crystalline: Lowering the Cost for Developers to Collect and Organize Information for Decision Making. In *CHI Conference on Human Factors in Computing Systems (CHI '22)*. ACM, New York, NY, USA, 1–16. doi:10.1145/3491102.3501968
- [86] Michael Xieyang Liu, Tongshuang Wu, Tianying Chen, Franklin Mingzhe Li, Aniket Kittur, and Brad A. Myers. 2024. Selenite: Scaffolding Online Sensemaking with Comprehensive Overviews Elicited from Large Language Models. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–26. doi:10.1145/3613904.3642149
- [87] Xinyi Lu, Kexin Phyllis Ju, Mitchell Dudley, Larissa Sano, and Xu Wang. 2026. AI-Mediated Feedback Improves Student Revisions: A Randomized Trial with FeedbackWriter in a Large Undergraduate Course. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)*. ACM, New York, NY, USA, 1–26. doi:10.1145/3772318.3791121
- [88] Sehi L'Yi, Astrid van den Brandt, Etowah Adams, Huyen N. Nguyen, and Nils Gehlenborg. 2024. Learnable and Expressive Visualization Authoring Through Blended Interfaces. Center for Open Science. doi:10.31219/osf.io/pjcn4
- [89] Ruofei Ma, Yu Zhao, Yuheng Shao, Yunjie Yao, and Quan Li. 2025. StratIncon Detector: Analyzing Strategy Inconsistencies Between Real-Time Strategy and Preferred Professional Strategy in MOBA Esports. In *Proceedings of the 30th International Conference on Intelligent User Interfaces (IUI '25)*. ACM, New York, NY, USA, 1314–1334. doi:10.1145/3708359.3712088
- [90] Shuai Ma, Ying Lei, Xinru Wang, Chengbo Zheng, Chuhan Shi, Ming Yin, and Xiaojuan Ma. 2023. Who Should I Trust: AI or Myself? Leveraging Human and AI Correctness Likelihood to Promote Appropriate Trust in AI-Assisted Decision-Making. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. ACM, New York, NY, USA, 1–19. doi:10.1145/3544548.3581058
- [91] Shuai Ma, Junling Wang, Yuanhao Zhang, Xiaojuan Ma, and April Yi Wang. 2025. DBOX: Scaffolding Algorithmic Programming Learning through Learner-LLM Co-Decomposition. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–20. doi:10.1145/3706598.3713748
- [92] Shuai Ma, Zijun Wei, Feng Tian, Xiangmin Fan, Jianming Zhang, Xiaohui Shen, Zhe Lin, Jin Huang, Radomir Měch, Dimitris Samaras, and Hongan Wang. 2019. SmartEye: Assisting Instant Photo Taking via Integrating User Preference with Deep View Proposal Network. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*. ACM, New York, NY, USA, 1–12. doi:10.1145/3290605.3300701
- [93] Wendy E. Mackay. 2000. Responding to cognitive overload: Co-adaptation between users and technology. *Intellectica* 30, 1 (2000), 177–193. doi:10.3406/intel.2000.1597
- [94] Allan MacLean, Richard Young, Victoria Bellotti, and Thomas Moran. 1991. Questions, Options, and Criteria: Elements of Design Space Analysis. *Human-Computer Interaction* 6, 3 (Sept. 1991), 201–250. doi:10.1207/s15327051hci0603&4\_2
- [95] Lauren Margulieux and Richard Catrambone. 2017. Using Learners' Self-Explanations of Subgoals to Guide Initial Problem Solving in App Inventor. In *Proceedings of the 2017 ACM Conference on International Computing Education Research (ICER '17)*. ACM, New York, NY, USA, 21–29. doi:10.1145/3105726.3106168
- [96] Ference Marton and Keith Trigwell. 2000. Variatio Est Mater Studiorum. *Higher Education Research & Development* 19, 3 (Nov. 2000), 381–395. doi:10.1080/07294360020021455
- [97] Damien Masson, Young-Ho Kim, and Fanny Chevalier. 2025. Textshop: Interactions Inspired by Drawing Software to Facilitate Text Editing. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–14. doi:10.1145/3706598.3713862
- [98] Justin Matejka, Tovi Grossman, and George Fitzmaurice. 2012. Citeology: Visualizing Paper Genealogy. In *CHI '12 Extended Abstracts on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 181–190. doi:10.1145/2212776.2212796
- [99] Marshall McLuhan. 1994. *Understanding Media: The Extensions of Man*. MIT Press, Cambridge, MA. <https://mitpress.mit.edu/9780262631594/understanding-media/> Originally published in 1964. Introduction by Lewis H. Lapham.
- [100] Bryan Min, Allen Chen, Yining Cao, and Haijun Xia. 2025. Malleable Overview-Detail Interfaces. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–25. doi:10.1145/3706598.3714164
- [101] Bryan Min, Sangho Suh, Jim Hollan, and Haijun Xia. 2026. Making Abstraction Concrete: A Design Space and Interaction Model of Abstraction in Interactive Systems. doi:10.48550/arXiv.2605.11344 arXiv:2605.11344 Preprint, version 1, May 11, 2026.
- [102] Bryan Min and Haijun Xia. 2025. Meridian: A Design Framework for Malleable Overview-Detail Interfaces. In *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (UIST '25)*. Association for Computing Machinery, New York, NY, USA, 1–14. doi:10.1145/3746059.3747654
- [103] Margaret Mitchell, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, and Timnit Gebru. 2019. Model Cards for Model Reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*. Association for Computing Machinery, New York, NY, USA. doi:10.1145/3287560.3287596
- [104] Yamato Miyatake. 2026. HCI Research Trails: An Interactive Citation Map for HCI Research. Interactive web visualization and open-source software. <https://hci-research-trails.vercel.app/> Source code: <https://github.com/mytkymt/paper-lineage>; accessed 2026-08-31.
- [105] Sina Mohseni, Niloofar Zarei, and Eric D. Ragan. 2021. A Multidisciplinary Survey and Framework for Design and Evaluation of Explainable AI Systems. *ACM Transactions on Interactive Intelligent Systems* 11, 3-4, Article 24 (aug 2021), 45 pages. doi:10.1145/3387166
- [106] Junho Myung, Hyunseung Lim, Hana Oh, Hyoungwook Jin, Nayeon Kang, So-Yeon Ahn, Hwajung Hong, Alice Oh, and Juho Kim. 2026. When Scaffolding Breaks: Investigating Student Interaction with LLM-Based Writing Support in Real-Time K-12 EFL Classrooms. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)*. ACM, New York, NY, USA, 1–18. doi:10.1145/3772318.3791517
- [107] Donald A. Norman. 1991. Cognitive Artifacts. In *Designing Interaction: Psychology at the Human-Computer Interface*, John M. Carroll (Ed.). Cambridge University Press, 17–38. [https://www.researchgate.net/publication/213799289\\_Cognitive\\_Artifacts](https://www.researchgate.net/publication/213799289_Cognitive_Artifacts)
- [108] Donald A. Norman. 2008. Signifiers, not affordances. *Interactions* 15, 6 (2008), 18–19. doi:10.1145/1409040.1409044
- [109] Kai Nylund, Jennifer Mankoff, and Venkatesh Potluri. 2025. Matplotlib: A Python Library for Adding Alt Text to Matplotlib Figures in Computational Notebooks. *Computer Graphics Forum* 44, 3 (May 2025). doi:10.1111/cgf.70119
- [110] Bruno Oliveira and Carla Teixeira Lopes. 2023. From 10 Blue Links Pages to Feature-Full Search Engine Results Pages - Analysis of the Temporal Evolution of SERP Features. In *Proceedings of the 2023 Conference on Human Information Interaction and Retrieval (CHIIR '23)*. ACM, New York, NY, USA, 338–345. doi:10.1145/3576840.3578307
- [111] Srishti Palani, Zijian Ding, Austin Nguyen, Andrew Chuang, Stephen MacNeil, and Steven P. Dow. 2021. CoNotate: Suggesting Queries Based on Notes Promotes Knowledge Discovery. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. ACM, New York, NY, USA, 1–14. doi:10.1145/3411764.3445618
- [112] Srishti Palani, Aakanksha Naik, Doug Downey, Amy X. Zhang, Jonathan Bragg, and Joseph Chee Chang. 2023. Relatedly: Scaffolding Literature Reviews with Existing Related Work Sections. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. ACM, New York, NY, USA, 1–20. doi:10.1145/3544548.3580841
- [113] Zachary A. Pardos, Matthew Tang, Ioannis Anastasopoulos, Shreya K. Sheel, and Ethan Zhang. 2023. OATutor: An Open-source Adaptive Tutoring System and Curated Content Library for Learning Sciences Research. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. ACM, New York, NY, USA, 1–17. doi:10.1145/3544548.3581574
- [114] John W. Payne, James R. Bettman, and Eric J. Johnson. 1993. *The Adaptive Decision Maker*. Cambridge University Press. doi:10.1017/cb09781139173933
- [115] Roy D. Pea. 2004. The Social and Technological Dimensions of Scaffolding and Related Theoretical Concepts for Learning, Education, and Human Activity. *Journal of the Learning Sciences* 13, 3 (July 2004), 423–451. doi:10.1207/s15327809jlst1303\_6

- [116] Xiaohan Peng, Janin Koch, and Wendy E. Mackay. 2025. FusAIIn: Composing Generative AI Visual Prompts Using Pen-based Interaction. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–20. doi:10.1145/3706598.3714027
- [117] Kevin Pu, K. J. Kevin Feng, Tovi Grossman, Tom Hope, Bhavana Dalvi Mishra, Matt Latzke, Jonathan Bragg, Joseph Chee Chang, and Pao Siangliulue. 2025. IdeaSynth: Iterative Research Idea Development Through Evolving and Composing Idea Facets with Literature-Grounded Feedback. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–31. doi:10.1145/3706598.3714057
- [118] Sadhana Puntambekar and Roland Hübscher. 2005. Tools for scaffolding students in a complex learning environment: What have we gained and what have we missed? *Educational psychologist* 40, 1 (2005), 1–12.
- [119] Chris Quintana, Brian J. Reiser, Elizabeth A. Davis, Joseph Krajcik, Eric Fretz, Ravit Golan Duncan, Eleni Kyza, Daniel Edelson, and Elliot Soloway. 2004. A Scaffolding Design Framework for Software to Support Science Inquiry. *Journal of the Learning Sciences* 13, 3 (July 2004), 337–386. doi:10.1207/s15327809jls1303\_4
- [120] Brian J. Reiser. 2004. Scaffolding Complex Learning: The Mechanisms of Structuring and Problematising Student Work. *Journal of the Learning Sciences* 13, 3 (July 2004), 273–304. doi:10.1207/s15327809jls1303\_2
- [121] Mohi Reza, Nathan M Laundry, Ilya Musabirov, Peter Dushniku, Zhi Yuan “Michael” Yu, Kashish Mittal, Tovi Grossman, Michael Liut, Anastasia Kuzminykh, and Joseph Jay Williams. 2024. ABScribe: Rapid Exploration & Organization of Multiple Writing Variations in Human-AI Co-Writing Tasks using Large Language Models. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–18. doi:10.1145/3613904.3641899
- [122] Bethany Rittle-Johnson, Abbey M. Loehr, and Kelley Durkin. 2017. Promoting self-explanation to improve mathematics learning: A meta-analysis and instructional design principles. *ZDM* 49, 4 (Jan. 2017), 599–611. doi:10.1007/s11858-017-0834-z
- [123] Sherry Ruan, Liwei Jiang, Justin Xu, Bryce Joe-Kun Tham, Zhengneng Qiu, Yeshuang Zhu, Elizabeth L. Murnane, Emma Brunskill, and James A. Landay. 2019. QuizBot: A Dialogue-based Adaptive Learning System for Factual Knowledge. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*. ACM, New York, NY, USA, 1–13. doi:10.1145/3290605.3300587
- [124] Marcel Ruoff, Brad A Myers, and Alexander Maedche. 2023. ONYX: Assisting Users in Teaching Natural Language Interfaces Through Multi-Modal Interactive Task Learning. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. ACM, New York, NY, USA, 1–16. doi:10.1145/3544548.3580964
- [125] Anara Sandygulova, Wafa Johal, Zhanel Zhexenova, Bolat Tleubayev, Aida Zhanatkyzy, Aizada Turarova, Zhansaula Telisheva, Anna CohenMiller, Thibault Asselborn, and Pierre Dillenbourg. 2020. CoWriting Kazakh: Learning a New Script with a Robot. In *Proceedings of the 2020 ACM/IEEE International Conference on Human-Robot Interaction (HRI '20)*. ACM, New York, NY, USA, 113–120. doi:10.1145/3319502.3374813
- [126] Mike Scaife and Yvonne Rogers. 1996. External cognition: how do graphical representations work? *International journal of human-computer studies* 45, 2 (1996), 185–213.
- [127] Tobias Schneider, Joana Hois, Alischa Rosenstein, Sabiha Ghellal, Dimitra Theofanou-Füllbier, and Ansgar R.S. Gerlicher. 2021. ExplAIIn Yourself! Transparency for Positive UX in Autonomous Driving. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. ACM, New York, NY, USA, 1–12. doi:10.1145/3411764.3446647
- [128] D.A. Schön. 1992. Designing as reflective conversation with the materials of a design situation. *Knowledge-Based Systems* 5, 1 (March 1992), 3–14. doi:10.1016/0950-7051(92)90020-g
- [129] Omar Shaikh, Valentino Emil Chai, Michele Gelfand, Diyi Yang, and Michael S. Bernstein. 2024. Rehearsal: Simulating Conflict to Teach Conflict Resolution. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–20. doi:10.1145/3613904.3642159
- [130] Jocelyn J Shen, Jennifer King Chen, Leah Findlater, and Griffin Dietz Smith. 2025. eaSEL: Promoting Social-Emotional Learning and Parent-Child Interaction through AI-Mediated Content Consumption. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–18. doi:10.1145/3706598.3713405
- [131] Bruce Sherin, Brian J. Reiser, and Daniel Edelson. 2004. Scaffolding Analysis: Extending the Scaffolding Metaphor to Learning Artifacts. *Journal of the Learning Sciences* 13, 3 (July 2004), 387–421. doi:10.1207/s15327809jls1303\_5
- [132] Donghoon Shin, Tze-Yu Chen, Gary Hsieh, and Lucy Lu Wang. 2025. What About My Design Context?: Exploring the Use of Generative AI to Support Customization of Translational Research Artifacts. In *Proceedings of the 2025 ACM Designing Interactive Systems Conference (DIS '25)*. ACM, New York, NY, USA, 1210–1227. doi:10.1145/3715336.3735686
- [133] Donghoon Shin, Lucy Lu Wang, and Gary Hsieh. 2024. From Paper to Card: Transforming Design Implications with Generative AI. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–15. doi:10.1145/3613904.3642266
- [134] Betsy Sparrow, Jenny Liu, and Daniel M. Wegner. 2011. Google effects on memory: Cognitive consequences of having information at our fingertips. *Science* 333, 6043 (2011), 776–778. doi:10.1126/science.1207745
- [135] Moritz Spiller, Ying-Hsang Liu, Md Zakir Hossain, Tom Gedeon, Julia Geissler, and Andreas Nürnberger. 2021. Predicting Visual Search Task Success from Eye Gaze Data as a Basis for User-Adaptive Information Visualization Systems. *ACM Transactions on Interactive Intelligent Systems* 11, 2 (May 2021), 1–25. doi:10.1145/3446638
- [136] Valeria Starkova and Iris Howley. 2026. CoAutoML: User Interface Framework for Machine Learning Novices using LLM-based AutoML and Test-Driven Machine Teaching. In *Proceedings of the 31st International Conference on Intelligent User Interfaces (IUI '26)*. ACM, New York, NY, USA, 755–782. doi:10.1145/3742413.3789153
- [137] Sangho Suh, Meng Chen, Bryan Min, Toby Jia-Jun Li, and Haijun Xia. 2024. Luminate: Structured Generation and Exploration of Design Space with Large Language Models for Human-AI Co-Creation. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–26. doi:10.1145/3613904.3642400
- [138] Eva Erdsone Toth, Daniel D. Suthers, and Alan M. Lesgold. 2002. “Mapping to know”: The effects of representational guidance and reflective assessment on scientific inquiry. *Science Education* 86, 2 (Feb. 2002), 264–286. doi:10.1002/sce.10004
- [139] Andrea C. Tricco, Erin Lillie, Wasifa Zarin, Kelly K. O’Brien, Heather Colquhoun, Danielle Levac, David Moher, Michah D. J. Peters, Tanya Horsley, Laura Weeks, Susanne Hempel, Elie A. Akl, Christine Chang, Jessie McGowan, Lesley Stewart, Lisa Hartling, Adrian Aldcroft, Michael G. Wilson, Chantelle Garritty, Simon Lewin, Christina M. Godfrey, Marilyn T. Macdonald, Etienne V. Langlois, Karla Soares-Weiser, Jo Moriarty, Tammy Clifford, Özge Tunçalp, and Sharon E. Straus. 2018. PRISMA Extension for Scoping Reviews (PRISMA-ScR): Checklist and Explanation. *Annals of Internal Medicine* 169, 7 (2018), 467–473. doi:10.7326/M18-0850
- [140] Tushar Udesi, Anna Khazenzon, Kabir Khan, Nick Breen, R. J. Corwin, Chris DiGiano, Kodi Weatherholtz, and Marek Zaluski. 2026. Methodologies for Improving the Quality of AI Tutoring in K-12 Education. In *Artificial Intelligence in Education (Lecture Notes in Computer Science, Vol. 16582)*. Springer, Cham, 201–214. doi:10.1007/978-3-032-29755-6\_14
- [141] Janneke van de Pol, Monique Volman, and Jos Beishuizen. 2010. Scaffolding in Teacher–Student Interaction: A Decade of Research. *Educational Psychology Review* 22, 3 (April 2010), 271–296. doi:10.1007/s10648-010-9127-6
- [142] David Wadden, Kejian Shi, Jacob Morrison, Alan Li, Aakanksha Naik, Shruti Singh, Nitzan Barzilay, Kyle Lo, Tom Hope, Luca Soldaini, Shannon Zejiang Shen, Doug Downey, Hannaneh Hajishirzi, and Arman Cohan. 2025. SciRIF: A Resource to Enhance Language Model Instruction-Following over Scientific Literature. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng (Eds.). Association for Computational Linguistics, Suzhou, China, 6072–6109. doi:10.18653/v1/2025.emnlp-main.310
- [143] Thiemo Wambsganss, Tobias Kueng, Matthias Soellner, and Jan Marco Leimeister. 2021. ArgueTutor: An Adaptive Dialog-Based Learning System for Argumentation Skills. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. ACM, New York, NY, USA, 1–13. doi:10.1145/3411764.3445781
- [144] Chenglong Wang, Yu Feng, Rastislav Bodik, Isil Dillig, Alvin Cheung, and Amy J Ko. 2021. Falx: Synthesis-Powered Visualization Authoring. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. ACM, New York, NY, USA, 1–15. doi:10.1145/3411764.3445249
- [145] Leijie Wang, Kathryn Yurechko, Pranati Dani, Quan Ze Chen, and Amy X. Zhang. 2025. End User Authoring of Personalized Content Classifiers: Comparing Example Labeling, Rule Writing, and LLM Prompting. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–21. doi:10.1145/3706598.3713691
- [146] Minhong Wang, Bian Wu, Paul A. Kirschner, and J. Michael Spector. 2018. Using cognitive mapping to foster deeper learning with complex problems in a computer-based environment. *Computers in Human Behavior* 87 (Oct. 2018), 450–458. doi:10.1016/j.chb.2018.01.024
- [147] Zifeng Wang, Lang Cao, Qiao Jin, Joey Chan, Nicholas Wan, Behdad Afzali, Hyun-Jin Cho, Chang-In Choi, Mehdi Emamverdi, Manjot K. Gill, Sun-Hyung Kim, Yijia Li, Yi Liu, Yiming Luo, Hanley Ong, Justin F. Rousseau, Irfan Sheikh, Jenny J. Wei, Ziyang Xu, Christopher M. Zallek, Kyungsang Kim, Yifan Peng, Zhiyong Lu, and Jimeng Sun. 2025. A foundation model for human-AI collaboration in medical literature mining. *Nature Communications* 16, 1 (2025), 8361. doi:10.1038/s41467-025-62058-5
- [148] Florian Weber, Thiemo Wambsganss, Seyed Parsa Neshaei, and Matthias Soellner. 2024. LegalWriter: An Intelligent Writing Support System for Structured and Persuasive Legal Case Writing for Novice Law Students. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New

- York, NY, USA, 1–23. doi:10.1145/3613904.3642743
- [149] Jacob O. Wobbrock, Shaun K. Kane, Krzysztof Z. Gajos, Susumu Harada, and Jon Froehlich. 2011. Ability-based design: Concept, principles and examples. *ACM Transactions on Accessible Computing* 3, 3, Article 9 (2011), 27 pages. doi:10.1145/1952383.1952384
- [150] Claes Wohlin. 2014. Guidelines for snowballing in systematic literature studies and a replication in software engineering. In *Proceedings of the 18th International Conference on Evaluation and Assessment in Software Engineering*. Association for Computing Machinery, New York, NY, USA, 1–10. doi:10.1145/2601248.2601268
- [151] David Wood, Jerome S. Bruner, and Gail Ross. 1976. The Role of Tutoring in Problem Solving. *Journal of Child Psychology and Psychiatry* 17, 2 (April 1976), 89–100. doi:10.1111/j.1469-7610.1976.tb00381.x
- [152] Austin P. Wright, Omar Shaikh, Haekyu Park, Will Epperson, Muhammed Ahmed, Stephane Pinel, Duen Horng (Polo) Chau, and Diyi Yang. 2021. RE-CAST: Enabling User Recourse and Interpretability of Toxicity Detection Models with Interactive Visualization. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW1 (April 2021), 1–26. doi:10.1145/3449280
- [153] Tongshuang Wu, Ellen Jiang, Aaron Donsbach, Jeff Gray, Alejandra Molina, Michael Terry, and Carrie J Cai. 2022. PromptChainer: Chaining Large Language Model Prompts through Visual Programming. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI EA '22). Association for Computing Machinery, New York, NY, USA, Article 359, 10 pages. doi:10.1145/3491101.3519729
- [154] Tongshuang Wu, Michael Terry, and Carrie Jun Cai. 2022. AI Chains: Transparent and Controllable Human-AI Interaction by Chaining Large Language Model Prompts. In *CHI Conference on Human Factors in Computing Systems* (CHI '22). ACM, New York, NY, USA, 1–22. doi:10.1145/3491102.3517582
- [155] Meng Xia, Reshika Palaniyappan Velumani, Yong Wang, Huamin Qu, and Xiaojuan Ma. 2021. QLens: Visual Analytics of Multi-step Problem-solving Behaviors for Improving Question Design. *IEEE Transactions on Visualization and Computer Graphics* 27, 2 (Feb. 2021), 870–880. doi:10.1109/tvcg.2020.3030337
- [156] Weitao You, Yinyu Lu, Zirui Ma, Nan Li, Mingxu Zhou, Xue Zhao, Pei Chen, and Lingyun Sun. 2025. DesignManager: An Agent-Powered Copilot for Designers to Integrate AI Design Tools into Creative Workflows. *ACM Transactions on Graphics* 44, 4 (July 2025), 1–26. doi:10.1145/3730919
- [157] J.D. Zamfirescu-Pereira, Eunice Jun, Michael Terry, Qian Yang, and Bjoern Hartmann. 2025. Beyond Code Generation: LLM-supported Exploration of the Program Design Space. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (CHI '25). ACM, New York, NY, USA, 1–17. doi:10.1145/3706598.3714154
- [158] Siyu Zha, Yujia Liu, Chengbo Zheng, Jiaqi Xu, Fuze Yu, Jiangtao Gong, and Yingqing Xu. 2025. Mentigo: An Intelligent Agent for Mentoring Students in the Creative Problem Solving Process. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (CHI '25). ACM, New York, NY, USA, 1–22. doi:10.1145/3706598.3713952
- [159] Ashley Ge Zhang, Yan Chen, and Steve Oney. 2023. VizProg: Identifying Misunderstandings By Visualizing Students' Coding Progress. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (CHI '23). ACM, New York, NY, USA, 1–16. doi:10.1145/3544548.3581516
- [160] Jiaje Zhang and Donald A. Norman. 1994. Representations in Distributed Cognitive Tasks. *Cognitive Science* 18, 1 (Jan. 1994), 87–122. doi:10.1207/s15516709cog1801\_3
- [161] Xinlei Zhang, Takashi Miyaki, and Jun Rekimoto. 2020. WithYou: Automated Adaptive Speech Tutoring With Context-Dependent Speech Recognition. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (CHI '20). ACM, New York, NY, USA, 1–12. doi:10.1145/3313831.3376322
- [162] Yu Zhang, Jingwei Sun, Li Feng, Cen Yao, Mingming Fan, Liuxin Zhang, Qianying Wang, Xin Geng, and Yong Rui. 2024. See Widely, Think Wisely: Toward Designing a Generative Multi-agent System to Burst Filter Bubbles. In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (CHI '24). ACM, New York, NY, USA, 1–24. doi:10.1145/3613904.3642545
- [163] Yuzhe Zhou, Prithvi Manjunatha Babu, Udayan Pandey, Alejandra J. Magana, and Tianyi Li. 2026. ExPeerience: Towards AI-Assisted Learnersourcing to Bridge Conceptual Understanding and Problem Solving in Database Programming Education. In *Proceedings of the 31st International Conference on Intelligent User Interfaces* (IUI '26). ACM, New York, NY, USA, 1178–1201. doi:10.1145/3742413.3789139
- [164] Zhongyi Zhou, Jing Jin, Vrushank Phadnis, Xiuxiu Yuan, Jun Jiang, Xun Qian, Kristen Wright, Mark Sherwood, Jason Mayes, Jingtao Zhou, Yiyi Huang, Zheng Xu, Yinda Zhang, Johnny Lee, Alex Olwal, David Kim, Ram Iyengar, Na Li, and Ruofei Du. 2025. InstructPipe: Generating Visual Blocks Pipelines with Human Instructions and LLMs. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (CHI '25). ACM, New York, NY, USA, 1–22. doi:10.1145/3706598.3713905
- [165] Jonathan Zong, Isabella Pedraza Pineros, Mengzhu (Katie) Chen, Daniel Hajas, and Arvind Satyanarayan. 2025. Semantic Scaffolding: Augmenting Textual Structures with Domain-Specific Groupings for Accessible Data Exploration. arXiv:2506.15883. <https://arxiv.org/abs/2506.15883>
- [166] Alexandra Zytek, Dongyu Liu, Rhema Vaithianathan, and Kalyan Veeramachaneni. 2022. Sibyl: Understanding and Addressing the Usability Challenges of Machine Learning In High-Stakes Decision Making. *IEEE Transactions on Visualization and Computer Graphics* 28, 1 (Jan. 2022), 1161–1171. doi:10.1109/tvcg.2021.3114864

## A EVIDENCE EXTRACTION AND CODING

This appendix holds what a reader needs to check or reuse the coding: the fields of each evidence card, how the closed and open fields were coded, the agreement checks, the data behind each analysis, how sources for the outcome comparisons were chosen, and how cited sources were matched.

### A.1 Card Fields

Stream 1 mechanism cards (Table 5) describe the design, and theory-lineage cards (Table 6) describe its use of prior work. Each Stream 2 source has a knowledge-claim card (Table 7). Table 9 distinguishes the available evidence for each analysis. The sectioned layout follows Model Cards [103], and AI-assisted design-card generation is a precedent for turning source findings into cards [133].

**Table 5: Stream 1 mechanism card: work and human activity first, then scaffold design and evaluation.**

| Card section                   | Fields recorded                                                                                                                                                                       |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Workflow changes (RQ1)         | Process before and after support; mechanism credited; attribution basis; source of the baseline.                                                                                      |
| Human activity (RQ1)           | Activity externalized to the tool or representation; activity retained by people; work still required.                                                                                |
| Scaffold design (RQ2)          | Author term; facet and value, with their merged codes; what makes each mechanism separate; supporting passage and page.                                                               |
| Learning and support (RQ2)     | Learning measurement, direction, and evidence; reliance; changes in support and their grounds; observation horizon.                                                                   |
| Measures and comparisons (RQ2) | Recorded outcome layer (process, artifact, learning); author claim, claim scope, contribution type, study design, participants, comparison, statistics, and claim–evidence alignment. |
| Conditions and limitations     | Author-stated limitations and boundary conditions.                                                                                                                                    |
| Recorded codes                 | Work domain, intent, expertise, capability gap, support properties, AI involvement, and technology era.                                                                               |
| Source                         | Paper ID, DOI, venue, year, page count, the PDF read, the full-text decision, a map of relevant pages, and codebook notes.                                                            |

**Table 6: Stream 1 theory-lineage card: how the paper uses the sources it cites (RQ2).**

| Card section                | Fields recorded                                                                                                            |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------|
| 0. Identity and position    | Link to the mechanism card; overall position toward prior work; position summary; human-facing rationale; lineage depth.   |
| 1. Documented sources (RQ2) | One entry per cited work: citation identity, relation to the focal paper, use, enactment, and supporting passage and page. |
| 2. What was read            | Introduction and related-work pages; design or evaluation pages checked for application of a cited idea.                   |
| 3. Synthesis note           | A short account of what the paper’s use of prior work shows.                                                               |

**Table 7: Stream 2 card: one to eight field-level claims, linked to the research questions they inform.**

| Card section             | Fields recorded                                                                                                                                                                                                      |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0. Identity and position | Card and work IDs; publication details; discovery routes; contribution types; position summary; overall uncertainty.                                                                                                 |
| 1. Page-grounded claims  | One entry per claim: author wording and reviewer interpretation; question links; applicability; affected human work; outcome evidence; evidence basis; counterevidence and limits; supporting excerpts; uncertainty. |
| 2. Evidence store        | Verbatim excerpts with pages and locators; links recording which claim each excerpt supports, its role, and the reviewer paraphrase.                                                                                 |

### A.2 Closed Fields

Closed fields take values from a fixed list. Intent records whether the support targets immediate performance or sensemaking, learning or skill development, or both. Measurement records what the evaluation assessed, so an intention to support learning stays apart from a

**Table 8: Blind open-code comparison on exactly matched observations. Codes gives existing / blind code counts; Rand is raw pair agreement, and pair F1 compares the pairs grouped together. Unmatched counts existing observations outside the comparison. Papers overlap across fields, so the 1,571 matched observations are not 1,571 distinct papers.**

| Field                | Obs. (papers) | Unmatched | Codes    | Rand  | ARI  | Pair F1 |
|----------------------|---------------|-----------|----------|-------|------|---------|
| Externalized to tool | 110 (110)     | 18        | 64 / 46  | 96.2% | .167 | .186    |
| Facet                | 198 (48)      | 15        | 132 / 76 | 98.0% | .132 | .142    |
| Process before       | 125 (48)      | 15        | 79 / 34  | 95.8% | .445 | .462    |
| Process after        | 419 (160)     | 0         | 262 / 60 | 97.0% | .079 | .088    |
| Value                | 695 (144)     | 0         | 517 / 95 | 98.1% | .082 | .087    |
| Learning evidence    | 24 (24)       | 24        | 5 / 16   | 50.4% | .108 | .217    |

measure of it; the direction of a result and its observation horizon are recorded with the measure. Support properties are coded separately and can co-occur: structuring, problematizing, contingency, adaptation, fading, responsibility transfer, and distributed support. Contingency responds to the person’s activity or need. Fading reduces support as competence grows; a fixed reduction on a schedule is not by itself fading. Responsibility transfer concerns who performs or regulates the activity, not whether the interface remains available. Technology is classified once per paper, from the AI in its system, an experimental condition, training, or evaluation, so a paper classified as generative AI can still contain mechanisms that involve no generation.

### A.3 Open Fields

The 21 open fields hold 142,877 observations in the papers’ own terms. Equivalent descriptions within a field were merged into 1,290 codes, and the codes were grouped into 223 categories and 92 families across 18 fields, with subcategories only where they add a distinction. Each grouping keeps links to its observations, and fields are never pooled. Codes that record missing information, such as an unspecified baseline or an unreported measure, stay apart from substantive categories. When one free-text value lists several limitations, its single code summarizes them rather than listing each. Appendix B defines the families and categories of the six fields the framework uses.

### A.4 Agreement Checks

To test the open-code grouping, a separate model pass coded extracted observations with the existing codes hidden, and we compared the two groupings field by field (Table 8). An observation entered the comparison only when its identifier, source card, and field value matched exactly. Two codings need not use the same words, so we compared partitions: whether each pair of observations shares a code in each coding. The adjusted Rand index (ARI) corrects pair agreement for chance, and positive-pair F1 summarizes overlap among the pairs either coding groups together; neither treats one coding as correct.

Raw agreement is high because both codings keep most pairs apart: for facet, 19,083 of 19,503 pairs are apart in both and 32 are together in both. The blind pass used fewer codes than the existing coding in five fields and more in learning evidence. It reported medium or high uncertainty on most assignments in five fields (for example, 659 of 695 values) and on 7 of 24 in learning evidence. The learning-evidence row covers 24 of 48 existing observations; the other 24 share a repeated not-measured value and fell outside the comparison’s selection rule.

A separate check asked only whether AI was involved: three fresh model sessions labeled 532 title-and-abstract records added in September and gave the same label to 529 (99.4%;  $\kappa = .988$ ).

### A.5 Validation Status of the Readings

For part of the evidence, the records of the original reading runs have not been fully validated, and we mark that evidence for provisional use. On 26 September 2026, such evidence appeared in 319 of the 1,076 learning papers and 1,252 of the 4,086 papers behind the main figures. The counts mark incomplete validation records, not errors, and the marked papers stay in the analyses.

### A.6 Data Used for Each Analysis

Each analysis uses the papers coded for the fields it needs (Table 9).

All 2,847 open-coded papers have complete domain and retained-work codes; 532 lack complete facet–value codes. The closed-field population includes 44 papers outside the open-field profile and two with unresolved support-property coding.

### A.7 Learning Measurement

Among the 2,891 closed-field papers, 301 (10.4%) measured learning. Two-thirds of those (199, 66.1%) measured it within a single session; 35 used repeated sessions, 38 followed people over weeks, and 25 measured learning in deployed use.

**Table 9: Data behind each analysis. Counts refer to the stated unit.**

| Data              | Count and unit                                                                                                       | Analysis                                                             |
|-------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Stream 1 corpus   | 4,638 papers on 28 September 2026; 475 non-AI comparison papers outside the systematic counts                        | Corpus counts (Section 4)                                            |
| Main figures      | 4,133 papers (4,162 records) with usable coding                                                                      | Annual, technology, and mechanism figures                            |
| Closed fields     | 2,891 papers coded by 28 August 2026; 301 measure learning                                                           | Intent, learning-measurement, and process counts                     |
| Open fields       | 142,877 observations in 21 fields from the 3,295 papers coded by 28 August 2026, of which 2,847 remain in the corpus | Design space and framework values; counts of papers carrying a value |
| Complete mappings | 2,315 papers with complete facet-value codes; 2,311 also with remaining-work codes                                   | Mechanism-by-work matrix (Appendix C)                                |
| Theory lineage    | 1,697 PDF-matched lineage cards; 43,733 recorded source-paper pairs                                                  | Roles of prior work (Section 6.5)                                    |
| Stream 2          | 512 sources; 2,867 claims                                                                                            | Prior findings and theory                                            |
| Citation network  | 4,751 papers from both streams                                                                                       | Citation atlas and co-citation map (Section 6.5)                     |

## A.8 Selection of Sources for Outcome Comparison

The study comparisons in Section 6 draw on sources chosen for their outcome evidence. A learning-scope assessment of 4,714 records proposed 1,226 within scope. We appraised the 422 of these whose cards recorded that learning was measured (twelve more had been appraised earlier and were analyzed separately), plus 14 performance cases found among 53 descriptions that needed interpretation: 436 records, or 434 distinct PDFs. The 190 sources flagged as having a documented capability comparison entered the detailed synthesis.

## A.9 Source Matching and Coverage

*A.9.1 Source-use coding.* Each theory-lineage record links one cited work to its role in one citing paper’s design, argument, or measure (Section 6.5), so a work cited by two papers yields two records.

*A.9.2 Citation metadata.* The bibliographic map joins the two streams by DOI or Semantic Scholar paper ID and counts a paper found in both once. It uses Semantic Scholar data retrieved on 9 September 2026 for 4,331 Stream 1 papers and 448 Stream 2 works (28 in both). Reference lists for 3,233 papers name 106,523 cited sources; two with conflicting identities are withheld. Of the remaining 106,521, 5,748 are cited by at least five review papers and 157 by at least 40 (Figure 15). A reference list shows that a paper cites a work; its lineage card shows how the paper uses it.

The co-citation map counts each citing paper once per source and keeps sources with at least five citers and pairs with at least three shared citers. It weights a pair by association strength: the pair’s shared citers times the number of papers, divided by the product of the two sources’ citer counts. Louvain clustering (seed 42) forms the neighbourhoods, and a maximum spanning forest supplies the drawn links; stem, branches, and leaves mark how widely a source is shared. Less-cited sources sit near their most shared neighbourhood without joining it. The interactive atlas supports zooming into a neighbourhood and inspecting each source.<sup>2</sup>

*A.9.3 Matching and unavailable sources.* Cited works without a DOI are matched by title and year. Title variants can split one work into several identities, which overcounts distinct works and undercounts shared references; descriptive labels and conflicting identities stay out of the shared-reference map. Papers without a reference list, and full texts we could not obtain or read, are outside the source-use analysis.

## B THE FRAMEWORK’S CODEBOOK

The framework combines mechanism design with five dimensions of human work and evidence. Tables 10 and 11 define those five dimensions; Appendix C.1 gives the mechanism categories and examples. Counts use the 28 August 2026 coding populations in Appendix A.6; a paper may carry several values. Each group names the paper denominator for its percentages. Daggers identify absent or unspecified values, as defined in each row.

### B.1 What follows for the person, the work and the evidence

<sup>2</sup>Interactive citation atlas: the interactive citation atlas .

**Table 10: The three dimensions about the person: what the person still does, keeps, and hands to the tool, with the share and number of papers carrying each value. A dagger marks the absence of a value.**

| Dimension and value                                | Definition                                                                                                                                                                                                                                                                                                                    | Share | Papers |
|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| <b>What the person still does</b> (n=2,842 papers) |                                                                                                                                                                                                                                                                                                                               |       |        |
| The task the tool did not do                       | Work of the task itself that the system does not do: performing the substantive task, making sense of what is shown, and checking and repairing what comes back.                                                                                                                                                              | 51.1% | 1,451  |
| New work the tool created                          | Obligations that exist only because the system exists: feeding and instructing it, and operating and coping with it.                                                                                                                                                                                                          | 29.9% | 851    |
| The whole task kept by the person                  | The person feeds the system and judges what comes back, across three or more stages, with no stage removed.                                                                                                                                                                                                                   | 10.6% | 300    |
| Final authority kept by design                     | The final decision, the reliance decision, values-laden judgement and accountability stay with the person as a matter of design or law, not as a step in a workflow.                                                                                                                                                          | 8.3%  | 237    |
| Duty asserted, nothing specified†                  | The value says responsibility remains with the person without naming any obligation.                                                                                                                                                                                                                                          | 0.1%  | 3      |
| <b>What the person keeps</b> (n=2,847 papers)      |                                                                                                                                                                                                                                                                                                                               |       |        |
| Judgement and decision                             | Interpreting and inferring, deciding and authorising, setting direction, bringing knowledge.                                                                                                                                                                                                                                  | 48.0% | 1,367  |
| Doing and revising the work                        | Doing the work, repairing and revising, steering the loop, improving the system.                                                                                                                                                                                                                                              | 35.7% | 1,016  |
| Checking the output                                | Checking output for errors, calibrating reliance, judging fit to goal, verifying against outside evidence.                                                                                                                                                                                                                    | 9.1%  | 260    |
| Responsibility, social and self-directed work      | Coordinating and deliberating with others, consent and disclosure, ethical judgment, communicating outward, holding accountability, care work; and learning, reflection and sustained attention to the system.                                                                                                                | 6.9%  | 197    |
| Not stated†                                        | The paper does not say what the person retains.                                                                                                                                                                                                                                                                               | 0.2%  | 7      |
| <b>What moves into the tool</b> (n=2,326 papers)   |                                                                                                                                                                                                                                                                                                                               |       |        |
| Computation and judgement                          | The tool predicts, simulates, fits models, calculates, optimises, estimates uncertainty, computes geometry; and attributes behaviour, classifies, detects errors, critiques, scores, infers a person's state, ranks and verifies.                                                                                             | 30.5% | 709    |
| Representation and translation                     | The tool gives external form, converts between formats and languages, lays material out, summarises, verbalises, encodes data visually, coordinates views, compiles specifications; and finds groupings, patterns, similarities, aggregates and comparisons.                                                                  | 23.6% | 549    |
| Decision, guidance and action                      | The tool recommends, selects the next step or decides outright; supplies templates, procedures, decompositions, reference material, instruction and prompts; propagates judgments, does clerical work, operates environments, enforces constraints, fabricates; stands in for people, coordinates them, relays communication. | 15.6% | 364    |
| Perception, search and memory                      | The tool senses, extracts, recognises, segments, transcribes and flags; finds and narrows material; records, stores, recalls and reminds; watches and tracks.                                                                                                                                                                 | 15.0% | 349    |
| Content generation                                 | The tool generates new artifacts (alternatives, images, prose, code, 3D forms, audio) and responses (explanations, questions, answers, utterances, conversational replies, revisions of the person's material).                                                                                                               | 14.9% | 346    |
| Offloading that failed or was withheld             | The offloading failed or was incomplete, or the tool deliberately stopped at a proposal and left the judgment with the person.                                                                                                                                                                                                | 0.3%  | 8      |
| Nothing offloaded, or not stated†                  | The value states that nothing moved to the tool, or names no operation at all.                                                                                                                                                                                                                                                | 0.0%  | 1      |



**Table 11: The two dimensions about the work and the evidence: what happens to the work and how learning is evidenced, with the share and number of papers carrying each value. A dagger marks the absence of a value.**

| Dimension and value                               | Definition                                                                                                                                                                                                                                                                                                               | Share | Papers |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| <b>What happens to the work</b> (n=2,643 papers)  |                                                                                                                                                                                                                                                                                                                          |       |        |
| The person's work changes form                    | The person still works, differently: reading new representations, navigating and searching, steering and specifying at a higher level, reviewing and selecting instead of producing, checking and repairing, following a guided procedure, judging and reflecting on new grounds, coordinating through shared artifacts. | 92.2% | 2,437  |
| A step becomes automatic                          | A step the person used to do is now done, or mostly done, by the system: automatic extraction or generation, exception-only attention, the same task made cheaper, automatic checking of the person's own work, effort freed for other work.                                                                             | 26.4% | 697    |
| Study condition or proposal, not practice         | The value describes a study arm, protocol or manipulation, or a proposed or recommended change that was not observed in use.                                                                                                                                                                                             | 16.1% | 425    |
| Work newly possible                               | The person can now do something they could not: work in a new setting or moment, at a larger scale or scope, or a task that was out of reach.                                                                                                                                                                            | 4.6%  | 121    |
| New work created by the system                    | Labour that did not exist before the system: setting up, configuring and maintaining it, added steps or obligations, burden or unproductive activity it caused.                                                                                                                                                          | 4.0%  | 105    |
| Work unchanged                                    | The practice did not change: the person keeps the core task and the system only informs it, under-uses or works around it, or nothing changed at all.                                                                                                                                                                    | 2.0%  | 54     |
| <b>How learning is evidenced</b> (n=2,800 papers) |                                                                                                                                                                                                                                                                                                                          |       |        |
| A proxy measured instead                          | Capability stood in for by a proxy: subjective judgements about the system, usability instruments, qualitative accounts, attitudes or engagement; in-session task performance, behaviour or rated work products; or the quality of the system's own output, including the machine's learning.                            | 37.6% | 1,054  |
| Learning measured                                 | An objective capability measure was taken: immediately with support present, later or with support withdrawn, or with a null, mixed or adverse result, or compromised by confounding or deliberate suppression.                                                                                                          | 9.4%  | 263    |
| Learning self-reported only                       | A learning or skill gain is asserted on self-report alone, or gestured at in remarks while the paper says learning itself was not evaluated.                                                                                                                                                                             | 4.6%  | 130    |
| Learning not measured†                            | Nothing about the person's learning, transfer, retention or capability was measured: the value names no measure and no substitute, the authors say so themselves or defer it, there were no participants at all, or learning is the stated purpose yet never evaluated.                                                  | 48.3% | 1,353  |

Where prior work is used

1,642 distinct papers · 1,697 PDF-matched lineage cards

Most shared identifier-linked sources

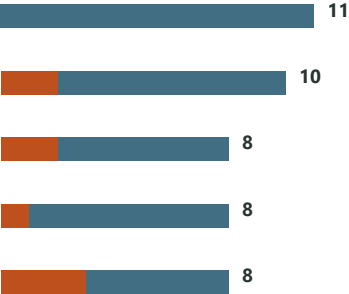
Manifold: A Model-Agnostic Framework for Interpretation and Diagnosis of Machine Learning Models  
Rulematrix: Visualizing and understanding classifiers with rules

The eyes have it: a task by data type taxonomy for information visualizations

Kori: Interactive synthesis of text and charts in data documents

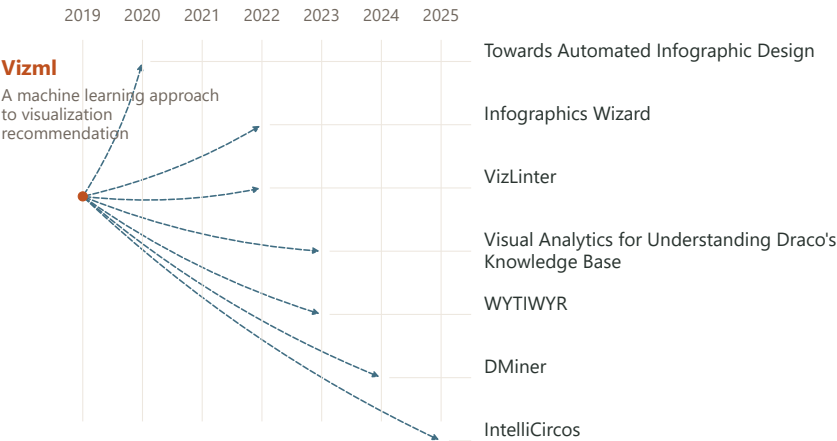
NL4DV: A Toolkit for Generating Analytic Specifications for Data Visualization from Natural Language Queries

Distinct papers with recorded use



Follow a source into later designs

Highest-ranked source with an unambiguous recorded year; every use is shown.

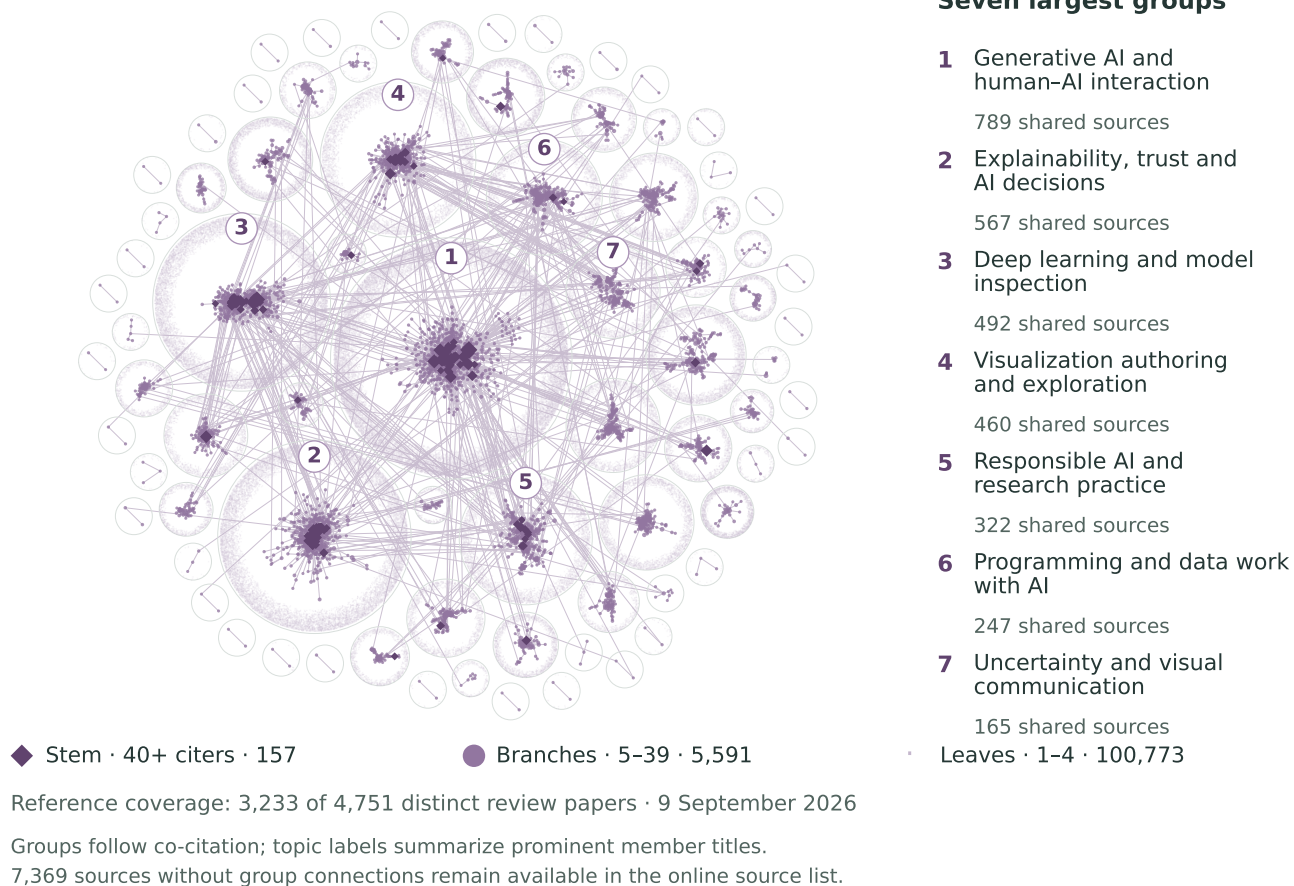


Solid / orange: the design takes something concrete from the source  
Dashed / blue: cited for framing, motivation, comparison or evaluation

**Figure 14: How papers use prior work.** Across 1,697 lineage cards for papers in the analysis, the upper panel ranks sources with a DOI or other identifier by the number of papers that use them. Orange marks papers whose design takes something concrete from the source (a mechanism, interface element, algorithm, representation or interaction); blue marks papers that cite it for framing, motivation, comparison or evaluation. Two language-model coders applied this one definition to every pair shown ( $\kappa = 0.90$ ). Below, the top-ranked source with a known publication year connects to later designs, with solid arrows for the first kind of use and dashed arrows for the second. The lineage records link 43,733 source–paper pairs to passages and cards.

## Shared sources across the two review streams

106,521 cited sources · 5,748 cited by five or more review papers



**Figure 15: Shared-reference neighborhoods.** Reference metadata covers 3,233 of 4,751 papers, from bibliographic data of 9 September 2026, before the corpus was limited to AI papers. Numbers label the seven largest groups of sources cited together.

## C DETAILED MECHANISM AND EVIDENCE PROFILES

*Coding and populations.* This appendix uses the coding of the mechanism cards completed by 28 August 2026, restricted to the systematic corpus (Section 4): 2,847 open-coded papers and 2,891 closed-field papers. Appendix A.6 lists the population behind each profile.

The complete facet catalogue supports the main mechanism analysis (Section 6); complementary evidence profiles support the workflow analysis (Section 5).

### C.1 Category Maps and Mechanism Examples

The eighteen category maps cover all 110 mechanism facets in five families. Branches group design operations that systems may combine. Schematics explain the choices; source figures show cited implementations. Variations are recorded implementations, not settings available in every system. Map order and size carry no frequency or effectiveness scale.

Search the full definitions. Links beneath each map open its source mechanisms.

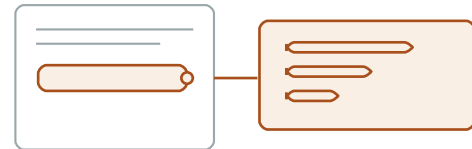
### C.1.1 Explanation and visibility. What can a person inspect and understand?

#### FC-C01 / Explanation and visibility

## Explaining system behaviour

### Which aspect of the system is made inspectable?

Makes a system's outputs, behaviour, uncertainty or current state inspectable, so people can understand what it is doing and judge its results.



#### DESIGN CHOICES

#### RETAINED DISTINCTIONS

#### A particular result



##### FC001

output rationale exposed

##### FC002

feature attribution for an output

#### Process and sequence



##### FC006

live process-state visibility

##### FC008

stepwise reasoning or execution trace

#### Model and internals



##### FC003

global model-behaviour summary

##### FC007

internal or latent state surfaced

#### Reliability and limits



##### FC004

uncertainty and confidence exposed

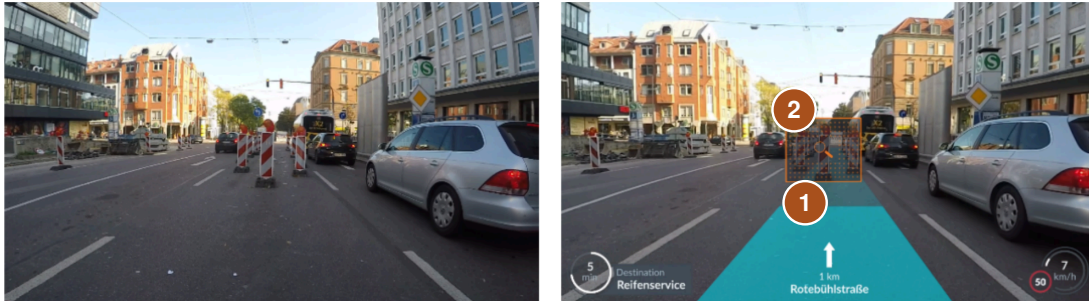
##### FC005

capability and knowledge-boundary disclosure

Recorded form · FC001: Rationale for a specific machine output (VA001).

Recorded form · FC002: Feature or factor contribution to an outcome (VA002).

**Figure 16: Explaining system behaviour. Which aspect of the system is made inspectable? Separate the explanation of one result, the running process, the model across cases, and the limits of what it can establish. Category definitions; Browse category mechanisms. Recorded forms: Rationale for a specific machine output; Feature or factor contribution to an outcome.**

**FC-C01 / SOURCE ILLUSTRATION****Explaining system behaviour****PAPER ILLUSTRATION · FC006**

1. Intended route overlaid on the road 2. Detected obstacle marked in the scene

Live overlays expose the vehicle's planned route and detected objects in the road scene.

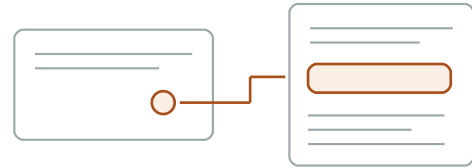
ExplAIn Yourself! Transparency for Positive UX in Autonomous Driving · Figure 2, PDF page 4

Schneider et al.; CC BY; cropped; numbered annotations added by this review.

**Figure 17: Explaining system behaviour: source illustration. Figure 2, PDF page 4 in [127]. Live overlays expose the vehicle's planned route and detected objects in the road scene. Inspect the illustrated mechanism.**

**FC-C02 / Explanation and visibility****Sources and evidence****How can a result be traced or checked?**

Connects a result to its sources, supporting evidence or underlying cases, so people can trace where it came from and check its basis.

**DESIGN CHOICES****RETAINED DISTINCTIONS****Origin and derivation****FC009**

output-to-source provenance link

**FC012**

analytic-history provenance capture

**Underlying cases****FC010**

aggregate-to-instance drill-down

**Evidence for a claim****FC011**

evidence presented for verification

---

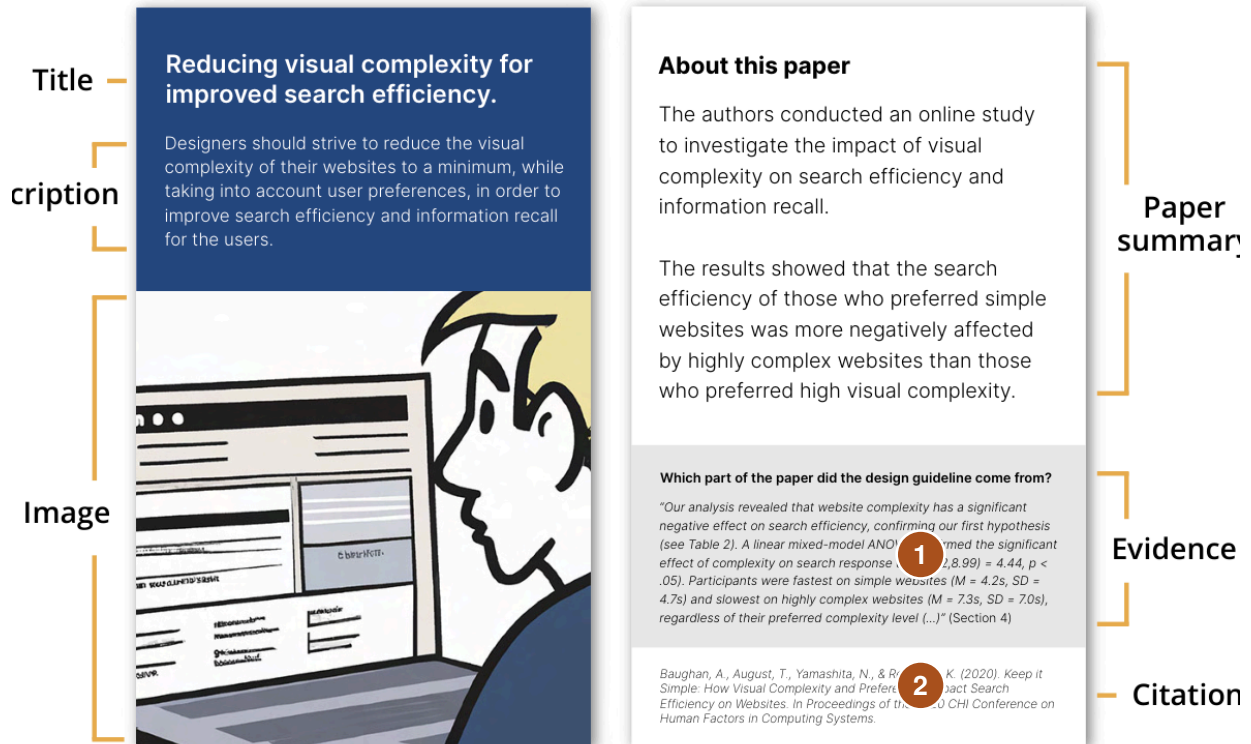
Recorded form · FC010: Drill-down from summary to underlying instances (VA012).

**Figure 18: Sources and evidence. How can a result be traced or checked?** Tracing origin and derivation, opening underlying cases, and assessing evidence answer different checking questions. Category definitions; Browse category mechanisms. Recorded forms: Drill-down from summary to underlying instances.

## FC-C02 / SOURCE ILLUSTRATION

## Sources and evidence

## PAPER ILLUSTRATION · FC009



1. Supporting passage 2. Citation back to the paper

The design card places a supporting source passage and its citation beside the implication.

From Paper to Card: Transforming Design Implications with Generative AI · Figure 1, PDF page 2

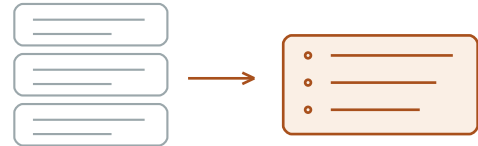
Shin et al.; CC BY; cropped; numbered annotations added by this review.

**Figure 19: Sources and evidence: source illustration. Figure 1, PDF page 2 in [133]. The design card places a supporting source passage and its citation beside the implication. Inspect the illustrated mechanism.**



**FC-C03 / Explanation and visibility****Summarising and labelling content****How is content condensed, interpreted or recast?**

Condenses or interprets content through summaries, labels, profiles or a change of representation, so people can grasp what it means.

**DESIGN CHOICES****RETAINED DISTINCTIONS****Condense content****FC107**

summarisation and condensation

**Characterise an item****FC105**

machine-generated semantic labelling

**FC109**

entity or concept profile

**Change its form****FC108**

semantic transformation into another form

---

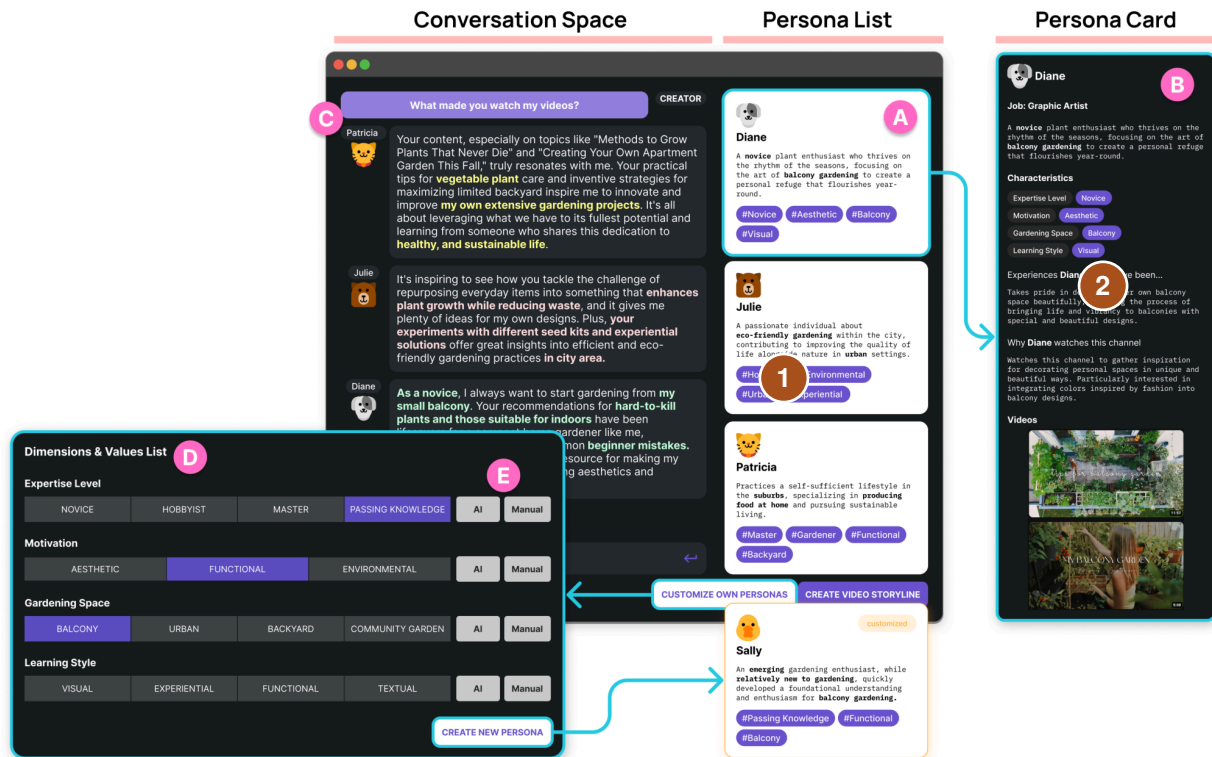
Recorded form · FC107: Condensed summary of a single item (VA097).

**Figure 20: Summarising and labelling content. How is content condensed, interpreted or recast? Distinguish reducing a body of content, characterising an item, and changing representational kind. Category definitions; Browse category mechanisms. Recorded forms: Condensed summary of a single item.**

## FC-C03 / SOURCE ILLUSTRATION

## Summarising and labelling content

## PAPER ILLUSTRATION · FC109



1. Compact audience profiles 2. Expanded persona details

Audience segments become profile cards, with detailed experiences and motivations available on selection.

Proxona: Supporting Creators' Sensemaking and Ideation with LLM-Powered Audience Personas · Figure 2, PDF page 6

Choi et al.; CC BY; cropped; numbered annotations added by this review.

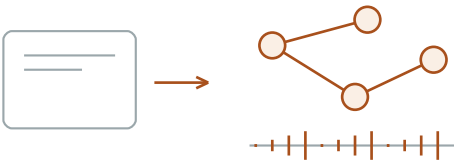
**Figure 21: Summarising and labelling content: source illustration. Figure 2, PDF page 6 in [31]. Audience segments become profile cards, with detailed experiences and motivations available on selection. Inspect the illustrated mechanism.**

FC-C04 / Explanation and visibility

Visual and sensory representation

Through which form can a person perceive meaning?

Expresses meaning through visual form, language, sound, touch or physical action, including translations and adaptations that make it accessible.



DESIGN CHOICES

RETAINED DISTINCTIONS

Perceptual carrier

|   |                                                       |                                             |                                                |
|---|-------------------------------------------------------|---------------------------------------------|------------------------------------------------|
| — | <b>FC085</b><br>visual encoding and graphical mapping | <b>FC083</b><br>non-visual sensory encoding | <b>FC086</b><br>embodied or physical enactment |
|---|-------------------------------------------------------|---------------------------------------------|------------------------------------------------|

Translation of meaning

|   |                                         |                                                          |
|---|-----------------------------------------|----------------------------------------------------------|
| — | <b>FC084</b><br>cross-modal translation | <b>FC089</b><br>verbal description of perceptual content |
|---|-----------------------------------------|----------------------------------------------------------|

Adaptation and reframing

|   |                                                       |                                                            |
|---|-------------------------------------------------------|------------------------------------------------------------|
| — | <b>FC088</b><br>accessible or simplified presentation | <b>FC087</b><br>metaphorical or symbolic re-representation |
|---|-------------------------------------------------------|------------------------------------------------------------|

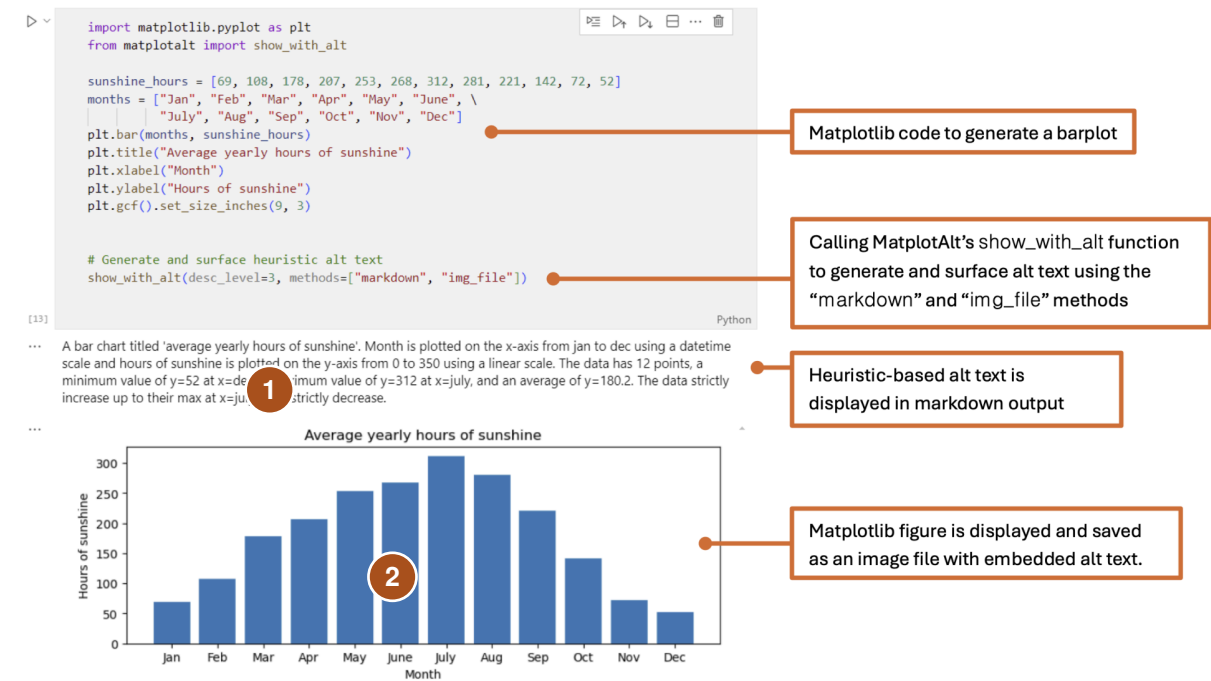
Recorded form · FC083: Spoken or audio rendering of content (VA071).

Figure 22: Visual and sensory representation. Through which form can a person perceive meaning? Separate the perceptual carrier, translation between carriers, and adaptations that change how the meaning can be grasped. Category definitions; Browse category mechanisms. Recorded forms: Spoken or audio rendering of content.

FC-C04 / SOURCE ILLUSTRATION

Visual and sensory representation

PAPER ILLUSTRATION · FC088



1. Description surfaced as text 2. The same chart remains visible

A notebook presents a chart and its generated description together; the library also supports alternative output placements.

Matplotlib: A Python Library for Adding Alt Text to Matplotlib Figures in Computational Notebooks · Figure 1, PDF page 1

Nylund et al.; CC BY; cropped; numbered annotations added by this review.

Figure 23: Visual and sensory representation: source illustration. Figure 1, PDF page 1 in [109]. A notebook presents a chart and its generated description together; the library also supports alternative output placements. Inspect the illustrated mechanism.

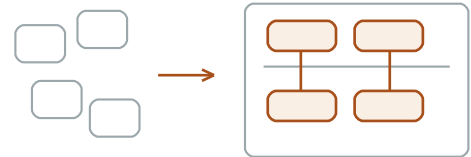
### C.1.2 Organisation and comparison. How can a person organise and compare material?

#### FC-C05 / Organisation and comparison

## Organising and structuring

### Which relationships give material its structure?

Arranges information into relationships, groups, levels or sequences, so people can understand its structure and move between the whole and its parts.



#### DESIGN CHOICES

#### RETAINED DISTINCTIONS

#### Links and containment

##### FC015

relational or network map

##### FC013

hierarchical or nested organisation

#### Groups and categories

##### FC016

similarity grouping  
and clustering

##### FC017

categorisation  
scheme or controlled  
vocabulary

##### FC106

human semantic  
coding and  
categorisation

#### Whole, parts and detail

##### FC014

overview or  
aggregate gist

##### FC018

decomposition into  
parts

##### FC019

multiscale detail and  
semantic zoom

#### Position and alignment

##### FC020

temporal  
arrangement of  
material

##### FC021

spatial arrangement  
of material

##### FC022

cross-representation  
correspondence

#### Quantitative structure

##### FC023

distributional or statistical structure display

Recorded form · FC015: Relational graph or dependency network (VA019).

**Figure 24: Organising and structuring. Which relationships give material its structure? Relations, category membership, granularity, position and statistical structure organise different aspects of the same material and may be combined. Category definitions; Browse category mechanisms. Recorded forms: Relational graph or dependency network.**

## FC-C05 / SOURCE ILLUSTRATION

## Organising and structuring

## PAPER ILLUSTRATION · FC018



6. Research question node 7. Linked solution node

An idea is decomposed into linked, editable research-question, solution, evaluation and contribution nodes.

Iterative Research Idea Development Through Evolving and Composing Idea Facets with Literature-Grounded Feedback · Figure 4, PDF page 8

Pu et al.; CC BY; cropped; numbered annotations added by this review.

**Figure 25: Organising and structuring: source illustration. Figure 4, PDF page 8 in [117]. An idea is decomposed into linked, editable research-question, solution, evaluation and contribution nodes. Inspect the illustrated mechanism.**

FC-C06 / Organisation and comparison

Comparing alternatives

What is placed in a common comparison frame?

Places alternatives, cases or states in a common frame so people can examine their similarities, differences and trade-offs.

|       | A           | B           |
|-------|-------------|-------------|
| _____ | <div></div> | _____       |
| _____ | _____       | <div></div> |
| _____ | <div></div> | _____       |

DESIGN CHOICES

RETAINED DISTINCTIONS

Separate candidates

FC024

side-by-side alternative comparison

Changed states or versions

FC025

contrastive or counterfactual comparison

FC028

change-over-time comparison

Standards and populations

FC026

benchmark or peer-referenced comparison

FC027

subgroup or population comparison

Recorded form · FC024: Side-by-side comparison of alternatives (VA023).

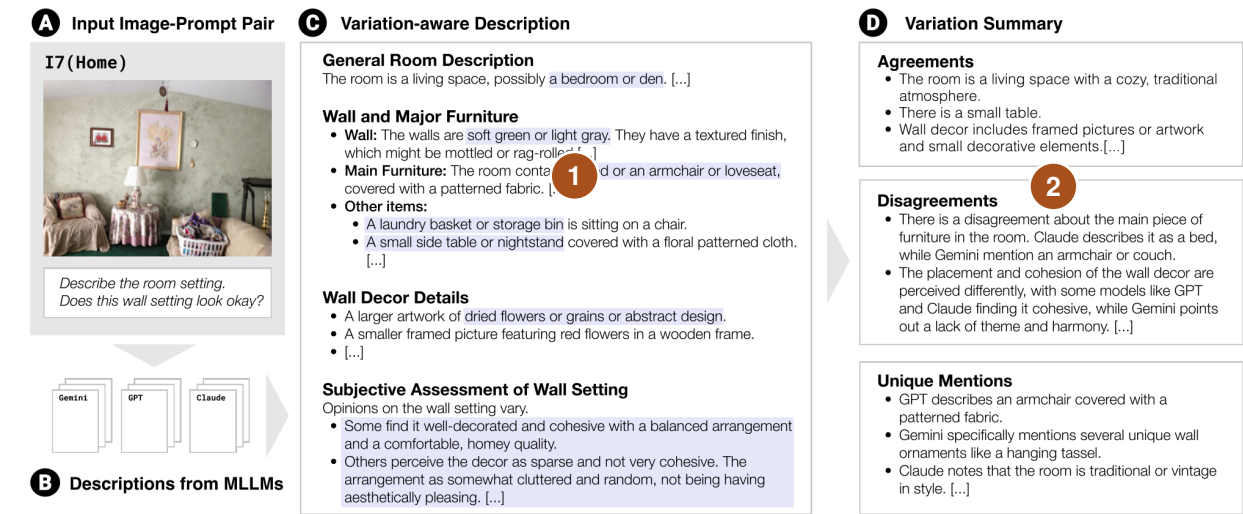
Recorded form · FC025: Before-and-after or version comparison (VA025).

Figure 26: Comparing alternatives. What is placed in a common comparison frame? The comparand distinguishes candidate selection, variation of the same object, and comparison against groups or reference points. Category definitions; Browse category mechanisms. Recorded forms: Side-by-side comparison of alternatives; Before-and-after or version comparison.

FC-C06 / SOURCE ILLUSTRATION

Comparing alternatives

PAPER ILLUSTRATION · FC024



1. Detailed merged description 2. Agreements and disagreements

The interface aligns multiple image descriptions in a detailed narrative and a separate discrepancy summary.

Surfacing Variations to Calibrate Perceived Reliability of MLLM-generated Image Descriptions · Figure 2, PDF page 5

Chen et al.; CC BY; cropped; numbered annotations added by this review.

Figure 27: Comparing alternatives: source illustration. Figure 2, PDF page 5 in [26]. The interface aligns multiple image descriptions in a detailed narrative and a separate discrepancy summary. Inspect the illustrated mechanism.



**FC-C07 / Organisation and comparison****Search, filtering and ranking****How is relevant material found and brought into view?**

Helps people find relevant material and control what is in view through search, navigation, filters, rankings and scope choices.

The diagram illustrates a search interface. At the top is a search bar with a magnifying glass icon. Below it is a filter bar with a plus icon. Below the filter bar are two result lists, each with a header and a list of items.

**DESIGN CHOICES****RETAINED DISTINCTIONS****Define the request****FC030**

query construction and search specification

**FC033**

analytic scope setting

**Select and order****FC029**

relevance filtering of a corpus

**FC032**

ranked or prioritised surfacing

**Navigate and look up****FC034**

navigation and wayfinding through structure

**FC031**

on-demand lookup of related information

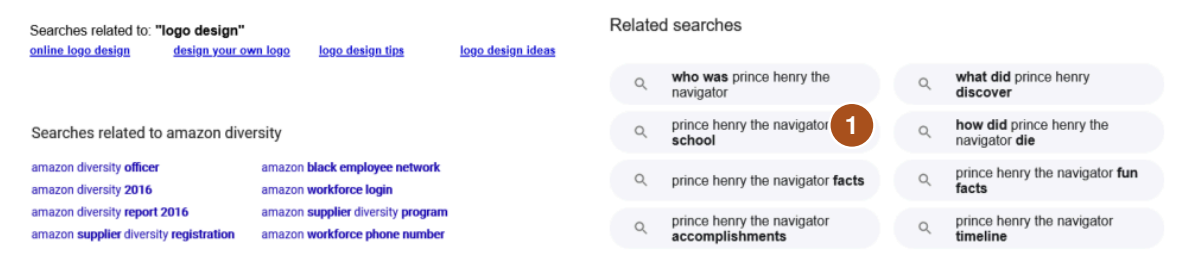
Recorded form · FC030: Filter, query or search over a set (VA028).

**Figure 28: Search, filtering and ranking. How is relevant material found and brought into view? Defining the request, selecting or ordering results, and moving through material are distinct retrieval operations. Category definitions; Browse category mechanisms. Recorded forms: Filter, query or search over a set.**

FC-C07 / SOURCE ILLUSTRATION

# Search, filtering and ranking

PAPER ILLUSTRATION · FC030



1. Follow-on query formulations

Related-search suggestions provide alternative formulations for continuing a search.

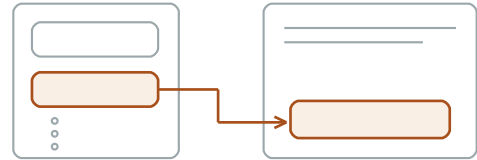
From 10 Blue Links Pages to Feature-Full Search Engine Results Pages - Analysis of the Temporal Evolution of SERP Features · Figure 16, PDF page 5

Oliveira et al.; CC BY; cropped; numbered annotations added by this review.

**Figure 29: Search, filtering and ranking: source illustration. Figure 16, PDF page 5 in [110]. Related-search suggestions provide alternative formulations for continuing a search. Inspect the illustrated mechanism.**

**FC-C08 / Organisation and comparison****Notes, history and memory****What is kept available for continuing the work?**

Keeps information and work history available through notes, annotations, memory aids and replay, so people can revisit and continue their work.

**DESIGN CHOICES****RETAINED DISTINCTIONS****Express and capture****FC077**

externalising internal structure into an artifact

**FC080**

capture of activity as it happens

**Mark and arrange****FC078**

annotation onto material

**FC082**

reorganisable working workspace

**Recall and revisit****FC079**

external memory and recall cue

**FC081**

replay and history navigation

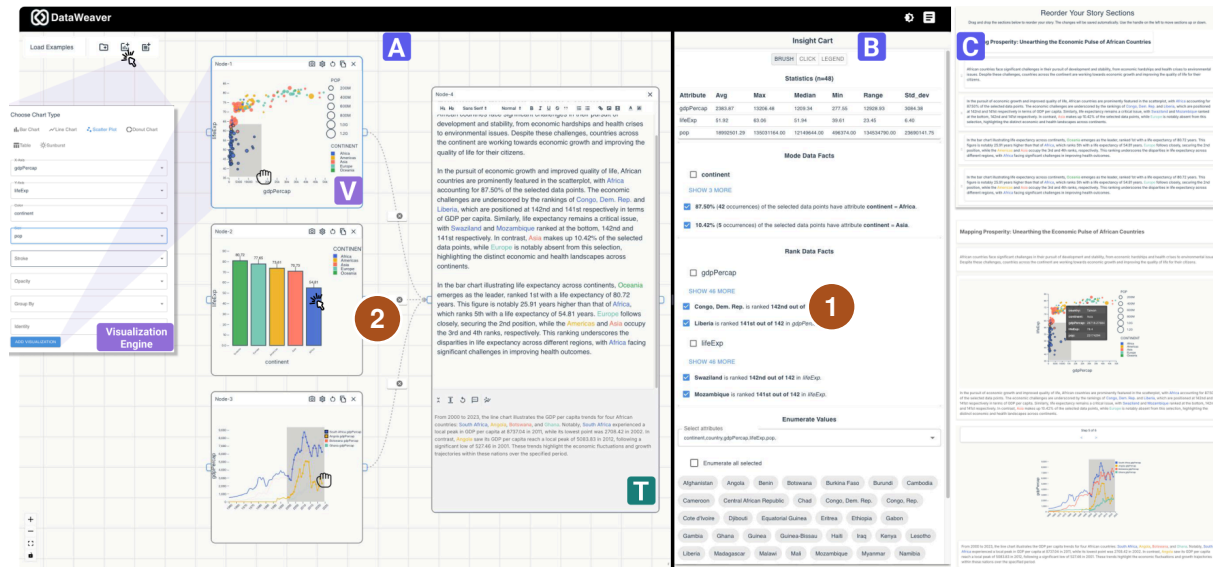
Recorded form · FC078: Annotation and comment capture on content (VA048).

**Figure 30: Notes, history and memory. What is kept available for continuing the work? Separate making or capturing a record, arranging material at hand, and returning to an earlier state. Category definitions; Browse category mechanisms. Recorded forms: Annotation and comment capture on content.**

## FC-C08 / SOURCE ILLUSTRATION

## Notes, history and memory

## PAPER ILLUSTRATION · FC082



1. Facts held in the Insight Cart 2. Connected story components

An Insight Cart holds candidate facts for later selection into a connected data story.

DATAWEAVER: Authoring Data-Driven Narratives through the Integrated Composition of Visualization and Text · Figure 2, PDF page 5

Fu et al.; CC BY; cropped; numbered annotations added by this review.

**Figure 31: Notes, history and memory: source illustration. Figure 2, PDF page 5 in [43]. An Insight Cart holds candidate facts for later selection into a connected data story. Inspect the illustrated mechanism.**

### C.1.3 User control and expression. What can a person express, change or try?

#### FC-C09 / User control and expression

## Goals and constraints

### How does a person specify what the system should do?

Lets people state what they want through goals, rules, criteria, examples or structured inputs that the system can act on.

#### DESIGN CHOICES

#### RETAINED DISTINCTIONS

#### Words and demonstrations

##### FC053

natural-language or conversational specification

##### FC054

example or demonstration specification

#### Structured and executable input

##### FC057

structured input form or schema

##### FC058

authoring a formal or executable representation

#### Conditions and criteria

##### FC055

explicit constraint or rule specification

##### FC056

criteria and rubric specification

Recorded form · FC055: Rule, criterion or rubric authoring (VA042).

Figure 32: Goals and constraints. How does a person specify what the system should do? Distinguish conveying intent, providing a structured or executable specification, and declaring constraints or evaluation criteria. Category definitions; Browse category mechanisms. Recorded forms: Rule, criterion or rubric authoring.

## FC-C09 / SOURCE ILLUSTRATION

## Goals and constraints

## PAPER ILLUSTRATION · FC054

☐ Show only Comments labeled as "Remove" 30 / 30 labeled examples; 14 example(s) labeled as "Remove".

Those people who pull out guns just because they get upset are called gang banger thug losers. 99% of their guns are illegally possessed. Legal gun owners aren't taking out their guns during an argument at parties.

Remove **1** Keep

So I should get rid of my gun because some idiot killed.

Remove Keep

Idiot sandwich 🥪

Remove Keep

Load More Examples Create **2** your Filter

1. Keep or remove this example 2. Train the filter from labels

Situated keep/remove judgments supply examples of the content a person wants the classifier to retain.

End User Authoring of Personalized Content Classifiers: Comparing Example Labeling, Rule Writing, and LLM Prompting · Figure 2, PDF page 6

Wang et al.; CC BY; cropped; numbered annotations added by this review.

**Figure 33: Goals and constraints: source illustration. Figure 2, PDF page 6 in [145]. Situated keep/remove judgments supply examples of the content a person wants the classifier to retain. Inspect the illustrated mechanism.**

**FC-C10 / User control and expression****Editing and steering****Where can a person intervene in the work?**

Lets people change a representation, revise an output or redirect a process through edits, settings, parameters and reversible actions.

**DESIGN CHOICES****RETAINED DISTINCTIONS****Object and output****FC047**

direct manipulation of a semantic object

**FC048**

editable machine output

**Settings and quantities****FC050**

configuration of system behaviour

**FC049**

continuous parameter or dial control

**Direction and commitment****FC051**

steering an ongoing process

**FC052**

reversibility and safe experimentation

Recorded form · FC049: Parameter, threshold or weight controls (VA039).

**Figure 34: Editing and steering. Where can a person intervene in the work?** Editing an object, configuring behaviour, steering the current process and controlling commitment operate at different points. Category definitions; Browse category mechanisms. Recorded forms: Parameter, threshold or weight controls.

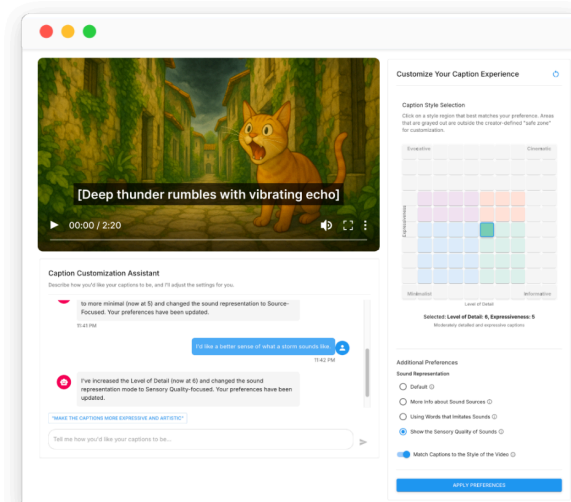
## FC-C10 / SOURCE ILLUSTRATION

## Editing and steering

## PAPER ILLUSTRATION · FC050

1

Overall interface



2

Style Grid for *Level of Detail* (x-axis) and *Expressiveness* (y-axis). Creator-defined anchors—shown as red dots—specify the bounds of the caption transformation space.

## Caption Preference Assistant

Describe how you'd like your captions to be, and I'll adjust the settings for you.

to more minimal (now at 5) and changed the sound representation to Source-Focused. Your preferences have been updated.

11:41 PM

I'd like a better sense of what a storm sounds like.

11:42 PM

I've increased the Level of Detail (now at 6) and changed the sound representation mode to Sensory Quality-focused. Your preferences have been updated.

"MAKE THE CAPTIONS MORE EXPRESSIVE AND ARTISTIC"

Tell me how you'd like your captions to be...

## Sound Representation

- ☐ Default
- ☐ More Info about Sound Sources
- ☐ Using Words that Imitates Sounds
- ☒ Show the Sensory Quality of Sounds
- ☐ Match Captions to the Style of the Video

3

Controls for *Sound Representation* and *Genre/Style Alignment*.

4

Interface where users set preferences by chatting with an AI assistant using natural language prompts.

5. Choose how the sound is represented

Viewers switch between sound sources, imitations and sensory descriptions for the same captioned material.

CapTune: Adapting Non-Speech Captions With Anchored Generative Models · Figure 4, PDF page 8

Huang et al.; CC BY; cropped; numbered annotations added by this review.

**Figure 35: Editing and steering: source illustration.** Figure 4, PDF page 8 in [60]. Viewers switch between sound sources, imitations and sensory descriptions for the same captioned material. Inspect the illustrated mechanism.



FC-C11 / User control and expression

Generating and choosing alternatives

How are possibilities offered and shaped?

Provides candidate outputs, continuations or a space of possibilities that people can explore, constrain and select from.



DESIGN CHOICES

RETAINED DISTINCTIONS

Present a choice space

FC035

candidate or option set to choose from

FC038

option-space landscape or design space

Generate or continue

FC036

generated alternatives to explore

FC110

completion and continuation of content

Bound generation

FC037

user-constrained generation

Recorded form · FC036: Gallery of generated alternatives to choose among (VA033).

Recorded form · FC110: Completion or continuation of what the person began (VA036).

Figure 36: Generating and choosing alternatives. How are possibilities offered and shaped? Separate presenting a choice space, producing new content, and bounding the generation process. A continuation is not necessarily a set of alternatives. Category definitions; Browse category mechanisms. Recorded forms: Gallery of generated alternatives to choose among; Completion or continuation of what the person began.

## FC-C11 / SOURCE ILLUSTRATION

## Generating and choosing alternatives

## PAPER ILLUSTRATION · FC110

Double-click the snippet to edit it directly. **THIS IS SUGGESTED TEXT**

Adjust role

Explain interaction × ⓘ

Adjust formatting

heading drophead body **note**

Get recommendations

Highlight interactive elements and guide the audience's interaction flow using moderately complex sentences. Use imperative sentences to instruct users on actions they can perform. Depending on the complexity, aim for 10–25 words, a readability level appropriate for someone with a 6th to 10th grade education, and a word variety of 50–70%.

Word count **5**

Word variety **40%**  ⓘ

Readability score **69**  ⓘ

Your text is likely to be understood by someone who has at least a 8th-grade education.

Generate text

[↻ Re-generate](#) [✂ Shorten](#) [📄 Simplify](#)

Add context to incorporate into generation

What should we consider when generating this text?

[🗑 Delete snippet](#)

## 1. Regenerate, shorten or simplify

A text snippet exposes explicit actions for generating a revised version while the author continues composing.

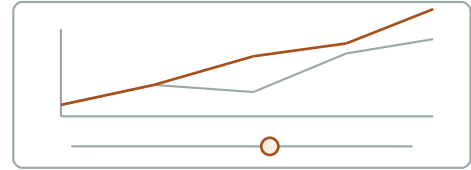
Plume: Scaffolding Text Composition in Dashboards · Figure 5, PDF page 11

Lisnic et al.; CC BY; cropped; numbered annotations added by this review.

**Figure 37: Generating and choosing alternatives: source illustration.** Figure 5, PDF page 11 in [84]. A text snippet exposes explicit actions for generating a revised version while the author continues composing. Inspect the illustrated mechanism.

**FC-C12 / User control and expression****What-if exploration and rehearsal****What can be tried before acting in the real setting?**

Lets people try hypothetical changes, probe a system's behaviour or practise in a simulated setting and inspect what happens.

**DESIGN CHOICES****RETAINED DISTINCTIONS****Probe the system****FC098**

system-behaviour probing

**Construct a scenario****FC097**

scenario construction and what-if exploration

**Rehearse an activity****FC096**

simulated environment for rehearsal

---

Recorded form · FC096: Practice, rehearsal or simulation environment (VA058).

**Figure 38: What-if exploration and rehearsal. What can be tried before acting in the real setting? Separate learning how a system responds, constructing a hypothetical world, and rehearsing a person's activity. Category definitions; Browse category mechanisms. Recorded forms: Practice, rehearsal or simulation environment.**

## FC-C12 / SOURCE ILLUSTRATION

## What-if exploration and rehearsal

## PAPER ILLUSTRATION · FC096



1. Dialogue history 2. Teacher question entry

Preservice teachers rehearse classroom discussion with represented students and a persistent dialogue history.

Pattern analysis of ambitious science talk between preservice teachers and AI-powered student agents · Figure 1, PDF page 3

Barrett et al.; CC BY; cropped; numbered annotations added by this review.

**Figure 39: What-if exploration and rehearsal: source illustration. Figure 1, PDF page 3 in [9]. Preservice teachers rehearse classroom discussion with represented students and a persistent dialogue history. Inspect the illustrated mechanism.**

C.1.4 *Guidance and feedback*. How is the next action or reflection supported?

**FC-C13 / Guidance and feedback**

**Recommendations and guidance**

**What helps a person decide where to look or what to do?**

Directs attention or suggests an action, resource or procedure, including corrective advice, consequence previews and warnings.



**DESIGN CHOICES**

**RETAINED DISTINCTIONS**

**Notice and hold back**

|   |                                                         |                                       |
|---|---------------------------------------------------------|---------------------------------------|
| — | <b>FC043</b><br>attention direction and salience cueing | <b>FC044</b><br>risk alert or warning |
|---|---------------------------------------------------------|---------------------------------------|

**Preview a consequence**

|   |                                                 |  |
|---|-------------------------------------------------|--|
| — | <b>FC046</b><br>prospective consequence preview |  |
|---|-------------------------------------------------|--|

**Choose an action or resource**

|   |                                          |                                                |
|---|------------------------------------------|------------------------------------------------|
| — | <b>FC040</b><br>next-step recommendation | <b>FC042</b><br>item or content recommendation |
|---|------------------------------------------|------------------------------------------------|

**Perform or repair**

|   |                                               |                                                    |
|---|-----------------------------------------------|----------------------------------------------------|
| — | <b>FC045</b><br>procedural how-to instruction | <b>FC041</b><br>corrective or remediation guidance |
|---|-----------------------------------------------|----------------------------------------------------|

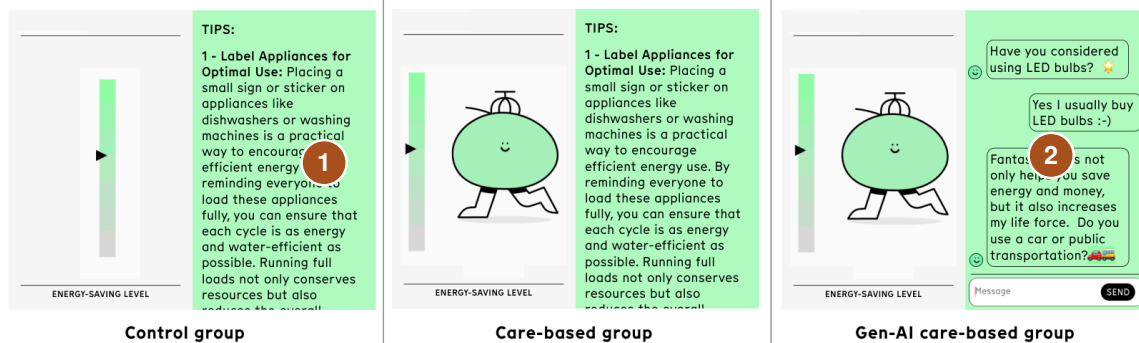
Recorded form · FC046: Preview of the consequence before committing (VA037).

**Figure 40: Recommendations and guidance. What helps a person decide where to look or what to do? Attention and warnings, previews, recommendations, and procedural or corrective guidance contribute different information to an action. Category definitions; Browse category mechanisms. Recorded forms: Preview of the consequence before committing.**

## FC-C13 / SOURCE ILLUSTRATION

## Recommendations and guidance

## PAPER ILLUSTRATION · FC040



1. Fixed tips 2. Conversational condition

The study contrasts fixed energy-saving tips with a conversational condition that elicits household context.

Care-Based Eco-Feedback Augmented with Generative AI: Fostering Pro-Environmental Behavior through Emotional Attachment · Figure 6, PDF page 6

Berney et al.; CC BY; cropped; numbered annotations added by this review.

**Figure 41: Recommendations and guidance: source illustration.** Figure 6, PDF page 6 in [12]. The study contrasts fixed energy-saving tips with a conversational condition that elicits household context. Inspect the illustrated mechanism.

FC-C14 / Guidance and feedback

Feedback and checking

What basis is provided for judging the work?

Shows how work is progressing or how a result meets a goal, criterion or reference. It includes feedback on attempts, problem diagnosis, quality assessments and checks.



DESIGN CHOICES

RETAINED DISTINCTIONS

Attempt and progress

FC065

formative outcome feedback

FC066

progress and status tracking

Problem, quality and status

FC063

Identifying problems in the work

FC064

evaluative judgment or quality rating

FC067

correctness or status marking

Check against a reference

FC068

verification and checking against a reference

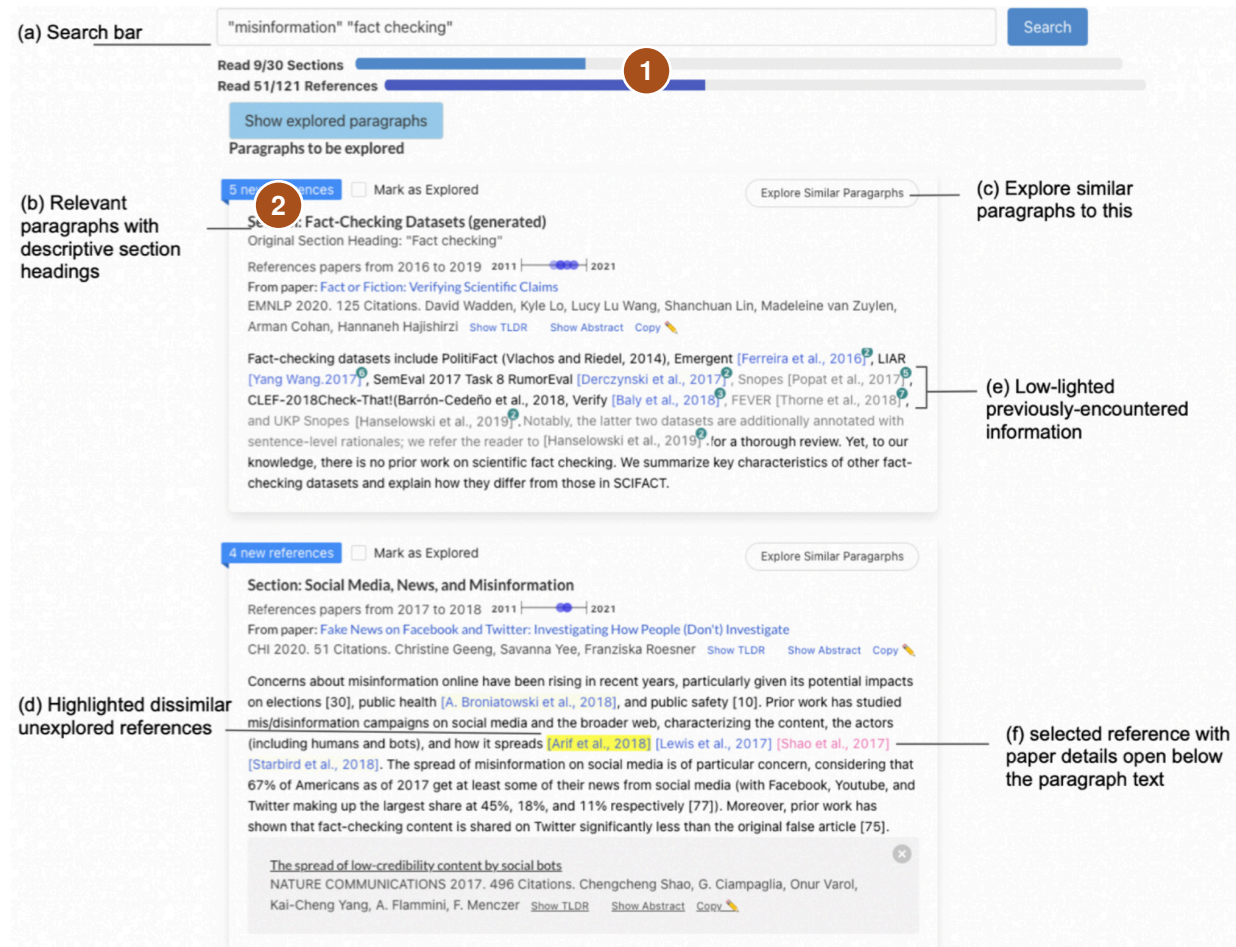
Recorded form · FC063: Error and failure diagnosis (VA007).

Figure 42: Feedback and checking. What basis is provided for judging the work? Separate feedback about an attempt or progress, appraisals and status marks, and an active check against a reference. Category definitions; Browse category mechanisms. Recorded forms: Error and failure diagnosis.

## FC-C14 / SOURCE ILLUSTRATION

## Feedback and checking

## PAPER ILLUSTRATION · FC066



1. Paragraph and reference coverage 2. Unread-reference count

Persistent coverage bars and local unread-reference counts make progress through the literature visible.

Relatedly: Scaffolding Literature Reviews with Existing Related Work Sections · Figure 1, PDF page 2

Palani et al.; CC BY; cropped; numbered annotations added by this review.

**Figure 43: Feedback and checking: source illustration. Figure 1, PDF page 2 in [112]. Persistent coverage bars and local unread-reference counts make progress through the literature visible. Inspect the illustrated mechanism.**



**FC-C15 / Guidance and feedback****Instruction and paced help****What help is available, and when is it available?**

Provides explanations, examples and orientation, and controls when or how much help is available through staged disclosure, learning paths or changing support.

**DESIGN CHOICES****RETAINED DISTINCTIONS****Explain, exemplify and orient**

**FC101**  
conceptual  
explanation of  
domain meaning

**FC102**  
worked example or  
exemplar

**FC103**  
orientation and  
onboarding

**Change amount or release**

**FC099**  
graduated or faded  
support

**FC100**  
progressive  
disclosure of detail

**FC039**  
deliberate restriction  
or withdrawal of  
support

**Sequence learning sessions**

**FC104**  
curriculum and learning-path structuring

Recorded form · FC099: Graduated help that escalates or fades by response (VA093).

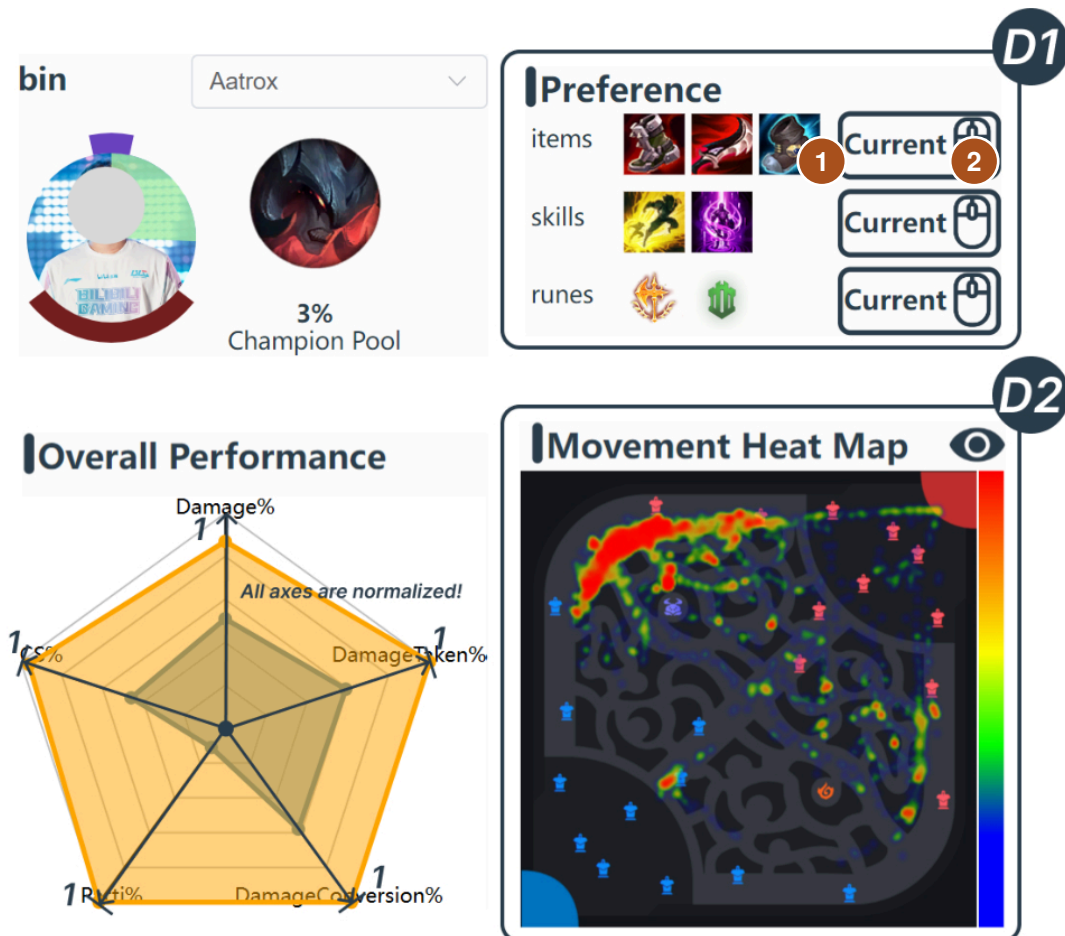
Recorded form · FC102: Worked or contrastive examples as instruction (VA057).

**Figure 44: Instruction and paced help. What help is available, and when is it available? Instructional content, the amount or timing of support, and progression across sessions are distinct design choices. Fading is not automatically evidence of learning. Category definitions; Browse category mechanisms. Recorded forms: Graduated help that escalates or fades by response; Worked or contrastive examples as instruction.**

## FC-C15 / SOURCE ILLUSTRATION

## Instruction and paced help

## PAPER ILLUSTRATION · FC102



1. Historical equipment choices 2. Switch to current choices

Historical equipment, skill and rune choices can be inspected beside controls revealing the current match.

StratIncon Detector: Analyzing Strategy Inconsistencies Between Real-Time Strategy and Preferred Professional Strategy in MOBA Esports · Figure 8, PDF page 12

Ma et al.; CC BY; cropped; numbered annotations added by this review.

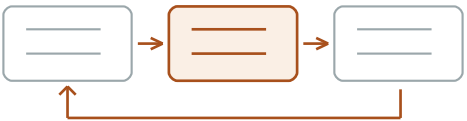
**Figure 45: Instruction and paced help: source illustration.** Figure 8, PDF page 12 in [89]. Historical equipment, skill and rune choices can be inspected beside controls revealing the current match. Inspect the illustrated mechanism.

FC-C16 / Guidance and feedback

Structuring the workflow

How are stages, responsibilities and continuity arranged?

Organises work into stages, loops, roles and decision points, with ways to coordinate actions, hand work over, recover from problems and resume later.



DESIGN CHOICES

RETAINED DISTINCTIONS

Sequence and recurrence

|   |                                           |                                         |
|---|-------------------------------------------|-----------------------------------------|
| — | <b>FC069</b><br>staged or phased workflow | <b>FC070</b><br>iterative revision loop |
|---|-------------------------------------------|-----------------------------------------|

Responsibility and routing

|   |                                                   |                                                  |                                                      |
|---|---------------------------------------------------|--------------------------------------------------|------------------------------------------------------|
| — | <b>FC071</b><br>human decision checkpoint or gate | <b>FC072</b><br>division of labour across agents | <b>FC074</b><br>orchestration and routing of actions |
|---|---------------------------------------------------|--------------------------------------------------|------------------------------------------------------|

Continuity and recovery

|   |                                       |                                                 |                                                     |
|---|---------------------------------------|-------------------------------------------------|-----------------------------------------------------|
| — | <b>FC073</b><br>handoff of work state | <b>FC075</b><br>interaction repair and recovery | <b>FC076</b><br>resumable and persistent work state |
|---|---------------------------------------|-------------------------------------------------|-----------------------------------------------------|

Recorded form · FC075: System move that clarifies, confirms or repairs (VA038).

Figure 46: Structuring the workflow. How are stages, responsibilities and continuity arranged? Separate progression through work, allocation of actions and authority, and continuity across interruptions or handoffs. Category definitions; Browse category mechanisms. Recorded forms: System move that clarifies, confirms or repairs.

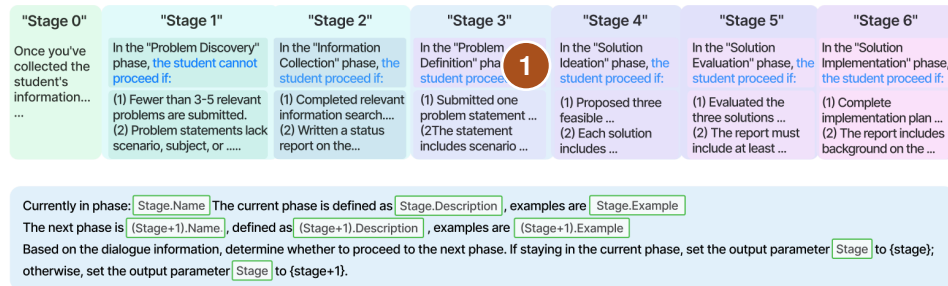
## FC-C16 / SOURCE ILLUSTRATION

## Structuring the workflow

## PAPER ILLUSTRATION · FC074

## Controller Agent Prompts

## [Stage Decision Prompt]



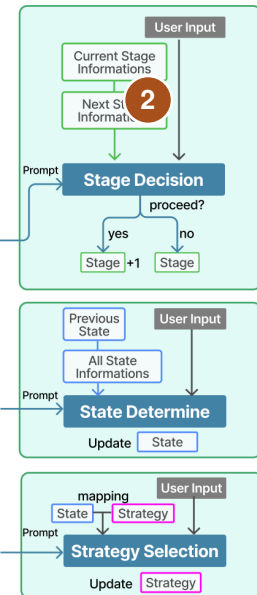
## [State Determine Prompt]

If not moving to the next phase, the student's status may be: `State.Name` `State.Description`  
 The student's previous status list is `State`. Remove all '0' and '1'; and based on the new dialogue record, update and output the student's status list state.  
 If moving to the next phase, the student's status will be "start," and update and output the student's status list as `state=[0]`.

## [Strategy Selection Prompt]

The current student statuses and corresponding strategies are as follows: `State.Name` `State.StrategyMapping` `Strategy.name` `Strategy.Description`  
 Based on the dialogue information, output the most urgent problem ID as `urge_state` and the corresponding best strategy ID as `strategy`

## Controller Agent Pipeline



1. Stage-specific completion criteria 2. Decision to advance or remain

Stage criteria and a controller decision determine whether the creative problem-solving workflow advances.

Mentigo: An Intelligent Agent for Mentoring Students in the Creative Problem Solving Process · Figure 2, PDF page 9

Zha et al.; CC BY; cropped; numbered annotations added by this review.

Figure 47: Structuring the workflow: source illustration. Figure 2, PDF page 9 in [158]. Stage criteria and a controller decision determine whether the creative problem-solving workflow advances. Inspect the illustrated mechanism.

**FC-C17 / Guidance and feedback****Questions and reflection prompts****Which part of a person's thinking is elicited?**

Asks people to express their goals, questions, explanations or reflections, making their own reasoning available to the work.

A diagram of a prompt box. It consists of a rounded rectangle with a light gray border. Inside, there is a smaller rounded rectangle with a light orange border containing a question mark and two horizontal lines. Below this is another rounded rectangle with a light gray border containing a single horizontal line.

**DESIGN CHOICES****RETAINED DISTINCTIONS****Needs and intentions****FC059**

intent and need elicitation

**Account and reasoning****FC062**

narrative and reasoning elicitation

**Questions and hypotheses****FC061**

inquiry and hypothesis prompting

**Reflection on thinking****FC060**

reflective and metacognitive prompting

---

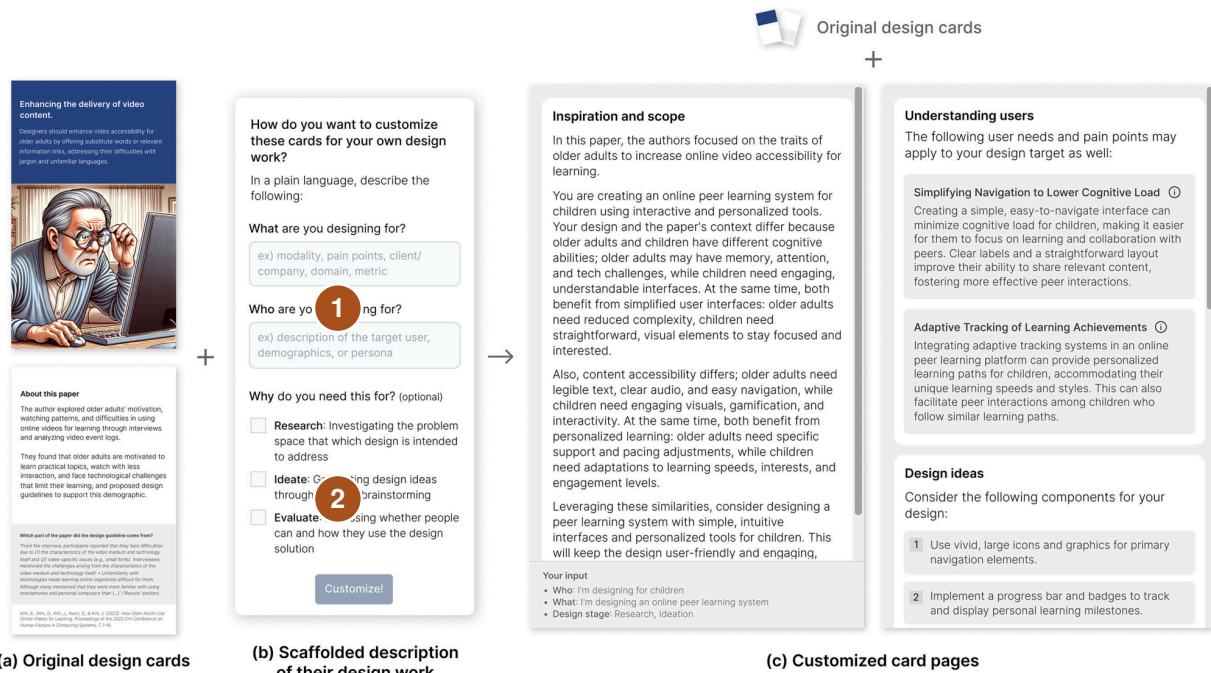
Recorded form · FC060: Reflection or self-assessment prompt (VA056).

**Figure 48: Questions and reflection prompts. Which part of a person's thinking is elicited? The object of the prompt distinguishes needs, an account of reasoning, inquiry about the material, and reflection on one's own thinking. Category definitions; Browse category mechanisms. Recorded forms: Reflection or self-assessment prompt.**

## FC-C17 / SOURCE ILLUSTRATION

## Questions and reflection prompts

## PAPER ILLUSTRATION · FC059



1. Who and what is being designed 2. Select the design stage

A guided form elicits design context before the system customizes research cards.

What About My Design Context?: Exploring the Use of Generative AI to Support Customization of Translational Research Artifacts · Figure 1, PDF page 7

Shin et al.; CC BY; cropped; numbered annotations added by this review.

**Figure 49: Questions and reflection prompts: source illustration. Figure 1, PDF page 7 in [132]. A guided form elicits design context before the system customizes research cards. Inspect the illustrated mechanism.**

### C.1.5 Collaboration and perspective. How can people share work and perspectives?

#### FC-C18 / Collaboration and perspective

## Shared views and perspectives

### How do contributions and viewpoints become shared?

Makes people's contributions and viewpoints available to one another through shared representations, structured discussion, role framing and control over what is disclosed.



#### DESIGN CHOICES

#### RETAINED DISTINCTIONS

##### Shared work and awareness

###### FC090

shared co-constructed representation

###### FC091

visibility of others' state and judgment

##### Viewpoints and exchange

###### FC092

perspective-taking and viewpoint framing

###### FC094

role or persona framing of the interaction

###### FC093

structured deliberation and perspective exchange

##### Control of disclosure

###### FC095

disclosure and privacy control

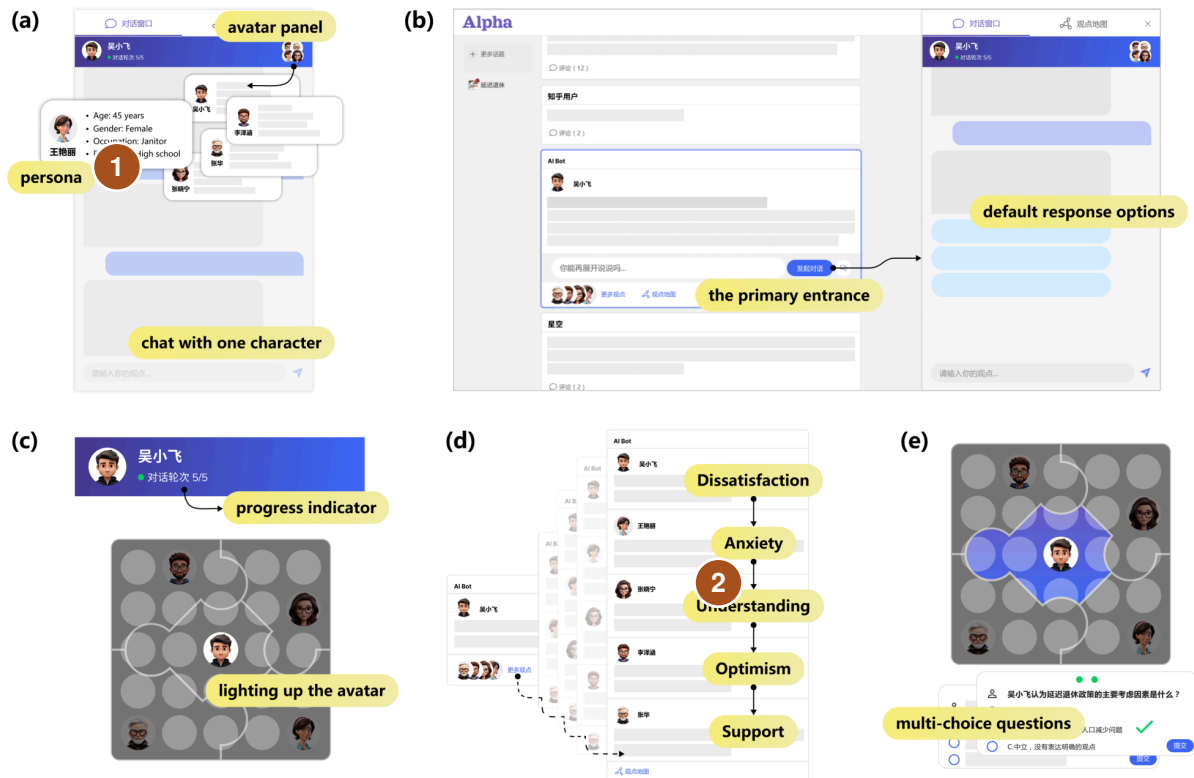
Recorded form · FC090: Shared workspace for joint work (VA065).

**Figure 50: Shared views and perspectives. How do contributions and viewpoints become shared? Separate shared artifacts and awareness, the framing and exchange of perspectives, and control of disclosure. Category definitions; Browse category mechanisms. Recorded forms: Shared workspace for joint work.**

## FC-C18 / SOURCE ILLUSTRATION

## Shared views and perspectives

## PAPER ILLUSTRATION · FC092



1. Distinct stakeholder profiles 2. Progressive viewpoint sequence

Personas expose distinct stakeholder positions; separate controls support dialogue and staged exposure.

See Widely, Think Wisely: Toward Designing a Generative Multi-agent System to Burst Filter Bubbles · Figure 3, PDF page 7

Zhang et al.; CC BY; cropped; numbered annotations added by this review.

**Figure 51: Shared views and perspectives: source illustration. Figure 3, PDF page 7 in [162]. Personas expose distinct stakeholder positions; separate controls support dialogue and staged exposure. Inspect the illustrated mechanism.**

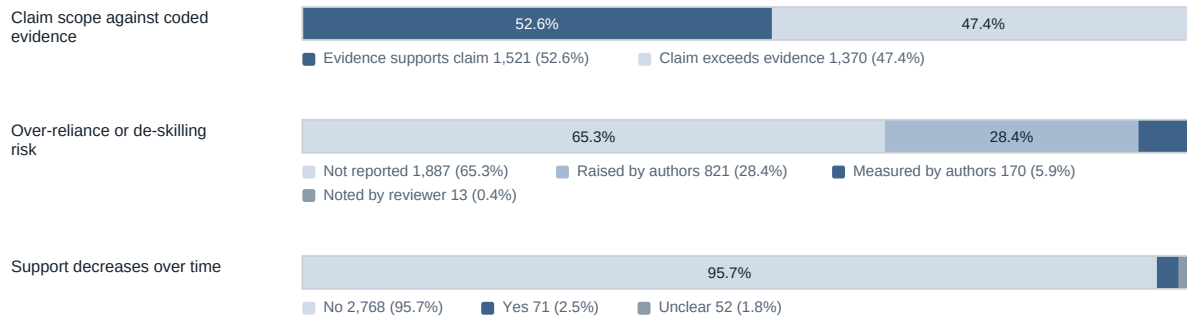
## C.2 Complementary Evidence Profiles

The learning-by-family matrix uses the open-code classification and its 2,761-paper matched population. It is distinct from the closed-field learning count of 301 among 2,891 included papers.



### Claim support and developmental risk indicators

Each row is a mutually exclusive coding over the same 2,891 included cards.



**Figure 52: Claim scope exceeds coded evidence in nearly half of included cards, over-reliance is raised far more often than it is measured, and decreasing support is rare. Each row is a mutually exclusive coding over the same 2,891 included cards, and every segment prints its own count and share. A claim-evidence gap records the distance between a card's stated claim and its linked evidence; it is not an assessment of the underlying paper. Examples, one or two cards per value: claim within evidence: InstructPipe [164].**

### Support profiles within learning-evidence groups

share of the row's papers carrying each family; 2,761 papers coded in both fields

|                                                     | what the interface makes reachable |                       |                            |                             |                               |
|-----------------------------------------------------|------------------------------------|-----------------------|----------------------------|-----------------------------|-------------------------------|
|                                                     | Organisation and comparison        | Guidance and feedback | Explanation and visibility | User control and expression | Collaboration and perspective |
| <b>learning evidence</b>                            |                                    |                       |                            |                             |                               |
| Learning not measured<br>n = 1,334 papers           | 73.8%<br>985                       | 66.9%<br>893          | 64.2%<br>856               | 52.5%<br>701                | 11.4%<br>152                  |
| A proxy measured instead<br>n = 1,039 papers        | 71.3%<br>741                       | 74.1%<br>770          | 61.5%<br>639               | 50.0%<br>520                | 14.6%<br>152                  |
| Learning measured<br>n = 258 papers                 | 60.9%<br>157                       | 82.9%<br>214          | 57.4%<br>148               | 41.5%<br>107                | 14.3%<br>37                   |
| Learning self-reported only<br>n = 130 papers       | 71.5%<br>93                        | 84.6%<br>110          | 63.8%<br>83                | 54.6%<br>71                 | 20.0%<br>26                   |
| <b>all papers in the matrix</b><br>n = 2,761 papers | 71.6%                              | 72.0%                 | 62.5%                      | 50.7%                       | 13.3%                         |

colour: the row's share against the column's share over all papers; dark is 8 points or more above, pale 8 or more below

**Figure 53: Guidance and feedback appear in 83% of capability-measured papers. Rows: the learning-evidence families, including the papers that measured no capability, with the number of papers in each. Columns: the facet families. Each cell is the share of the row's papers carrying the column family, with the count; colour is the cell's distance from the column's share over all papers in the matrix, given in the last row. The model-assisted grouping is descriptive.**

## C.3 Context and Evidence Profiles

The 2,891 closed-field papers record 2,071 empirical, 691 artifact, 123 methodological, four dataset, and two theoretical contributions. Contribution labels do not identify research methods or target users; study designs and participants are reported in Section 5.2.

In the design space, the domain mix shows no simple annual shift: technical, analytic, and creative professions account for 54.8% in 2019 and 55.5% in 2025, ranging from 53.4% to 61.6% across these years. Annual totals also reflect reading coverage and the incomplete 2026 publication window, so they do not estimate field-wide growth.

Study design is recorded for 2,071 papers: 691 within-subject designs (33.4%), 328 single-condition usability studies (15.8%), and 119 field deployments (5.7%). The 2,072 entries include one paper with two designs; percentages use papers. Usability studies reveal practical difficulties; deployments can expose work missed by short comparisons.

What is made reachable, against what the person still does

share of the row's papers in each column; 2,311 papers coded in both fields

| what the interface makes reachable              | what the person still does   |                           |                                   |                                |
|-------------------------------------------------|------------------------------|---------------------------|-----------------------------------|--------------------------------|
|                                                 | The task the tool did not do | New work the tool created | The whole task kept by the person | Final authority kept by design |
| Organisation and comparison<br>n = 1,639 papers | 52.2%<br>856                 | 29.1%<br>477              | 11.4%<br>187                      | 7.1%<br>117                    |
| Guidance and feedback<br>n = 1,679 papers       | 49.9%<br>838                 | 30.1%<br>506              | 11.1%<br>187                      | 8.8%<br>147                    |
| Explanation and visibility<br>n = 1,452 papers  | 53.2%<br>772                 | 28.0%<br>407              | 10.1%<br>146                      | 8.7%<br>126                    |
| User control and expression<br>n = 1,170 papers | 42.0%<br>491                 | 35.8%<br>419              | 14.2%<br>166                      | 7.9%<br>93                     |
| Collaboration and perspective<br>n = 311 papers | 56.3%<br>175                 | 19.0%<br>59               | 12.9%<br>40                       | 11.9%<br>37                    |
| all papers in the matrix<br>n = 2,311 papers    | 51.4%                        | 30.2%                     | 10.1%                             | 8.2%                           |

colour: the row's share against the column's share over all papers; dark is 8 points or more above, pale 8 or more below

**Figure 54: The profiles share an ordering, with different shares of new work. The matrix uses the 2,311 systematic papers with complete facet–value and remaining-work assignments. Rows: the facet families, with the number of papers carrying each. Columns: what the person still does. Each cell is the share of the row’s papers whose remaining work falls in the column family, with the count. Papers can belong to several mechanism-family rows; each has one remaining-work summary category in this analysis. Colour is the cell’s distance from the column’s share over all papers in the matrix, given in the last row. The model-assisted grouping is descriptive.**

The participant field also covers 2,071 papers: mixed groups account for 540 (26.1%), target practitioners 499 (24.1%), and crowdworkers 251 (12.1%); 382 (18.4%) are unspecified. Participants need not represent a system’s intended users, and missing descriptions do not establish broad applicability.

*C.3.1 Most process shifts rest on an author’s attribution.* Of 8,045 process shifts across 2,891 papers, 6,102 (75.8%) are author-attributed, 1,746 (21.7%) use measured contrasts, and 197 are reviewer inferences. Baselines also differ: 1,940 use a control condition, 869 state none, and the remainder refer to prior practice or an author’s assertion. Describing a changed workflow does not establish its effect size.

C.4 Mechanism Families and Remaining Work

The detailed profile complements the control/work comparison in Section 5.2.

D ANNUAL TRENDS AND VENUE RATES

These tests and counts support the comparisons in Section 5 and Section 6. Trends use AI-involved papers with usable codes for each field. Technology follows the AI a paper builds or uses, independently of publication year: generative AI includes hybrids and earlier neural generators; traditional AI/ML includes predictive, discriminative, and symbolic systems. The technology projection covers 3,488 classified papers; 645 remain unassigned. Each annual percentage uses papers with assigned technology and usable coding for that dimension in the same year. Count summaries combine 2019–2026; the annual lines separate the years. Alias disagreements are withheld for the affected field. Adjusted contrasts account for publication year and venue as described below.

D.1 Research-context trends

We model binary context membership with year and venue fixed effects, using venue-clustered standard errors with a finite-sample correction and Student’s *t* reference distribution. The linear probability coefficient estimates average within-venue change in percentage points per year. Holm adjustment controls the family-wise error rate across the 16 directional tests. Intervals are pointwise 95% confidence intervals. Separate categorical-year tests check temporal variation and departures from a straight trend; these are available with the analysis data. Results describe the coded review frame.

**Table 13: User control after omitting each major venue and on shared year-venue cells. Gaps are GenAI minus traditional AI/ML in percentage points. Venue-omission gaps are adjusted for year and venue, with pointwise 95% confidence intervals and Holm correction over all 51 outcomes in each fit. Shared-cell gaps are point estimates. Equal-venue gaps also have 95% venue-bootstrap intervals. The one-paper threshold keeps 82.8% of the coded papers, the five-paper threshold 40.8%.**

| Venue omission      | Gap [95% CI]           | Holm $p$                |
|---------------------|------------------------|-------------------------|
| Primary             | +13.8 [+8.9, +18.7]    | < .001                  |
| Omit CHI            | +12.8 [+6.9, +18.8]    | 0.005                   |
| Omit CSCW           | +13.6 [+8.4, +18.8]    | < .001                  |
| Omit EMNLP          | +14.3 [+9.4, +19.3]    | < .001                  |
| Omit IUI            | +13.9 [+8.5, +19.3]    | < .001                  |
| Omit TVCG           | +12.2 [+7.5, +16.9]    | < .001                  |
| Omit VIS            | +14.5 [+9.7, +19.4]    | < .001                  |
| Shared-cell weight  | At least one per group | At least five per group |
| Pooled papers       | +11.4                  | +17.3                   |
| Within-cell overlap | +12.7                  | +17.1                   |
| Equal cells         | +8.5                   | +16.6                   |
| Equal venues        | +8.4                   | +15.1                   |
| 95% interval        | [-2.1, +18.9]          | [+9.3, +21.1]           |
| Equal years         | +10.3                  | +18.8                   |

**Table 12: Venue-adjusted annual changes in research contexts, in percentage points per year. Primary estimates use 2019–2025; the last column includes available 2026 papers. Sensitivity intervals and adjusted tests are provided online.**

| Context                                 | Annual change [95% CI] | Adjusted $p$ | With 2026 |
|-----------------------------------------|------------------------|--------------|-----------|
| AI and machine learning practice        | -0.59 [-1.17, -0.01]   | 0.651        | -0.64     |
| Data analysis and visualization         | -0.35 [-0.90, +0.20]   | 1.000        | -0.46     |
| Software engineering and programming    | -0.35 [-0.73, +0.03]   | 0.881        | -0.28     |
| Scientific research practice            | +0.35 [-0.03, +0.72]   | 0.881        | +0.39     |
| Information seeking and knowledge work  | +0.41 [+0.14, +0.68]   | 0.065        | +0.33     |
| Writing, reading and language           | +0.22 [-0.10, +0.55]   | 1.000        | +0.30     |
| Design practice                         | -0.16 [-0.59, +0.27]   | 1.000        | -0.04     |
| Creative and media production           | +0.17 [-0.33, +0.67]   | 1.000        | +0.25     |
| Education and learning                  | +0.11 [-0.32, +0.54]   | 1.000        | +0.09     |
| Health and clinical care                | +0.27 [-0.25, +0.80]   | 1.000        | +0.25     |
| Accessibility and disability            | +0.29 [+0.07, +0.52]   | 0.176        | +0.09     |
| Everyday life and personal decisions    | -0.28 [-0.60, +0.04]   | 0.881        | -0.08     |
| Online platforms and civic information  | -0.09 [-0.48, +0.30]   | 1.000        | -0.03     |
| Physical and embodied activity          | -0.18 [-0.61, +0.26]   | 1.000        | -0.11     |
| Business, industry and work             | +0.29 [-0.03, +0.60]   | 0.881        | +0.14     |
| Public sector, law and societal systems | -0.04 [-0.20, +0.12]   | 1.000        | -0.11     |

## D.2 Technology comparisons

We compare binary code membership using a linear probability model with technology, categorical publication year, and venue fixed effects. Primary estimates use 2019–2025, and sensitivity models restrict the analysis to 2023–2025 or add partial 2026. Venue-clustered standard errors use a finite-sample correction and a Student  $t$  reference with one fewer degree of freedom than the number of venues, and model intervals are pointwise 95% confidence intervals. Holm correction covers all 51 context, activity, family, and category contrasts jointly, with each sensitivity window and venue omission corrected separately. We omit six major venues one at a time: CHI, CSCW, EMNLP, IUI, TVCG, and VIS, each of which supplies at least 5% of the coded papers for workflow, families, or categories in 2019–2025. The choice rests on coding denominators, not outcomes. Weighting checks reweight shared year-venue cells with at least one or at least five coded papers per technology by pooled paper count, within-cell overlap, equal cells, equal venues, or equal years. Intervals for equal-venue gaps come from 10,000 resamples of venues.

*User control.* User control and expression occurs in 634/1,072 GenAI papers (59.1%) and 624/1,381 traditional AI/ML papers (45.2%) in 2019–2025. Omitting each major venue gives gaps from +12.2 to +14.5 points, each with Holm-adjusted support (Table 13). On the 2,031 papers in shared year-venue cells, the gap is +12.7 with overlap weights and +8.4 when venues receive equal weight. Every weighting and cell threshold leaves it positive.

**Table 14: The other gaps given a direction in the text. Gaps are GenAI minus traditional AI/ML in percentage points. Primary gaps are adjusted for year and venue, with 95% confidence intervals. Equal-venue gaps use shared year–venue cells with at least one or at least five papers per technology, with 95% venue-bootstrap intervals. The two thresholds keep 82.7–85.4% and 40.3–46.6% of each coded population. Every gap keeps its sign under all five weightings at both thresholds.**

| Gap                                    | Primary [95% CI]     | Equal venues, papers per group |                     |
|----------------------------------------|----------------------|--------------------------------|---------------------|
|                                        |                      | At least one                   | At least five       |
| Tool generates new artifacts           | +16.6 [+12.7, +20.5] | +13.8 [+6.6, +22.3]            | +17.7 [+9.3, +29.6] |
| Tool generates responses and revisions | +9.4 [+5.7, +13.2]   | +10.2 [+4.7, +17.1]            | +7.6 [+0.8, +14.1]  |
| Tool calculates and models             | -11.5 [-15.5, -7.5]  | -16.2 [-25.8, -7.2]            | -16.2 [-25.6, -7.9] |
| Tool evaluates and classifies          | -9.7 [-14.6, -4.7]   | -2.6 [-12.7, +7.9]             | -13.2 [-21.5, -2.7] |
| Tool compares and finds patterns       | -6.7 [-9.7, -3.7]    | -7.0 [-14.3, -1.1]             | -1.2 [-9.5, +8.4]   |
| Generated alternatives to explore      | +4.8 [+3.0, +6.6]    | +4.3 [+0.9, +8.5]              | +1.5 [-1.7, +4.5]   |
| Completion and continuation of content | +3.0 [+1.4, +4.6]    | +2.1 [-0.6, +5.3]              | +2.1 [+0.6, +3.9]   |
| Person steers the loop                 | +3.4 [+1.2, +5.6]    | +6.7 [+2.4, +11.7]             | +0.3 [-5.0, +4.9]   |
| Person improves the system             | -5.9 [-7.7, -4.0]    | -5.3 [-9.3, -1.8]              | -6.4 [-10.9, -2.7]  |
| Organization and comparison (family)   | -8.2 [-12.5, -3.8]   | -12.2 [-20.4, -4.1]            | -8.1 [-17.5, +1.7]  |
| Ranked or prioritized surfacing        | -6.5 [-8.3, -4.7]    | -3.4 [-7.6, +0.3]              | -9.7 [-18.0, -3.7]  |

**Table 15: Mechanism-family trends, 2019–2025: 2,453 coded papers across 39 venues. Changes are percentage points per year; brackets show 95% confidence intervals.**

| Family                        | Average annual change [CI] | Holm <i>p</i> | Change in technology difference [CI] | Holm <i>p</i> |
|-------------------------------|----------------------------|---------------|--------------------------------------|---------------|
| Explanation and visibility    | +1.16[−0.70, +3.01]        | 1.000         | +1.47[−0.86, +3.80]                  | 1.000         |
| Organisation and comparison   | +0.08[−1.70, +1.86]        | 1.000         | +0.49[−2.04, +3.01]                  | 1.000         |
| User control and expression   | −1.02[−2.47, +0.42]        | 0.961         | +0.53[−1.12, +2.17]                  | 1.000         |
| Guidance and feedback         | +1.61[+0.72, +2.50]        | 0.007         | +1.91[−0.38, +4.20]                  | 0.692         |
| Collaboration and perspective | +1.13[+0.53, +1.73]        | 0.005         | +2.15[−0.02, +4.32]                  | 0.414         |

**Table 16: Mechanism-family trends, 2023–2025: 1,391 coded papers across 39 venues. Changes are percentage points per year; brackets show 95% confidence intervals.**

| Family                        | Average annual change [CI] | Holm <i>p</i> | Change in technology difference [CI] | Holm <i>p</i> |
|-------------------------------|----------------------------|---------------|--------------------------------------|---------------|
| Explanation and visibility    | −0.34[−5.73, +5.06]        | 1.000         | +1.44[−6.17, +9.05]                  | 1.000         |
| Organisation and comparison   | −2.86[−6.65, +0.94]        | 0.952         | −2.86[−9.68, +3.96]                  | 1.000         |
| User control and expression   | −3.41[−6.55, −0.27]        | 0.273         | −0.41[−6.16, +5.34]                  | 1.000         |
| Guidance and feedback         | +1.92[−2.69, +6.54]        | 1.000         | +3.49[−3.80, +10.77]                 | 1.000         |
| Collaboration and perspective | +2.84[+1.27, +4.41]        | 0.007         | +6.28[+1.67, +10.89]                 | 0.080         |

*Facets and work codes.* The facet and work-code gaps in the main text use the same model on each paper’s latest coding, with a normal reference for intervals. Of the 2,988 technology-classified papers from 2019–2025, 2,532 have facet codes (1,143 GenAI and 1,389 traditional AI/ML), 2,119 have codes for delegated work, and 2,526 have codes for the work people keep. Holm correction runs across the 101 facets coded on at least 40 papers and, separately, across the 24 work codes on at least 30 papers. A gap gets a direction in the text only if it has Holm-adjusted support, keeps its sign under all five weightings at both thresholds, and has an equal-venue interval that excludes zero at one threshold or both (Table 13 and Table 14). Fitted separately for the two models that coded most papers, the control gap is +13.9 (95% CI 8.8 to 19.0; 2,160 papers) and +11.1 (−1.1 to 23.3; 203 papers).

### D.3 Temporal change in mechanism families

We distinguish average annual change in family prevalence from annual change in the GenAI–traditional AI/ML difference. Both come from a linear probability model with technology, publication year, their interaction and venue fixed effects. Average change holds the technology mixture fixed at 43.75% GenAI (the original primary cohort’s reference share), separating it from shifts in the composition of the corpus. Positive gap change means movement toward a more positive GenAI difference; it need not enlarge the absolute gap. Venue-clustered intervals use finite-sample correction and a Student *t* reference. Holm correction covers all ten tests within each window; the 95% confidence intervals are pointwise. These are linear summaries of coded design prevalence, not tests of learning or causal technology effects. Partial 2026 is excluded.

## **D.4 Annual distributions**

Research contexts: volume and annual share

3,307 coded papers with assigned AI technology - categories overlap

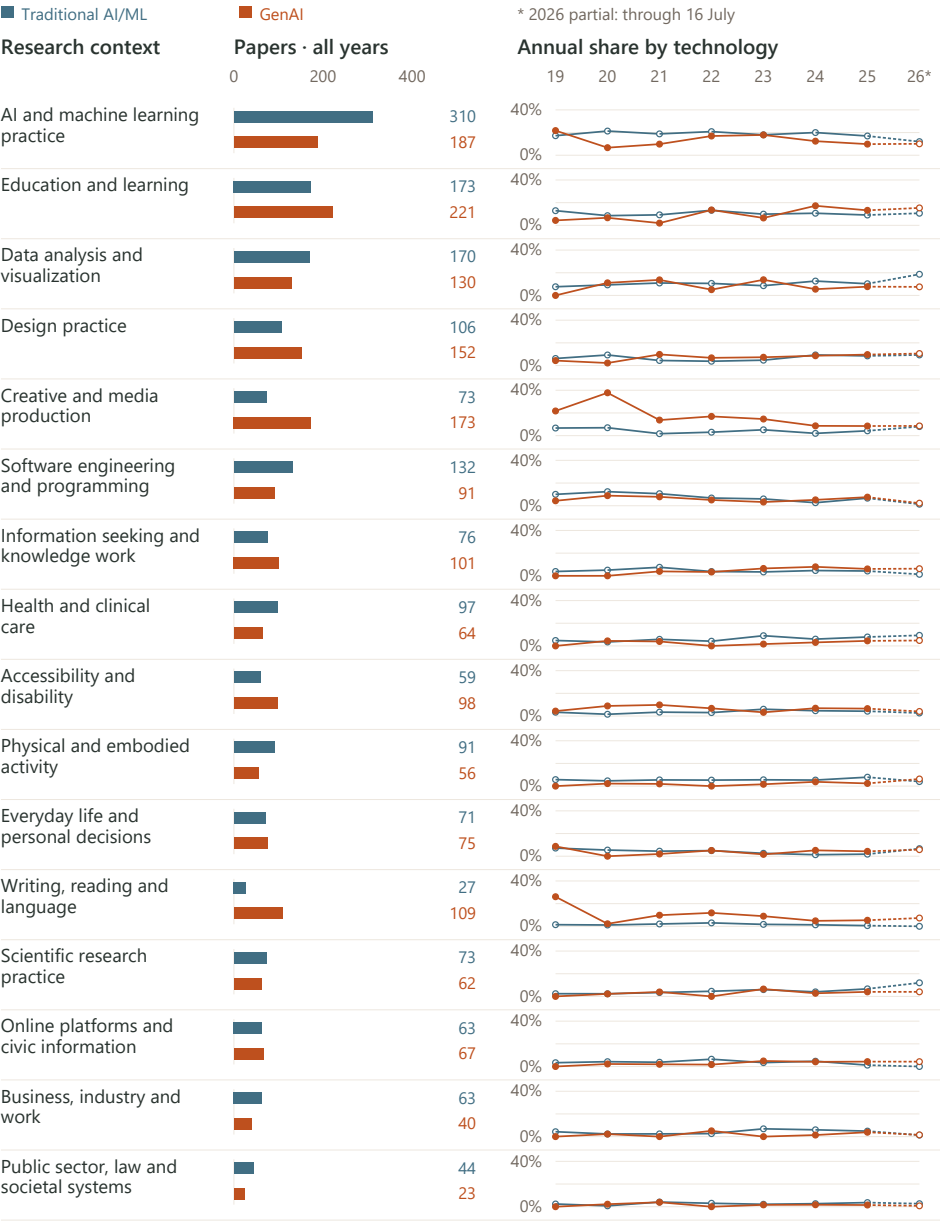
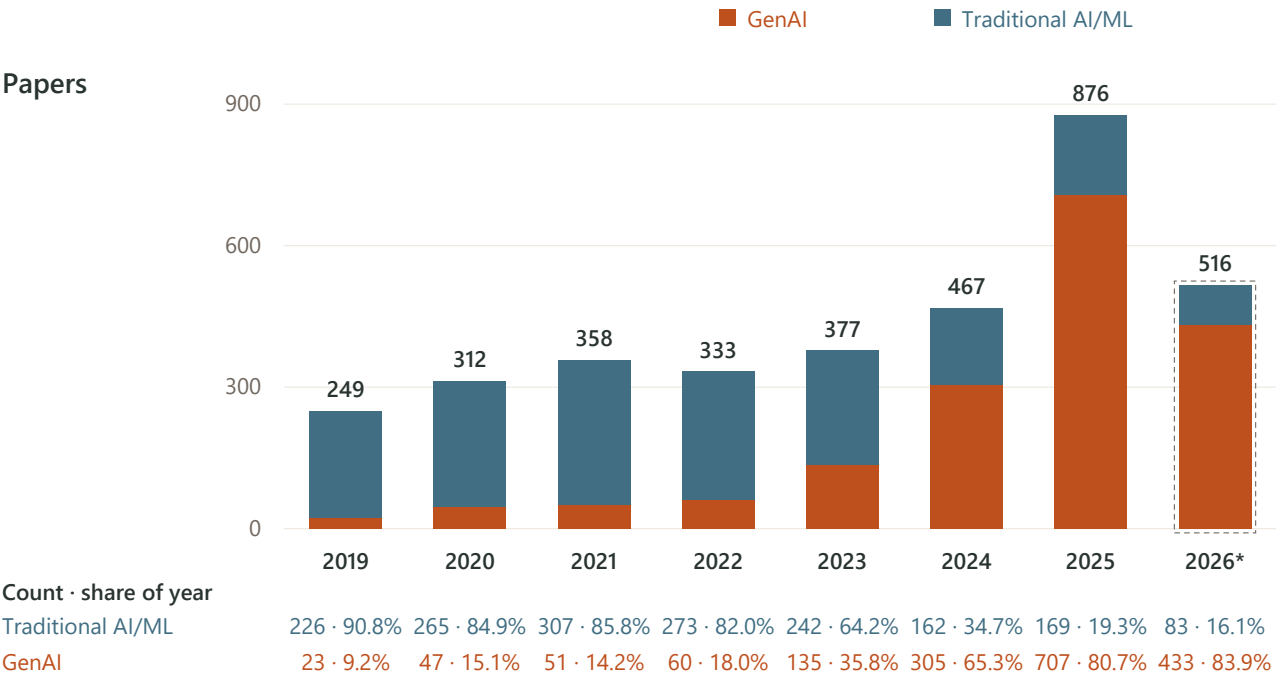


Figure 55: Research contexts by AI technology. Bars show paper counts across all years; lines show each context’s share of its technology group per year, with the coded papers behind those shares below. Contexts overlap. Shares are not adjusted; Table 12 gives the venue-adjusted tests. Only papers with usable context coding and a known technology are included.

# AI technologies by year

3,488 classified papers · bar height shows volume; percentages show composition

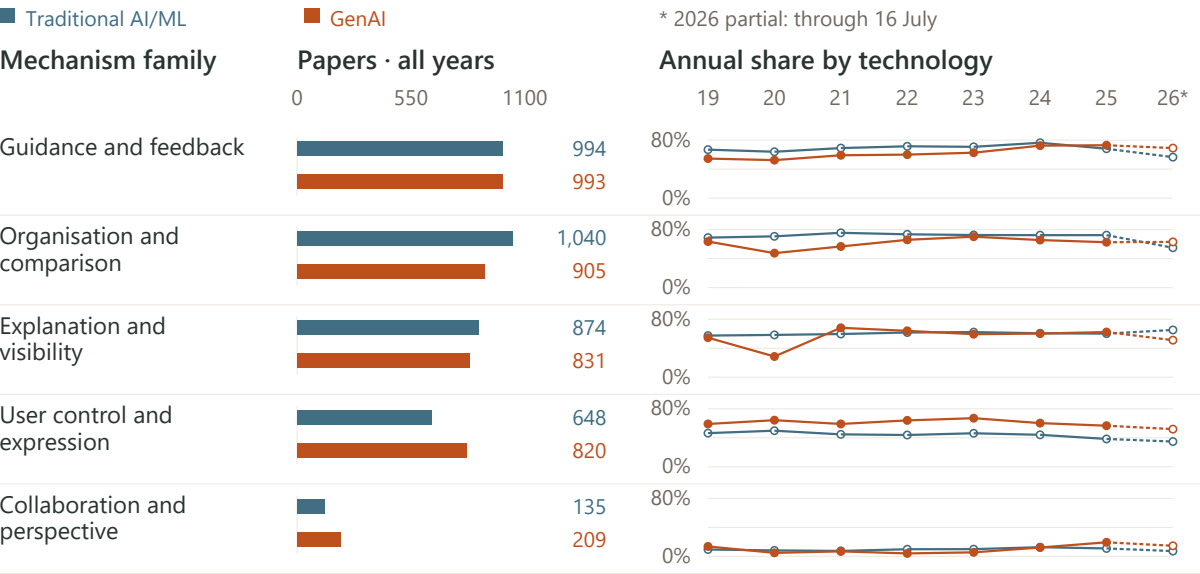


GenAI includes hybrids and earlier generative models. \* 2026 through 16 July; September update pending.

Figure 56: AI technologies by year. Stacked bar heights encode paper counts, with annual totals above. Rows below give each segment’s count and within-year percentage. Classification follows the technology used; 2026 is partial (collected in July). The figure omits 645 papers without a resolved technology classification.

## Mechanism families: volume and annual share

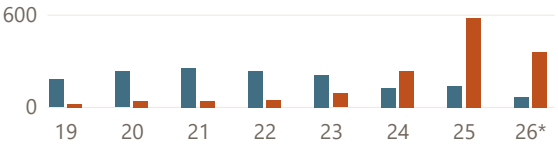
2,880 coded papers with assigned AI technology · categories overlap



### Papers behind the annual shares

Each bar counts coded papers available in that technology group and year.

Early GenAI groups are small.



GenAI includes earlier generators and hybrids. These distributions describe designs, not effectiveness.

**Figure 57: Mechanism families by AI technology.** Bars encode counts across all years; lines compare annual shares within each technology group. The lower bars show annual coding denominators. Families overlap. Only papers with usable family codes and a known technology are included. Shares are not adjusted; Table 15 gives the venue-adjusted trends.



## Mechanism categories: volume and annual share

2,874 coded papers with assigned AI technology · categories overlap

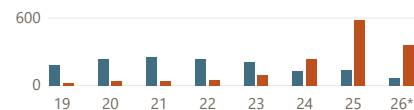


### Papers behind the annual shares

Each bar counts coded papers available in that technology group and year.

Early GenAI groups are small.

GenAI includes earlier generators and hybrids. These distributions describe designs, not effectiveness.



**Figure 58: Mechanism categories by AI technology.** Bars encode counts across all years; lines show annual shares within each technology group. Categories overlap, and 2026 covers papers published by 16 July. Shares are not adjusted for year or venue. Only papers with usable category codes and a known technology are included.

## D.5 Learning overlays

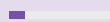
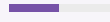
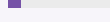
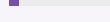
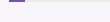
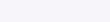
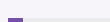
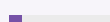
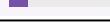
These counts cover the 3,417 papers with usable category coding, of the 4,133 papers in the analysis at 28 September 2026, including papers without an assigned AI technology. Learning scope means the paper is in the learning review; measured means the paper measured learning explicitly. Learning scope and Measured are counted separately within each category; missing or conflicting labels are omitted from the corresponding column.

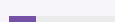
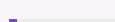
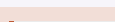
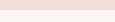
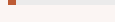
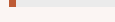
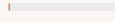
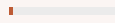
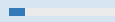
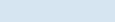
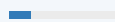
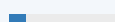
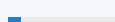
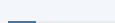
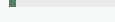
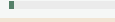
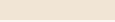
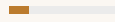
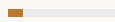
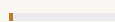
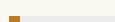
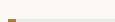
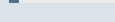
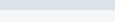
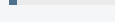
**Table 17: Papers per mechanism category: all papers, papers in learning scope, and papers that measured learning. Counts are distinct papers; categories overlap.**

| Category                            | Papers | Learning scope | Measured |
|-------------------------------------|--------|----------------|----------|
| Explaining system reasoning         | 976    | 213            | 97       |
| Linking to sources                  | 683    | 119            | 34       |
| Summarising and labelling content   | 423    | 107            | 38       |
| Visual and multimodal form          | 469    | 155            | 55       |
| Organising and structuring          | 1,382  | 293            | 94       |
| Comparing alternatives              | 731    | 166            | 61       |
| Search, filtering and ranking       | 776    | 177            | 71       |
| Notes, history and memory           | 612    | 170            | 68       |
| Specifying intent                   | 820    | 161            | 56       |
| Direct manipulation                 | 744    | 138            | 56       |
| Choosing among candidates           | 487    | 104            | 37       |
| What-if and probing                 | 171    | 79             | 32       |
| Recommending next steps             | 1,000  | 330            | 137      |
| Feedback and assessment             | 906    | 318            | 122      |
| Instruction and paced help          | 688    | 347            | 156      |
| Workflow structure                  | 660    | 173            | 70       |
| Prompting the person to reflect     | 506    | 242            | 91       |
| Shared views and perspective-taking | 410    | 163            | 58       |

## D.6 Venue candidates and current inclusions

**Table 18: Venue candidates and AI-involved inclusions in the 16 July 2026 frame, assessed through 28 September 2026. The 37,978 venue-year records represent 37,660 candidate identities. Inclusion (%) is Included divided by Initial candidates. Papers without AI, and papers whose identity or scope is unresolved, are not in Included. Venue memberships can overlap; group and overall totals count each identity once.**

| Venue                        | Initial candidates | Included | Inclusion%                                                                                  |
|------------------------------|--------------------|----------|---------------------------------------------------------------------------------------------|
| <b>HCI &amp; interaction</b> | 17,436             | 2,615    |  15.0% |
| 56.1% of included corpus     |                    |          |                                                                                             |
| CHI                          | 7,639              | 1,152    |  15.1% |
| UIST                         | 860                | 195      |  22.7% |
| IUI                          | 687                | 321      |  46.7% |
| DIS                          | 1,471              | 167      |  11.4% |
| CSCW                         | 2,412              | 212      |  8.8%  |
| C&C                          | 301                | 44       |  14.6% |
| TOCHI                        | 478                | 38       |  7.9%  |
| TiiS                         | 230                | 95       |  41.3% |
| EICS                         | 174                | 23       |  13.2% |
| HAI                          | 214                | 26       |  12.1% |
| ECSCW                        | 205                | 4        |  2.0%  |
| GROUP                        | 142                | 6        |  4.2%  |
| ASSETS                       | 387                | 58       |  15.0% |
| HRI                          | 630                | 96       |  15.2% |
| IVA                          | 222                | 39       |  17.6% |

| Venue                                                      | Initial candidates | Included | Inclusion%                                                                                  |
|------------------------------------------------------------|--------------------|----------|---------------------------------------------------------------------------------------------|
| CHIIR                                                      | 312                | 40       |  12.8%   |
| HCOMP                                                      | 112                | 28       |  25.0%   |
| IJHCS                                                      | 960                | 71       |  7.4%    |
| <b>Design &amp; creativity</b><br>2.2% of included corpus  | 2,502              | 102      |  4.1%    |
| ICCC                                                       | 422                | 28       |  6.6%    |
| Design Science                                             | 232                | 15       |  6.5%    |
| Design Studies                                             | 234                | 1        |  0.4%    |
| IDETC                                                      | 1,614              | 58       |  3.6%    |
| <b>Visualization</b><br>19.5% of included corpus           | 6,336              | 910      |  14.4%   |
| VIS                                                        | 1,772              | 352      |  19.9%   |
| EuroVis                                                    | 537                | 84       |  15.6%   |
| TVCG                                                       | 4,073              | 460      |  11.3%   |
| VL/HCC                                                     | 255                | 65       |  25.5%   |
| <b>Graphics</b><br>2.9% of included corpus                 | 2,385              | 137      |  5.7%    |
| SIGGRAPH                                                   | 1,673              | 104      |  6.2%    |
| SIGGRAPH Asia                                              | 712                | 33       |  4.6%    |
| <b>Learning &amp; education</b><br>4.9% of included corpus | 2,834              | 228      |  8.0%    |
| L@S                                                        | 181                | 34       |  18.8%   |
| LAK                                                        | 655                | 87       |  13.3%   |
| AIED                                                       | 1,168              | 41       |  3.5%    |
| IJAIED                                                     | 347                | 34       |  9.8%    |
| EDM                                                        | 483                | 32       |  6.6%    |
| <b>Software engineering</b><br>7.6% of included corpus     | 3,944              | 354      |  9.0%    |
| CHASE                                                      | 249                | 18       |  7.2%    |
| ICSE                                                       | 1,038              | 129      |  12.4%   |
| FSE                                                        | 1,297              | 88       |  6.8%    |
| ASE                                                        | 1,360              | 119      |  8.8%   |
| <b>AI &amp; language</b><br>6.8% of included corpus        | 2,223              | 319      |  14.3% |
| ACL                                                        | 731                | 133      |  18.2% |
| EMNLP                                                      | 794                | 146      |  18.4% |
| RecSys                                                     | 698                | 40       |  5.7%  |
| <b>Overall (distinct identities)</b>                       | 37,660             | 4,665    |  12.4% |

## E VOCABULARY GROWTH DURING READING

These curves show how each vocabulary grew as cards were read, so a reader can judge how complete the codes are. The vocabulary built from the first 3,295 mechanism cards gained 12 codes across the next 1,754 papers, and each new code was assigned to an existing category and family. All open fields were checked on 821 of those papers; the other batches checked mechanism facets, values, and domains (Figure 59).

### E.1 Reading the Curves

We count distinct values after each batch of ten cards, in reading order, and scale each curve to its own final vocabulary. Codes, categories, and families are drawn where those levels exist; categories and families were assigned after coding, so they can reach their final size early in a curve. Reference lists in the lineage cards are open-ended, while their relation options are fixed by the card format. A flat segment means that no new label appeared in those cards, not that unread papers would add none.

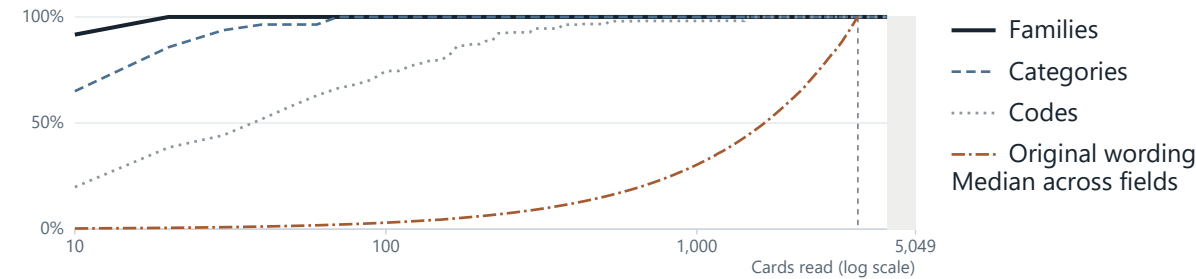
### E.2 Stream 1 mechanism cards: Interface and Outcome Fields

These grids show the variation the overview hides. Every panel states its field and final count, and lines are drawn only for the levels present in that field.

# Vocabulary growth across three card types

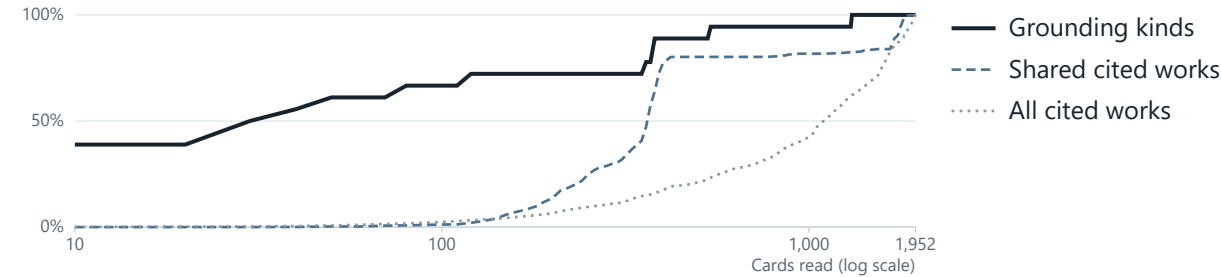
Share of the recorded vocabulary encountered; paper counts use a log scale

## Stream 1 · Mechanism cards · 5,049



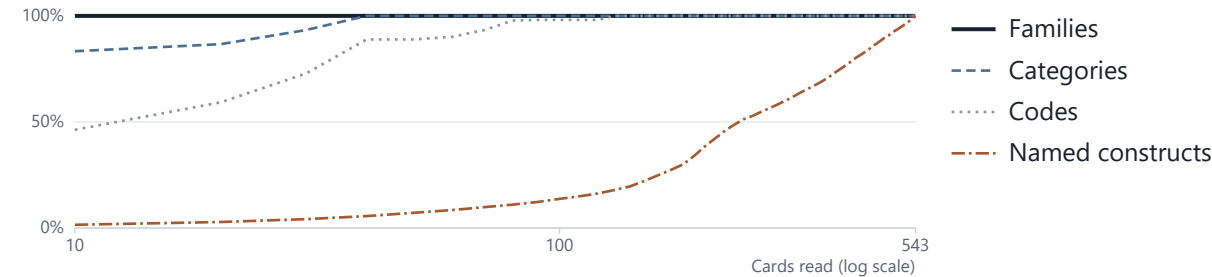
After 3,295: 12 new codes in 1,754 cards; field coverage varies in the shaded region.

## Stream 1 · Theory-lineage cards · 1,952



Cited works remain an open vocabulary; grounding kinds are a fixed schema.

## Stream 2 · Knowledge-claim cards · 543



Frozen coding covers 542 cards; named constructs cover all 543. Curves stop at their coverage.

**Figure 59: Vocabulary growth across the three card types.** Panels count cards in reading order on logarithmic axes: 5,049 mechanism cards, 1,952 theory-lineage cards, and 543 Stream 2 cards. The Stream 1 panels include cards of non-AI comparison papers. Shading marks uneven field coverage, and median curves stop before it.

## Codes, categories and families reached, by cards read, every field



Figure 60: Stream 1, mechanism cards: vocabulary growth in each open field. Each panel separates codes, categories, and families where present, in reading order.

E.3 Stream 1 theory-lineage cards: Cited Sources and Their Roles

The curve covers the 1,952 newly written lineage cards; reused cards appear in the gallery but not in this sequence.

Stream 1 theory-lineage cards: references and relations

Reading order on a log scale; each curve reaches 100% of its own recorded vocabulary.

Relations, positions and grounding kinds use fixed options: their flat tails are not evidence of open-code saturation.

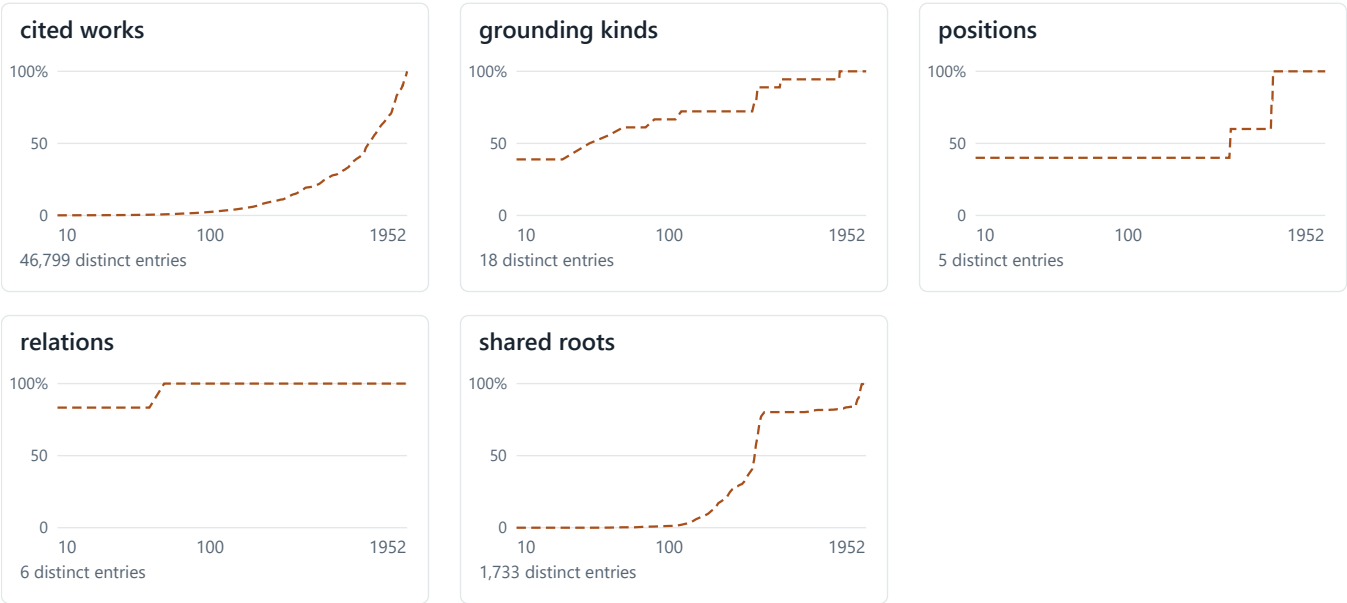


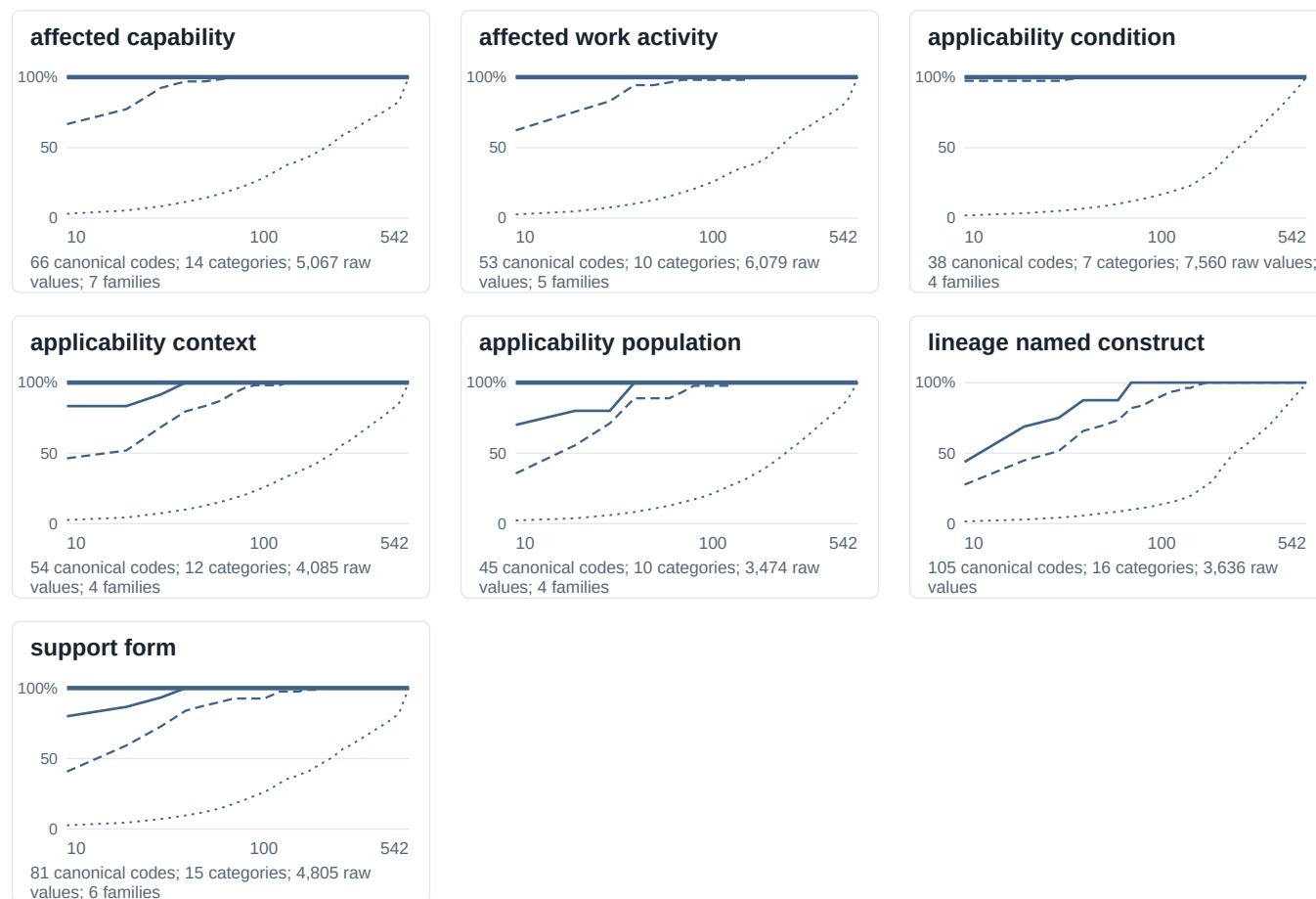
Figure 61: Stream 1, theory-lineage cards: reference lists keep growing after relation options are exhausted. Fixed options and open-ended lists are separate panels because they close in different ways.

## E.4 Stream 2: Claims and Conditions

### Stream 2: each coded field

Reading order on a log scale; each curve reaches 100% of its own recorded vocabulary.

Dotted: raw values. Dashed: canonical codes. Solid: categories. Thick: families, where present.



**Figure 62: Stream 2: vocabulary growth differs by coded field. Codes, categories, and families are drawn only where that level exists.**

Stream 2's seven open fields cover 542 cards and 35,699 observations. Of the 449 codes, 448 are used, and no new code entered in the last 16 coding batches. A flat curve can still hide a missing code: no population code names a studied unit that is or includes a machine, and 18 assignments flag this while using codes for people or artifacts.

## E.5 New Codes Within the First Coding

Within the first coding of each open field, the last new code appeared between 2.5% and 48.7% of the way through the fields whose coding order was kept (Figure 63). Three fields still have low-confidence assignments for more than a quarter of their observations, so a vocabulary can stop growing and remain too coarse.

Where did the recorded vocabulary stop expanding?

Colour ends at the last new code. The pale remainder reuses codes already encountered.

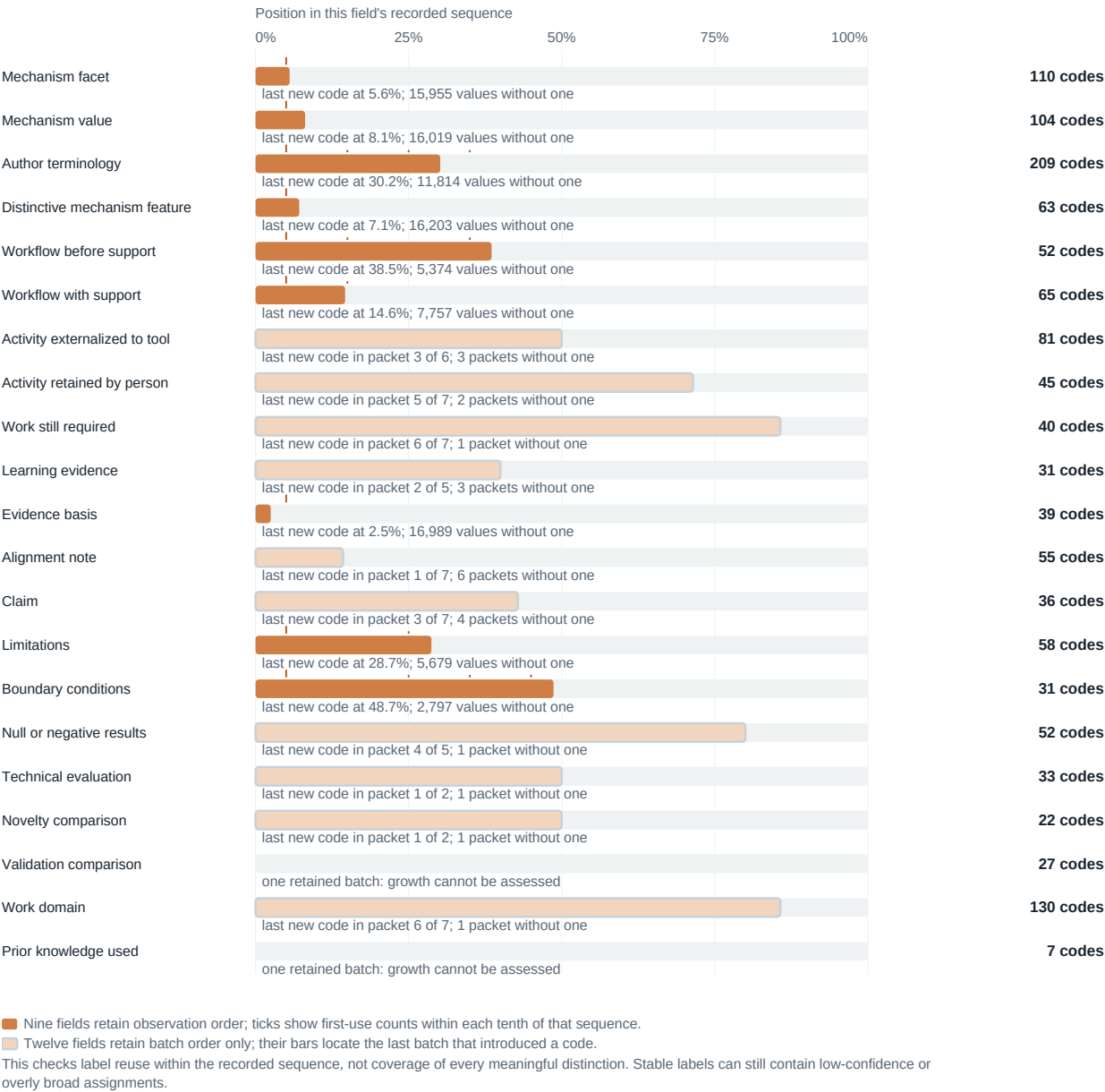


Figure 63: New-code creation within each field of the first coding. Bars locate the last new code in the recorded coding sequence; ticks show new codes by tenth of the sequence. Where only batch order survives, the figure reports batch resolution.